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Section 1: Introduction, purpose, hardware, design

SAIL, which stands for Security and AI Infrastructure Lab, is a self hosted infrastructure and automation environment designed to support hands-on learning in DevOps, cybersecurity, artificial intelligence, and systems operations. The environment uses Raspberry Pi systems and an Apple Silicon computer connected through a local home network to create a small scale computing environment where infrastructure, monitoring, automation, and AI related systems can be configured and operated together.

The purpose of the environment is to provide hands-on experience with deployment, monitoring, troubleshooting, automation, logging, security tooling, and operational maintenance through the use of real hardware and locally hosted services. Multiple systems and services run across the shared network, including containerized applications, monitoring tools, log collection systems, AI related services, and automation workflows. The environment also allows operational failures, alert generation, troubleshooting activities, and recovery procedures to be tested and documented over time. All systems connect through a local wired home network using Ethernet connections. The environment runs on existing hardware using free and locally hosted software and services.

SAIL Hardware

SAIL uses the following hardware:

Device	Purpose
Raspberry Pi 4 8GB x3	Kubernetes cluster nodes
MacBook	Administration workstation, AI services, documentation, and automation tools
1TB External SSD	Persistent storage for cluster data, logs, metrics, backups, and documentation assets
Ethernet cables	Network connectivity between systems
Network switch	Local wired network connection between devices

SAIL Storage Design

SAIL uses a single 1TB external SSD as centralized storage for the environment. The storage drive connects to pi-node-1 and stores long term operational and project data generated by the environment.

The centralized storage supports:

- Monitoring data
- Logs
- Kubernetes persistent storage
- AI model files
- Documentation

- Screenshots and diagrams
- Backup files
- Project assets

The Raspberry Pi systems maintain their own operating system storage while the external SSD attached to pi-node-1 stores larger persistent datasets and operational records generated by the environment.

SAIL Network Design

All SAIL systems operate on a local wired network inside the home environment. Each device connects through Ethernet to improve stability and simplify troubleshooting.

The systems use the following names:

System	Role
MacBook	Admin workstation, AI host, documentation, kubectl, Ansible, Git
pi-node-1	k3s control plane, schedulable worker node, and external storage host
pi-node-2	k3s worker
pi-node-3	k3s worker

SAIL Operational Goals

The primary purpose of SAIL is to provide a controlled environment for developing and evaluating operational, security, and AI-assisted investigation workflows.

The environment is intended to support:

- Kubernetes administration and operations
- Infrastructure monitoring and observability
- Centralized logging and event collection
- Runtime security monitoring
- Vulnerability assessment
- Configuration automation
- Failure testing and recovery exercises
- AI-assisted analysis of operational and security events
- Hands-on DevOps and cybersecurity learning

Rather than serving only as a Kubernetes cluster, SAIL functions as a platform for generating, detecting, investigating, and responding to infrastructure and security events. The environment allows operational failures, configuration issues, security alerts, and other events to be intentionally introduced and analyzed using monitoring, logging, security, and AI tooling.

The long-term objective is to develop repeatable workflows that demonstrate how modern infrastructure, security operations, and AI-assisted analysis can be combined within a self-hosted environment.

Section 2: Software choices and architecture decisions

Technology	Purpose	Reason for Selection
k3s/containerd	Kubernetes orchestration and container runtime	Lightweight Kubernetes platform with an integrated container runtime, well suited for Raspberry Pi systems
Ollama	Local AI model hosting	Runs AI models locally on Apple Silicon hardware
Open WebUI	AI web interface	Browser based access to local AI models
Prometheus	Metrics collection	Tracks system and service health
Grafana	Monitoring dashboards	Visualizes metrics and operational data
Loki	Log collection	Centralized log storage and searching
Falco	Runtime security monitoring	Detects suspicious activity in containers and Linux systems
Trivy	Vulnerability scanning	Scans containers and configurations for known issues
Ansible	System configuration automation	Reduces repetitive setup work across systems
Docker	Container image development and testing	Building/testing the Orbit image during development.
GitHub	Documentation and configuration tracking	Stores notes, configurations, diagrams, and troubleshooting records

Architecture Overview

The environment is organized around a Kubernetes cluster running on three Raspberry Pi systems. One Raspberry Pi serves as the k3s control plane and also participates as a worker node, while the remaining Raspberry Pi systems serve as worker nodes. Kubernetes schedules workloads across the available cluster nodes based on resource availability and configuration. The MacBook functions as an administration workstation and hosts AI related services, documentation assets, and management tools used to operate the environment. The environment uses a local wired network and centralized storage provided through an external SSD connected to pi-node-1. All services operate locally using existing hardware and free software.

Section 3: Operating system setup

The SAIL environment uses Raspberry Pi OS Lite (64-bit) on all Raspberry Pi systems. Raspberry Pi OS was selected because it is the operating system maintained specifically for Raspberry Pi hardware and provides broad compatibility with the tools planned for the environment, including k3s, containerd, Prometheus, Grafana, Loki, Falco, Trivy, Docker, and

Ansible. During the planning phase, alternative operating systems were evaluated, including Ubuntu Server. Both options appeared capable of supporting the planned architecture.

Raspberry Pi OS was ultimately selected because the primary objective of SAIL is to provide a stable platform for infrastructure, security, monitoring, automation, and AI-assisted investigation workflows. As the operating system designed specifically for the hardware used in the project, Raspberry Pi OS offers strong hardware support, extensive community documentation, lower resource overhead, and broad compatibility with the planned software stack.

Operating System Installation

The following are performed from a computer connected to the SAIL environment.

Step 1: Download Raspberry Pi Imager

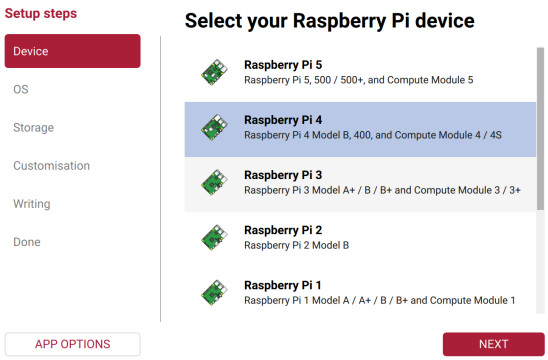
1. Open a web browser. Navigate to: https://www.raspberrypi.com/software/
2. Download Raspberry Pi Imager.
3. Install the application.
4. Launch Raspberry Pi Imager.

Step 2: Prepare the First MicroSD Card

1. Insert a MicroSD card into the computer.
2. Verify the operating system detects the card before proceeding.

Step 3: Select the Raspberry Pi Device

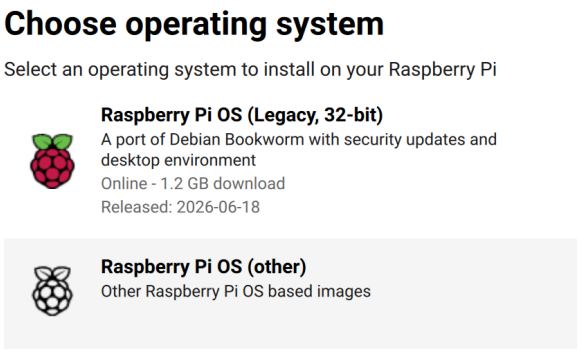
1. Within Raspberry Pi Imager:
    a. Select **Device**
    b. Select **Raspberry Pi 4**



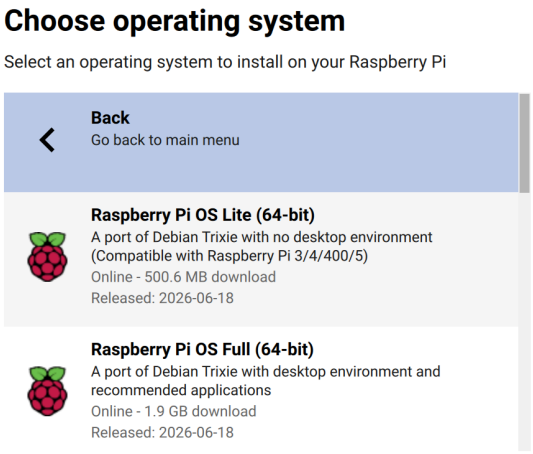
    c. Choose **NEXT**

Step 4: Select the Operating System

1. Select **Raspberry Pi OS (other)**

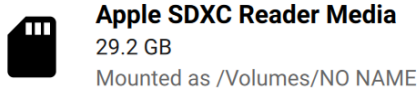


2. Select **Raspberry Pi OS Lite (64-bit)**



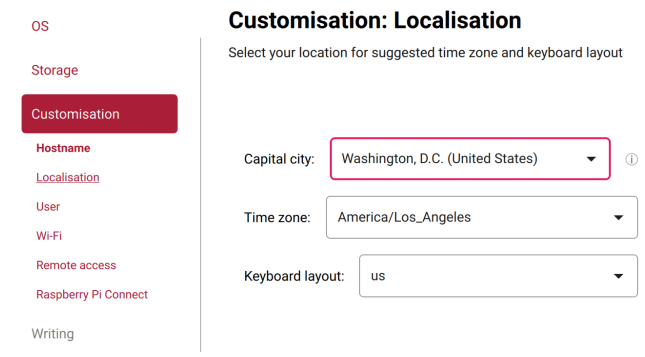
Step 5: Select Storage

1. Select the inserted MicroSD card

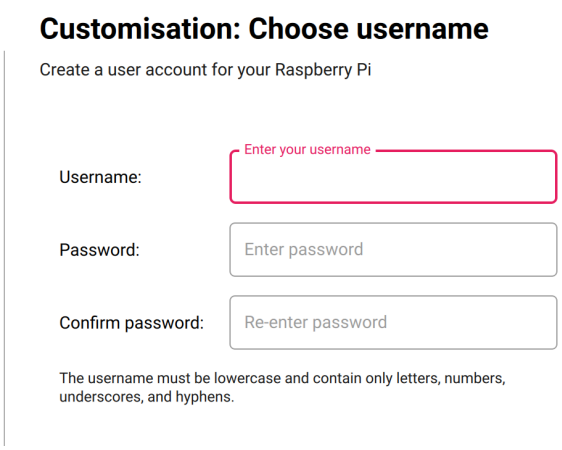


Step 6: Configure Operating System Customization

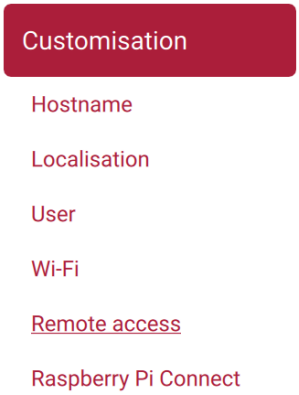
1. Enter Hostname: pi-node-1
2. Enter Capital city, Time zone, and keyboard layout



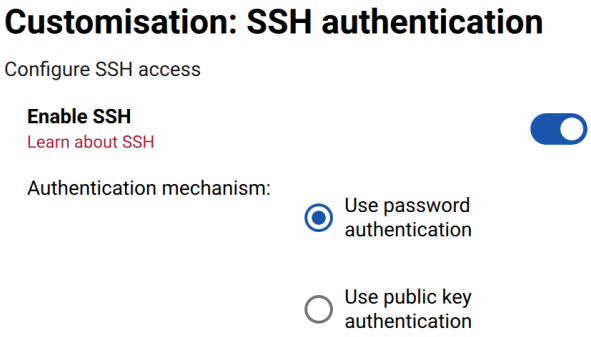
3. Username: <your username>
4. Password: <your password>



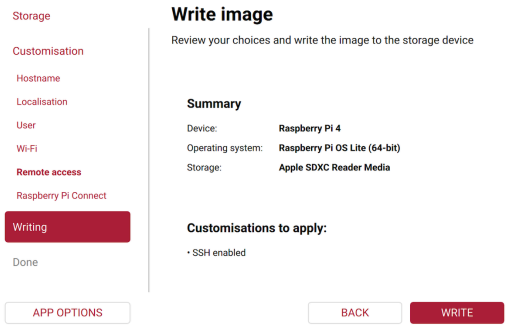
5. Click on Remote Access



6. Enable: SSH



7. Click writing, confirm settings and click **WRITE**



Step 7: Write the Image

1. Wait for image creation and verification to complete.
2. Label the MicroSD card: pi-node-1

Step 8: Create Remaining Node Images

1. Repeat Steps 2 through 7.
2. For the second card use: Hostname: pi-node-2
3. For the third card use: Hostname: pi-node-3
4. Label each completed MicroSD card.

Step 9: Install the MicroSD Cards

1. Insert the appropriate MicroSD card into each Raspberry Pi.
2. Verify that each Raspberry Pi receives the card matching its hostname.

Step 10: Boot the Cluster

1. Verify:
    a. Ethernet cables are connected
    b. The SSD is connected to pi-node-1
    c. Power supplies are connected
2. Apply power to the Raspberry Pi systems.
3. Allow several minutes for the initial boot process to complete.

Step 11: Identify Assigned IP Addresses

1. Log in to the router or gateway administration interface.
2. Locate the Raspberry Pi systems in the connected devices list.
3. Record for each node:
    a. Hostname
    b. MAC Address
    c. IP Address

Step 12: Verify SSH Connectivity

1. Open a terminal on the computer.
2. Connect to pi-node-1: ssh <username>@<ip-address>
    a. Example: ssh sailadmin@10.0.0.20
3. When prompted: Are you sure you want to continue connecting (yes/no/[fingerprint])?
    a. Type: yes
4. Enter the password configured during imaging.
5. Verify a successful login.
6. Repeat for pi-node-2 and pi-node-3.

Step 13: Update Raspberry Pi OS

1. Run: sudo apt update
   sudo apt full-upgrade -y
2. Wait for updates to complete.

Step 14: Reboot the Node

1. Run: sudo reboot
2. Wait approximately one minute.
3. Reconnect using SSH.
4. Repeat Steps 13 and 14 on all three nodes.

Step 15: Verify Hostname Configuration

1. Run: hostname
2. Verify the hostname matches the intended node name.

Step 16: Verify Operating System Installation

1. Run: cat /etc/os-release
2. Verify that Raspberry Pi OS Lite (64-bit) is installed.

Step 17: Record Baseline System Information

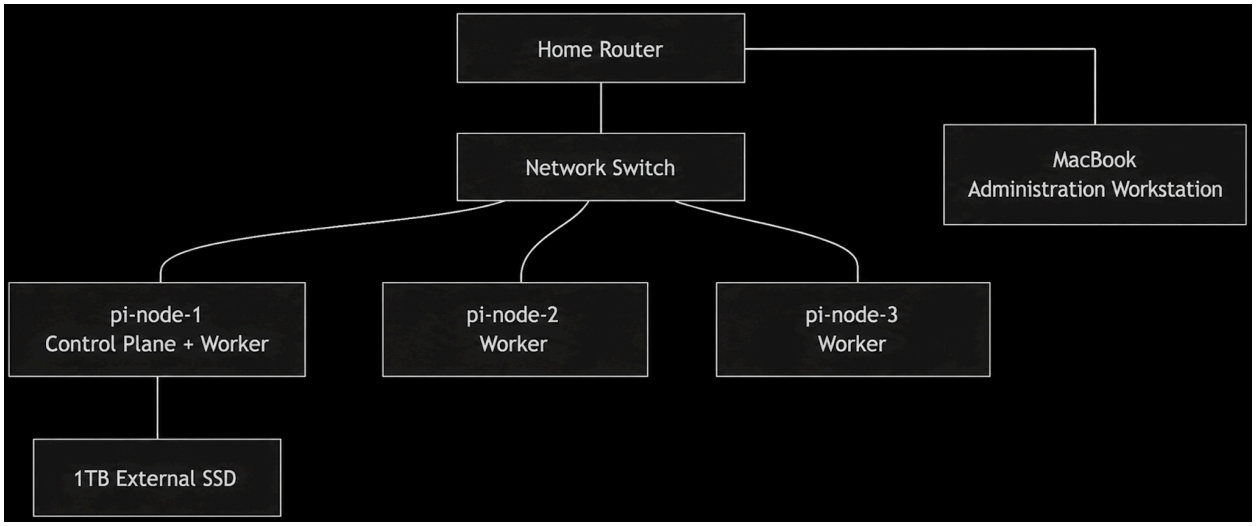
1. Run: free -h
2. Record available memory.
3. Run: df -h
4. Record available storage.This information is used later when evaluating cluster resource utilization and monitoring data.

Operating system installation is complete when:

- All three Raspberry Pi systems boot successfully
- SSH access is verified
- Operating system updates are installed
- Hostnames are configured correctly
- Network connectivity is verified
- Baseline system information has been recorded

Section 4: Physical setup and wiring

Network Topology Diagram



Hardware Layout

Physical Tower Layout

Top

Netgear GS105E

```
┌──────────────────────────┐
│    pi-node-1             │
├──────────────────────────┤
│    pi-node-2             │
├──────────────────────────┤
│    pi-node-3             │
├──────────────────────────┤
│   SSD hard drive         │
│   Anker Charger          │
└──────────────────────────┘
```

Bottom

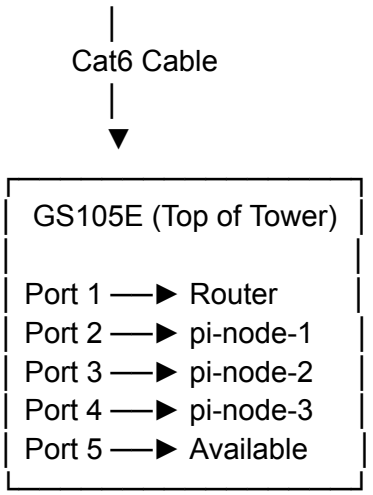
Rear

```
┌──────────────────────────┐
│    Fan 1                 │
├──────────────────────────┤
│    Fan 2                 │
└──────────────────────────┘
```

Hardware Wiring

Ethernet Wiring

Home Router



Power Wiring

Anker Charger

USB-C #1 ──────────► pi-node-1

USB-C #2 ──────────► pi-node-2

USB-C #3 ──────────► pi-node-3

pi-node-1 USB 3.0 ──► 1TB SSD

Fan 1

Red ──────────► pi-node-1 GPIO Pin 4

Black──────────► pi-node-1 GPIO Pin 6

Fan 2

Red ──────────► pi-node-3 GPIO Pin 4

Black──────────► pi-node-3 GPIO Pin 6

Ethernet Cable Construction

The SAIL environment uses custom Ethernet cables to connect the Raspberry Pi cluster nodes, network switch, and administrative workstation. Building custom cables allows cable lengths to be matched to the physical layout of the environment while providing hands-on experience with basic network infrastructure.

Materials

- Cat5e or Cat6 Ethernet cable

- RJ45 connectors
- RJ45 crimping tool
- Cable stripper
- Cable tester

Step 1: Measure Cable Length

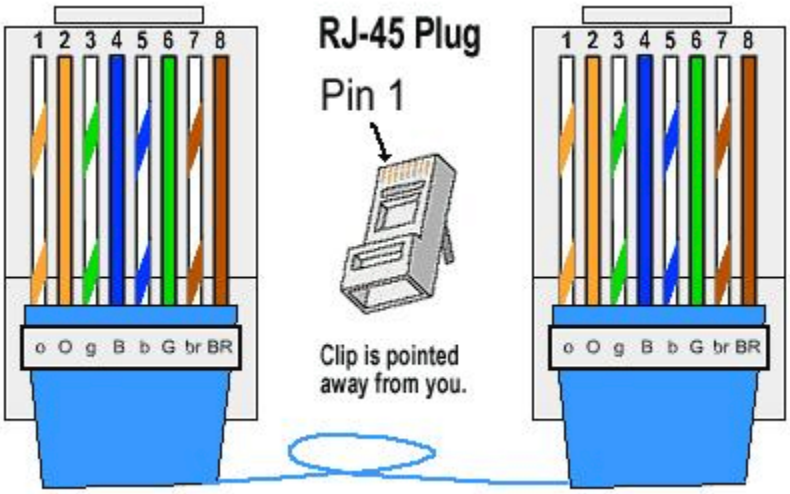
1. Measure the distance between the devices to be connected.
2. Add approximately 12 inches (30 cm) of extra cable to allow for routing and future adjustments.
3. Cut the cable to the desired length.

Step 2: Strip the Cable Jacket

1. Using the cable stripper, remove approximately 1 inch (2.5 cm) of the outer cable jacket from the end of the cable.
2. Take care to avoid damaging the internal conductors.

Step 3: Arrange Conductors

1. Untwist the wire pairs and arrange them according to the T568B standard:
    a. White/Orange
    b. Orange
    c. White/Green
    d. Blue
    e. White/Blue
    f. Green
    g. White/Brown
    h. Brown



2. Flatten the conductors and place them in the correct order.

Step 4: Trim Conductors

1. Hold the conductors firmly and trim them to an equal length approximately 0.5 inches (1.3 cm) from the cable jacket.

Step 5: Insert RJ45 Connector

1. Insert the conductors into the RJ45 connector while maintaining the T568B wire order.

2. Verify that all conductors reach the end of the connector and remain in the correct sequence.
3. Verify that the cable jacket extends into the connector body.

Step 6: Crimp Connector

1. Insert the connector into the crimping tool.
2. Fully compress the crimping tool to secure the conductors and strain relief.

Step 7: Repeat for Opposite End

1. Repeat Steps 2 through 6 using the same T568B wiring order.

Step 8: Test Cable

1. Connect both ends of the cable to a cable tester.
2. Verify continuity on all eight conductors.
3. Replace and recrimp any connector that fails testing.

Verification

1. Connect the completed cable between two network devices.
2. Verify link lights are present on both devices and confirm network connectivity.

Physical Environment Assembly

Materials

- Raspberry Pi 4 (8GB) x3
- Raspberry Pi tower or cluster case
- MicroSD cards
- 1TB external SSD
- Network switch
- Ethernet cable
- RJ45 connectors
- RJ45 crimping tool
- Cable tester
- Raspberry Pi power supplies

Step 1: Prepare the MicroSD Cards

1. Verify that the prepared and labeled MicroSD cards are available for installation.
2. Configure hostname, SSH, username/password, locale, etc.
3. Label each card (`pi-node-1`, `pi-node-2`, `pi-node-3`).

Step 2: Install Storage Media

1. Insert the prepared MicroSD cards into the Raspberry Pi systems.
2. Connect the 1TB external SSD to the node designated as pi-node-1.
3. Verify that all storage devices are seated securely.

Step 3: Assemble the Raspberry Pi Tower

1. Assemble the Raspberry Pi tower according to the Hardware Layout and Hardware Wiring diagrams in this section. Refer to the manufacturer's instructions only for the mechanical assembly of the tower.
2. Install all Raspberry Pi systems into the tower.

3. Install any supplied heatsinks, cooling fans, or mounting hardware.
4. Verify that all components are mounted securely and that cooling components operate without obstruction.

Step 4: Construct Ethernet Cables

1. Measure the required cable lengths based on the planned location of the tower and network switch.
2. Add approximately 12 inches (30 cm) to each measurement to allow for cable routing and future adjustments.
3. Construct and test Ethernet cables using the Ethernet Cable Construction procedure documented earlier in this section.
4. Label cables as desired.

Step 5: Install Network Infrastructure

1. Place the Raspberry Pi tower and network switch in their intended locations.
2. Connect all Raspberry Pi systems to the network switch.
3. Connect the network switch to the local network.
4. Verify that all network connections are fully seated.

Step 6: Connect Power

1. Connect power supplies to the Raspberry Pi systems and network switch.
2. Route cables to minimize strain on connectors and maintain airflow around equipment.

Step 7: Power On the Environment

1. Apply power to the network switch and Raspberry Pi systems.
2. Observe power indicators, network link lights, and cooling operation.
3. Allow the systems to complete their initial startup sequence.

Step 8: Verify Physical Infrastructure

Verify:

- Raspberry Pi tower is assembled correctly
- Storage devices are connected
- Ethernet cables pass testing
- Network link lights are active
- Cooling components are operating
- All systems receive power

Completion Criteria

Physical environment assembly is complete when:

- The Raspberry Pi tower is assembled
- Storage devices are installed
- Network cabling is connected and tested
- Network connectivity is established
- All systems power on successfully
- The environment is ready for operating system installation

Section 5: Network Configuration

Network Planning and IP Address Allocation

Before configuring persistent IP addresses, the local network configuration was reviewed to determine how the cluster nodes would be addressed and managed. Establishing an addressing plan early helps avoid conflicts with existing devices and simplifies SSH access, and cluster administration.

The first step was identifying the address currently assigned to the system. A quick review of assigned addresses was performed using:

```
ifconfig | grep "inet"
```

Because multiple addresses were returned, the active network adapter was identified using:

```
networksetup -listallhardwareports
```

After identifying the active network interface, its assigned address was obtained using:

```
ipconfig getifaddr en0
```

The resulting address was used to determine the address range used by the local network. Readers following this process should note the first three portions of the address, which typically indicate the local network range.

Next, the default gateway was identified using:

```
netstat -nr | grep default
```

The gateway address was entered into a web browser to access the router or gateway administration interface.

Within the administration interface, the local network settings were reviewed to identify:

- The gateway address
- The subnet mask
- The DHCP address range
- Available IP reservation features

The review confirmed that the gateway supported DHCP reservations. DHCP reservations allow specific devices to consistently receive the same IP address while still using DHCP for network configuration. This approach provides the convenience of DHCP while maintaining predictable addresses for cluster infrastructure. Based on the available address space, three addresses were selected and reserved for the Kubernetes nodes. Assigning predictable addresses simplifies administration, automation, and troubleshooting.

Planned DHCP Reservations

To provide predictable node addresses for SSH access, cluster administration, Ansible automation, and troubleshooting, DHCP reservations is configured on the local gateway.

System	Role	Reserved IP
pi-node-1	k3s control plane, worker, SSD host	10.0.0.20
pi-node-2	k3s worker	10.0.0.21
pi-node-3	k3s worker	10.0.0.22

The reservations can be created through the gateway administration interface after the Raspberry Pi systems have been connected to the network and obtained initial DHCP leases.

The reservation process consists of:

1. Log in to the gateway administration interface.
2. Navigate to Connected Devices.
3. Select Add Device With Reserved IP.
4. Identify the Raspberry Pi by hostname, MAC address, or current DHCP address and select it from the device list.
5. Assign the planned IP address.
6. Save the configuration.
7. Repeat for the remaining nodes.
8. Reboot the Raspberry Pi systems or renew their DHCP leases.

Once completed, the cluster nodes consistently receives the same addresses whenever they reconnect to the network.

If there is no direct access to the gateway, have the nodes request the addresses. Check what the automatically assigned addresses are with:

Node	Command	Result IP
pi-node-1	`ping pi-node-1.local`	10.0.0.155
pi-node-2	`ping pi-node-2.local`	10.0.0.119
pi-node-3	`ping pi-node-3.local`	10.0.0.149

To change them to the planned static IPs, start with node 1:

1. SSH into the node: `ssh sailadmin@pi-node-1.local`
2. ID the ethernet connection: nmcli connection show
    a. For all nodes, that connection is `Wired connection 1` with device `eth0`
3. Configure the static IP:
```
sudo nmcli connection modify "Wired connection 1" \
      ipv4.method manual \
      ipv4.addresses 10.0.0.20/24 \
      ipv4.gateway 10.0.0.1 \
      ipv4.dns "10.0.0.1"
```
4. Verify the saved configuration before activating it:
```
      nmcli -f ipv4.method,ipv4.addresses,ipv4.gateway,ipv4.dns
      connection show "Wired connection 1"
```
   Expected result for node 1:
```
      ipv4.method:       manual
      ipv4.addresses:    10.0.0.20/24
      ipv4.gateway:      10.0.0.1
      ipv4.dns:          10.0.0.1
```
5. Activate the new config: `sudo nmcli connection up "Wired connection 1"`
6. Verify the node's new address: `ip -4 addr show eth0`
```
      inet 10.0.0.20/24
      valid_lft forever preferred_lft forever
```
7. Repeat these steps with nodes 2 & 3
8. Exit ssh and verify all three from the computer:
```
      ping -c 2 10.0.0.20
      ping -c 2 10.0.0.21
      ping -c 2 10.0.0.22
```

IP Address Verification

After configuring the static IP addresses, verify that each node is reachable at its assigned address using the ping commands above. All three nodes should respond from their planned addresses: `10.0.0.20`, `10.0.0.21`, and `10.0.0.22`.

Shared Storage Configuration

The 1TB external SSD is physically connected to **pi-node-1** and provides centralized persistent storage for the SAIL environment. The SSD is shared with the other Raspberry Pi nodes using NFS so that Kubernetes workloads running on any node can access persistent storage over the local network.

Step 1: Identify the SSD on pi-node-1

a. Connect to **pi-node-1**: `ssh sailadmin@10.0.0.20` then `lsusb.` Note the Samsung HD is listed.

```
[sailadmin@pi-node-1:~ $ lsusb
 Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
 Bus 001 Device 002: ID 2109:3431 VIA Labs, Inc. Hub
 Bus 001 Device 007: ID 04e8:4001 Samsung Electronics Co., Ltd PSSD T7
 Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

b. List attached block devices and filesystems: `lsblk -f`

```
sailadmin@pi-node-1:~ $ lsblk -f
NAME        FSTYPE FSVER LABEL  UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       swap   1
sda
└─sda1      exfat  1.0   T7     56F6-5C57
mmcblk0
├─mmcblk0p1 vfat   FAT32 bootfs B2F0-82D2                           428.7M    15% /boot/firmware
└─mmcblk0p2 ext4   1.0   rootfs 15f4c6be-1102-4331-9904-f78e78afd1fd  23.1G    14% /
zram0       swap   1     zram0  3671c3ec-0be0-42a5-92d1-cb35a031fb42                [SWAP]
sailadmin@pi-node-1:~ $
```

c. Identify the 1TB external SSD by its size and device name. It is:

```
sda
└─sda1  exfat  1.0  T7  56F6-5C57
```

d. Verify the selected device is the external SSD; not the MicroSD card containing Pi OS.

```
[sailadmin@pi-node-1:~ $ ls -l /dev/sda /dev/sda1
brw-rw---- 1 root disk 8, 0 Aug 25 19:46 /dev/sda
brw-rw---- 1 root disk 8, 1 Aug 25 19:46 /dev/sda1
sailadmin@pi-node-1:~ $
```

e. Record the SSD partition name. It will typically resemble: `/dev/sda1` -**Confirmed**

```
/dev/sda   → whole Samsung T7 device
/dev/sda1  → partition 1 on the T7
```

Step 2: Prepare the SSD Filesystem

a. Check whether the SSD partition already contains a filesystem:

```
sudo blkid /dev/sda1
```

```
[sailadmin@pi-node-1:~ $ sudo blkid /dev/sda1
/dev/sda1: LABEL="T7" UUID="56F6-5C57" BLOCK_SIZE="512" TYPE="exfat" PARTUUID="3ee6e1cc-01"
sailadmin@pi-node-1:~ $
```

b. Format the SSD partition as `ext4`:

```
sudo mkfs.ext4 -L SAIL_STORAGE /dev/sda1
```

```
[sailadmin@pi-node-1:~ $ sudo mkfs.ext4 -L SAIL_STORAGE /dev/sda1
 mke2fs 1.47.2 (1-Jan-2025)
 /dev/sda1 contains a exfat file system labelled 'T7'
[Proceed anyway? (y,N) y
 Creating filesystem with 244190008 4k blocks and 61054976 inodes
 Filesystem UUID: bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a
 Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624, 11239424, 20480000, 23887872, 71663616, 78675968,
        102400000, 214990848

 Allocating group tables: done
 Writing inode tables: done
 Creating journal (262144 blocks): done
 Writing superblocks and filesystem accounting information: done

 sailadmin@pi-node-1:~ $
```

c. Verify the filesystem: `lsblk -f`

```
[sailadmin@pi-node-1:~ $ lsblk -f
NAME        FSTYPE FSVER LABEL       UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       swap   1
sda
└─sda1      ext4   1.0   SAIL_STORAGE bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a
mmcblk0
├─mmcblk0p1 vfat   FAT32 bootfs       B2F0-82D2                            428.7M    15% /boot/firmware
└─mmcblk0p2 ext4   1.0   rootfs       15f4c6be-1102-4331-9904-f78e78afd1fd   23.1G    14% /
zram0       swap   1     zram0        3671c3ec-0be0-42a5-92d1-cb35a031fb42                [SWAP]
sailadmin@pi-node-1:~ $
```

`ext4` provides a native Linux filesystem for the NFS storage.

verified:

- Partition: `/dev/sda1`
- Filesystem: **ext4**
- Label: **SAIL_STORAGE**
- New UUID: `bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a`

Step 3: Create the Permanent SSD Mount Point

a. Create the mount location: `sudo mkdir -p /mnt/sail-storage`

b. Obtain the SSD filesystem UUID: `sudo blkid /dev/sda1`

c. Record the value following `UUID=`bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a

d. Open `/etc/fstab`: `sudo nano /etc/fstab`

e. Add: `UUID=<ssd-uuid> /mnt/sail-storage ext4 defaults,nofail 0 2`

f. Replace `<ssd-uuid>` with the UUID recorded above.

g. Save and exit the file.

h. Mount the filesystems defined in `/etc/fstab`: `sudo mount -a`

```
[sailadmin@pi-node-1:~ $ sudo systemctl daemon-reload
[sailadmin@pi-node-1:~ $ sudo mount -a
 sailadmin@pi-node-1:~ $
```

i. Verify the SSD is mounted: `findmnt /mnt/sail-storage`

```
[sailadmin@pi-node-1:~ $ findmnt /mnt/sail-storage
TARGET            SOURCE    FSTYPE OPTIONS
/mnt/sail-storage /dev/sda1 ext4   rw,relatime,stripe=8191
sailadmin@pi-node-1:~ $
```

j. Verify its available capacity: `df -h /mnt/sail-storage`

```
[sailadmin@pi-node-1:~ $ df -h /mnt/sail-storage
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       916G  2.1M  870G   1% /mnt/sail-storage
```

Using the UUID rather than `/dev/sda1` prevents the mount from depending on the SSD receiving the same `/dev/sdX` name after every reboot. Confirmed: `/dev/sda1` is mounted at `/mnt/sail-storage`.

Step 4: Create the SAIL NFS Storage Directory

a. Create a directory specifically for Kubernetes persistent storage:

```
sudo mkdir -p /mnt/sail-storage/kubernetes
```

b. Set permissions suitable for the SAIL private cluster:

```
sudo chmod 0777 /mnt/sail-storage/kubernetes
```

c. Verify the directory: `ls -ld /mnt/sail-storage/kubernetes`

```
[sailadmin@pi-node-1:~ $ sudo mkdir -p /mnt/sail-storage/kubernetes
[sailadmin@pi-node-1:~ $ sudo chmod 0777 /mnt/sail-storage/kubernetes
[sailadmin@pi-node-1:~ $ ls -ld /mnt/sail-storage/kubernetes
 drwxrwxrwx 2 root root 4096 Aug 25 20:15 /mnt/sail-storage/kubernetes
```

The broad permissions allow Kubernetes workloads using different container UIDs/GIDs to create storage directories in this isolated lab environment.

Step 5: Install the NFS Server on pi-node-1

a. Update the package index: `sudo apt update`

```
Fetched 27.1 MB in 7s (4,082 kB/s)
98 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Repository 'http://deb.debian.org/debian trixie InRelease' changed its 'Version' value from '13.5' to '13.6'
sailadmin@pi-node-1:~ $
```

b. Install the NFS server: `sudo apt install -y nfs-kernel-server`

```
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+rpt1+deb13u3) ...
sailadmin@pi-node-1:~ $
```

c. Enable and start the service: `sudo systemctl enable --now nfs-kernel-server`

```
[sailadmin@pi-node-1:~ $ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nfs-kernel-server
```

d. Verify the service: `systemctl is-active nfs-kernel-server`

e. Confirm the result is: `active` - **Confirmed**

Step 6: Export the SAIL Storage Directory

a. Open the NFS exports configuration: `sudo nano /etc/exports`

b. Add: `/mnt/sail-storage/kubernetes 10.0.0.0/24(rw,sync,no_subtree_check)`

c. Save and exit the file. This means

- `/mnt/sail-storage/kubernetes` → directory on the T7 being shared.
- `10.0.0.0/24` → permits clients on the SAIL `10.0.0.x` subnet.
- `rw` → permits reads and writes.
- `sync` → requires writes to be committed before the server reports them complete.
- `no_subtree_check` → disables subtree verification, appropriate here because the exported directory is being used as persistent shared storage.

d. Apply the export configuration: `sudo exportfs -rav`

e. Verify the active export: `sudo exportfs -v`

f. Confirm `/mnt/sail-storage/kubernetes` is exported to `10.0.0.0/24`.

```
[sailadmin@pi-node-1:~ $ sudo nano /etc/exports
[sailadmin@pi-node-1:~ $ sudo exportfs -rav
exporting 10.0.0.0/24:/mnt/sail-storage/kubernetes
[sailadmin@pi-node-1:~ $ sudo exportfs -v
/mnt/sail-storage/kubernetes
                10.0.0.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
sailadmin@pi-node-1:~ $
```

This restricts access to systems on the SAIL local subnet rather than exposing the storage to arbitrary hosts. **Completed**

Step 7: Configure and Verify NFS Client Access

The `sail-nfs-client-setup.sh` script automates NFS client configuration across all three SAIL nodes by connecting to each over SSH, updating its package index, and installing `nfs-common` so Kubernetes can later mount NFS-backed persistent volumes. The script then temporarily mounts the SSD-backed NFS share on `pi-node-2` and `pi-node-3`, testing write access and cross-node visibility by creating a file on node 2, confirming node 3 can access it, and creating a second file from node 3. This provides a repeatable verification that the

shared-storage path from the cluster nodes through the NFS server on `pi-node-1` to the SSD is functioning as designed.

a. From the VSCode, navigate to the SAIL project directory containing the script:
```
cd <SAIL-project-directory>
```
b. Verify that the script is present: `ls -l sail-nfs-client-setup.sh`
c. Make the script executable: `chmod +x sail-nfs-client-setup.sh`
d. Run the script: `./sail-nfs-client-setup.sh`
e. Enter the `sailadmin` password when authentication requested.
f. Allow the script to complete for all three nodes. Pause if the script reports an error.
g. Confirm the final output reports successful verification of:
- `nfs-common` installation on all three nodes.
- NFS mounting and write access from `pi-node-2`.
- NFS mounting and write access from `pi-node-3`.
- Visibility from `pi-node-3` of the test file created by `pi-node-2`.

```
========================================
NFS client setup and verification complete
========================================
Verified:
  - nfs-common installed on all three nodes
  - pi-node-2 can mount the NFS share
  - pi-node-2 can write to shared storage
  - pi-node-3 can mount the NFS share
  - pi-node-3 can read data written by pi-node-2
  - pi-node-3 can write to shared storage
```

Step 8: Verify Shared Storage from pi-node-1

a. Connect to **pi-node-1**: `ssh sailadmin@10.0.0.20`
b. List the contents of the shared storage directory: `ls -l /mnt/sail-storage/kubernetes`
c. Confirm both test files exist:
```
pi-node-2-test
pi-node-3-test
```

```
[sailadmin@pi-node-1:~ $ ls -l /mnt/sail-storage/kubernetes
total 0
-rw-rw-r-- 1 sailadmin sailadmin 0 Aug 25 20:50 pi-node-2-test
-rw-rw-r-- 1 sailadmin sailadmin 0 Aug 25 20:50 pi-node-3-test
sailadmin@pi-node-1:~ $
```

d. Remove the test files:
```
sudo rm /mnt/sail-storage/kubernetes/pi-node-2-test
sudo rm /mnt/sail-storage/kubernetes/pi-node-3-test
```

Completion Criteria -> Completed

The shared storage configuration is complete when:

- The 1TB SSD is permanently mounted at `/mnt/sail-storage` on `pi-node-1`.
- The Kubernetes storage directory exists at `/mnt/sail-storage/kubernetes`.
- `pi-node-1` exports that directory using NFS.
- The NFS export is restricted to the SAIL `10.0.0.0/24` network.
- `pi-node-2` can mount and write to the NFS share.
- `pi-node-3` can mount and write to the NFS share.
- Files written through one node are visible through the other nodes.

- Test files and temporary mounts have been removed.

Environment Shutdown Procedure

The SAIL environment should be shut down cleanly before disconnecting power. This allows the Raspberry Pi systems to complete disk operations and shut down the operating system before power is removed.

Step 1: Shut Down pi-node-1

1. Open a terminal on the computer.
2. Connect to **pi-node-1**: `ssh sailadmin@10.0.0.20`
3. Shut down the node: `sudo shutdown -h now`
4. Wa it for the SSH connection to close.

Step 2: Shut Down pi-node-2

1. Connect to **pi-node-2**: `ssh sailadmin@10.0.0.21`
2. Shut down the node: `sudo shutdown -h now`
3. Wait for the SSH connection to close.

Step 3: Shut Down pi-node-3

1. Connect to **pi-node-3**: `ssh sailadmin@10.0.0.22`
2. Shut down the node: `sudo shutdown -h now`
3. Wait for the SSH connection to close.

Step 4: Disconnect Tower Power

1. Wait approximately 30 seconds after shutting down the final Raspberry Pi.
2. Unplug the tower from its power source.
3. Verify that the Raspberry Pi systems, cooling fans, and network switch are no longer receiving power.

Section 6: Kubernetes setup

The SAIL environment uses k3s to provide Kubernetes orchestration across the Raspberry Pi cluster. k3s was selected because it is a lightweight Kubernetes distribution designed for environments with limited computing resources while maintaining compatibility with the standard Kubernetes APIs and tooling. This allows SAIL to provide a realistic Kubernetes environment using Raspberry Pi hardware without the overhead of a full Kubernetes installation.

The cluster consists of one control plane node and two worker nodes. The control plane is hosted on pi-node-1, which also participates as a schedulable worker node. The remaining Raspberry Pi systems function as worker nodes that host workloads distributed by Kubernetes. This configuration provides a balance between efficient hardware utilization and a realistic multi-node cluster suitable for infrastructure management, monitoring, security testing, and AI-assisted operational analysis.

Kubernetes Installation

The following procedures prepare the Raspberry Pi systems for k3s, install k3s on the Raspberry Pi cluster, configure Kubernetes administration from a computer, configure persistent storage using the external SSD, and verify that the Kubernetes environment is functioning correctly.

Step 1: Verify Network Connectivity

a. Open a terminal on the computer.
b. Verify that each Raspberry Pi is accessible using SSH: `ssh <username>@10.0.0.20`
c. After successfully connecting, run: exit
d. Repeat for: `ssh <username>@10.0.0.21` and `ssh <username>@10.0.0.22`
e. Resolve any connectivity issues before continuing.

All working - completed

Step 2: Verify and Configure Required Raspberry Pi cgroups

The `sail-k3s-prerequisites.sh` script verifies that all three Raspberry Pi nodes have the memory cgroup configuration required by k3s. It checks each node over SSH, adds only missing parameters, verifies the resulting configuration, and identifies any nodes that must be rebooted before k3s installation.

a. From Terminal, navigate to the SAIL project directory containing the setup scripts:
`cd <SAIL-project-directory>/setup_scripts`
b. Make the prerequisite script executable: `chmod +x sail-k3s-prerequisites.sh`
c. Run the script: `./sail-k3s-prerequisites.sh`
d. Enter the `sailadmin` password when SSH or `sudo` requests authentication.
e. If the script reports that one or more nodes were modified, reboot **only the nodes identified by the script**: `ssh sailadmin@<node-ip> "sudo reboot"`
f. Wait for each rebooted node to become reachable again.
g. Run the prerequisite script again: `./sail-k3s-prerequisites.sh`
h. Confirm the final run reports:

All three nodes were already configured correctly.
No reboot is required by this script.

```
========================================
k3s prerequisite verification complete
========================================
All three nodes are configured correctly.
No reboot is required.
```

Step 3: Install and Assemble the k3s Cluster

The `sail-k3s-cluster-setup.sh` script installs the k3s control plane on `pi-node-1`, retrieves the cluster join token, installs the k3s agents on `pi-node-2` and `pi-node-3`, and verifies that all three systems successfully join the Kubernetes cluster. The script preserves the different server and worker installation roles while automating the repeated SSH and installation work, and it stops if installation or cluster readiness verification fails.

a. From the MacBook Terminal, navigate to the SAIL setup-scripts directory:
`cd <SAIL-project-directory>/setup_scripts`
b. Make the cluster setup script executable: `chmod +x sail-k3s-cluster-setup.sh`
c. Run the script: `./sail-k3s-cluster-setup.sh`
d. Enter the `sailadmin` password when SSH or `sudo` requests authentication.

e. Allow the script to install the k3s server on `pi-node-1`, retrieve the cluster join token, and install the worker agents on `pi-node-2` and `pi-node-3`.

f. Confirm the script reaches: `k3s cluster setup complete`

g. Confirm the final output reports that all three nodes joined the cluster and report **Ready**.

```
========================================
Verifying cluster
========================================
NAME        STATUS   ROLES           AGE     VERSION
pi-node-1   Ready    control-plane   14m     v1.36.3+k3s1
pi-node-2   Ready    <none>          7m49s   v1.36.3+k3s1
pi-node-3   Ready    <none>          6m51s   v1.36.3+k3s1

========================================
k3s cluster setup complete
========================================
Verified:
  - k3s server installed on pi-node-1
  - k3s agent installed on pi-node-2
  - k3s agent installed on pi-node-3
  - all three nodes joined the cluster
  - all three nodes report Ready
```

Step 4: Verify Cluster Information

a. Run: `sudo kubectl cluster-info`

b. Verify that the Kubernetes API server is accessible.

```
sailadmin@pi-node-1:~ $ sudo kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:6443
CoreDNS is running at https://127.0.0.1:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy
Metrics-server is running at https://127.0.0.1:6443/api/v1/namespaces/kube-system/services/https:metrics-server:https/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

Step 5: Verify Kubernetes System Pods

a. Run: `sudo kubectl get pods --all-namespaces`

b. Verify that all Kubernetes system pods report a status of **Running** or **Completed**.

```
sailadmin@pi-node-1:~ $ sudo kubectl get pods --all-namespaces
NAMESPACE     NAME                                      READY   STATUS      RESTARTS      AGE
kube-system   coredns-54996dc9b4-cmvhx                  1/1     Running     0             20m
kube-system   helm-install-traefik-crd-m5j9c            0/1     Completed   0             19m
kube-system   helm-install-traefik-fq8fn                0/1     Completed   1 (19m ago)   19m
kube-system   local-path-provisioner-58d557dc48-s7jg8   1/1     Running     0             20m
kube-system   metrics-server-6dc596dfb8-726qs           1/1     Running     0             20m
kube-system   svclb-traefik-a496dd55-cvjbp              2/2     Running     0             13m
kube-system   svclb-traefik-a496dd55-hp67j              2/2     Running     0             12m
kube-system   svclb-traefik-a496dd55-s66hl              2/2     Running     0             19m
kube-system   traefik-59b7647586-z6xsr                  1/1     Running     0             19m
```

Step 6: Install the Kubernetes NFS CSI Driver

a. While connected to **pi-node-1**, install the NFS CSI driver:

```
curl -skSL
https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/v4.13.4/depl
oy/install-driver.sh | sudo bash -s v4.13.4 --
```

b. Then verify:

```
sudo kubectl -n kube-system get pods -o wide -l app=csi-nfs-controller
sudo kubectl -n kube-system get pods -o wide -l app=csi-nfs-node
```

```
sailadmin@pi-node-1:~ $ curl -skSL https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/v4.13.4/deploy/install-driver.sh | sudo bash -s v4.13.4 --
Installing NFS CSI driver, version: v4.13.4 ...
serviceaccount/csi-nfs-controller-sa created
serviceaccount/csi-nfs-node-sa created
clusterrole.rbac.authorization.k8s.io/nfs-external-provisioner-role created
clusterrolebinding.rbac.authorization.k8s.io/nfs-csi-provisioner-binding created
clusterrole.rbac.authorization.k8s.io/nfs-external-resizer-role created
clusterrolebinding.rbac.authorization.k8s.io/nfs-csi-resizer-role created
csidriver.storage.k8s.io/nfs.csi.k8s.io created
deployment.apps/csi-nfs-controller created
daemonset.apps/csi-nfs-node created
NFS CSI driver installed successfully.
sailadmin@pi-node-1:~ $ sudo kubectl -n kube-system get pods -o wide -l app=csi-nfs-controller
NAME                                 READY   STATUS              RESTARTS   AGE   IP          NODE        NOMINATED NODE   READINESS GATES
csi-nfs-controller-65b4f9877-5qbjz   0/5     ContainerCreating   0          9s    10.0.0.22   pi-node-3   <none>           <none>
sailadmin@pi-node-1:~ $ sudo kubectl -n kube-system get pods -o wide -l app=csi-nfs-node
NAME                 READY   STATUS              RESTARTS   AGE   IP          NODE        NOMINATED NODE   READINESS GATES
csi-nfs-node-27nvf   0/3     ContainerCreating   0          15s   10.0.0.22   pi-node-3   <none>           <none>
csi-nfs-node-f76m8   0/3     ContainerCreating   0          15s   10.0.0.21   pi-node-2   <none>           <none>
csi-nfs-node-zccfd   0/3     ContainerCreating   0          15s   10.0.0.20   pi-node-1   <none>           <none>
```

c. Wait for the driver components to start.

d. Verify the NFS CSI pods:

```
sudo kubectl -n kube-system get pods -o wide | grep csi-nfs
```

e. Confirm the NFS CSI controller and node components report `Running`.

```
sailadmin@pi-node-1:~ $ sudo kubectl -n kube-system get pods -o wide | grep csi-nfs
csi-nfs-controller-65b4f9877-5qbjz       5/5     Running     0             86s   10.0.0.22   pi-node-3   <none>           <none>
csi-nfs-node-27nvf                       3/3     Running     0             85s   10.0.0.22   pi-node-3   <none>           <none>
csi-nfs-node-f76m8                       3/3     Running     0             85s   10.0.0.21   pi-node-2   <none>           <none>
csi-nfs-node-zccfd                       3/3     Running     0             85s   10.0.0.20   pi-node-1   <none>           <none>
```

The NFS CSI driver allows Kubernetes to dynamically provision persistent volumes from NFS storage.

Step 7: Configure Kubernetes Administration from the Computer

a. Connected to **pi-node-1**, make a temp copy of Kubernetes configuration for `sailadmin`:

```
sudo cp /etc/rancher/k3s/k3s.yaml /home/sailadmin/k3s.yaml
sudo chown sailadmin:sailadmin /home/sailadmin/k3s.yaml
```

b. Disconnect from **pi-node-1**: `exit`

c. On the computer terminal, verify that `kubectl` is installed: `kubectl version --client`

d. If `kubectl` is not installed, install it using the appropriate package manager for the computer's operating system.

e. Create the Kubernetes configuration directory: `mkdir -p ~/.kube`

f. If an existing kubeconfig exists, back it up before continuing:

```
if [ -f ~/.kube/config ]; then
    cp ~/.kube/config ~/.kube/config.backup
fi
```

g. Copy the k3s configuration from **pi-node-1**:

```
scp sailadmin@10.0.0.20:/home/sailadmin/k3s.yaml ~/.kube/config
```

h. Restrict access to the configuration file: `chmod 600 ~/.kube/config`

i. Open `~/.kube/config` in a text editor and change the Kubernetes server address from:

```
https://127.0.0.1:6443
```

to:

```
https://10.0.0.20:6443
```

j. Verify Kubernetes administration from the computer: `kubectl get nodes`

```
NAME        STATUS   ROLES           AGE   VERSION
pi-node-1   Ready    control-plane   45m   v1.36.3+k3s1
pi-node-2   Ready    <none>          38m   v1.36.3+k3s1
pi-node-3   Ready    <none>          37m   v1.36.3+k3s1
```

k. Confirm all three nodes report **Ready**.

l. Remove the temporary configuration copy from **pi-node-1**:

```
ssh sailadmin@10.0.0.20 "rm /home/sailadmin/k3s.yaml"
```

Going forward, Kubernetes administration can be performed directly from the computer, no ssh.

Step 8: Create the SAIL NFS StorageClass

a. Create the StorageClass definition in `sail-nfs-storageclass.yaml`:

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: sail-nfs
provisioner: nfs.csi.k8s.io
parameters:
  server: 10.0.0.20
  share: /mnt/sail-storage/kubernetes
reclaimPolicy: Retain
volumeBindingMode: Immediate
allowVolumeExpansion: true
mountOptions:
  - nfsvers=4.1
```

b. Apply the StorageClass: `kubectl apply -f sail-nfs-storageclass.yaml`

c. Verify it: `kubectl get storageclass`

d. Confirm `sail-nfs` appears and its provisioner is: `nfs.csi.k8s.io`

```
sail-nfs        nfs.csi.k8s.io   Retain   Immediate   true   21s
```

The StorageClass tells Kubernetes where the shared persistent storage is located and which storage driver should provision volumes from it.

Step 9: Verify Kubernetes Persistent Storage

a. Create a test PersistentVolumeClaim `sail-storage-test-pvc.yaml:`

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: sail-storage-test
spec:
  accessModes:
    - ReadWriteMany
  storageClassName: sail-nfs
  resources:
    requests:
      storage: 1Gi
```

b. Apply it: `kubectl apply -f sail-storage-test-pvc.yaml`

c. Verify the PVC: `kubectl get pvc sail-storage-test`

d. Confirm its status reaches: `Bound`

e. Create a temporary pod using the PVC with `sail-storage-test-pod.yaml`:

```
apiVersion: v1
kind: Pod
metadata:
  name: sail-storage-test
spec:
  containers:
    - name: test
      image: busybox
```

```
      command: ["sh", "-c", "echo SAIL-storage-test > /data/test.txt &&
sleep 3600"]
      volumeMounts:
        - name: sail-storage
          mountPath: /data
  volumes:
    - name: sail-storage
      persistentVolumeClaim:
        claimName: sail-storage-test
```

f. Apply the pod: `kubectl apply -f sail-storage-test-pod.yaml`

g. Verify the pod reaches `Running`: `kubectl get pod sail-storage-test`

h. Write test data to the persistent volume:

```
kubectl exec sail-storage-test -- sh -c 'echo SAIL-storage-test > /data/test.txt'
```

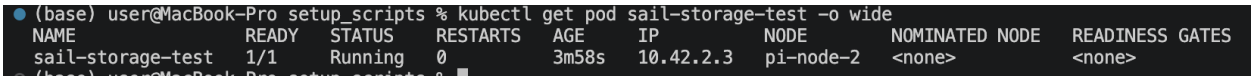
i. Verify the data written to persistent storage:

```
kubectl exec sail-storage-test -- cat /data/test.txt
```

Confirm: `SAIL-storage-test` - **Confirmed**

j. Determine which Kubernetes node is running the pod:

```
kubectl get pod sail-storage-test -o wide
```



k. Delete the pod: `kubectl delete pod sail-storage-test`

l. Recreate it using the same PVC:

```
kubectl apply -f sail-storage-test-pod.yaml
```

m. Wait until the recreated pod reaches `Running`:

```
kubectl get pod sail-storage-test
```

n. Verify the original data remains available:

```
kubectl exec sail-storage-test -- cat /data/test.txt
```

o. Confirm: `SAIL-storage-test` - **Confirmed**

Verifies that the data exists independently of the individual pod and is available after pod deletion and recreation, to support the later persistence requirement.

p. Delete the temporary pod: `kubectl delete pod sail-storage-test`

q. Delete the test PVC: `kubectl delete pvc sail-storage-test`

r. Remove the temporary test YAML files:

```
rm sail-storage-test-pod.yaml
rm sail-storage-test-pvc.yaml
```

s. Keep `sail-nfs-storageclass.yaml` because `sail-nfs` is part of the permanent SAIL Kubernetes configuration.

t. Verify the StorageClass remains available: `sudo kubectl get storageclass`

u. Confirm `sail-nfs` remains listed. **Confirmed**

That confirms:

- the `sail-nfs` StorageClass can provision a PVC;
- the test pod can mount and write to the NFS-backed volume;
- the data remained after the pod was deleted and recreated;
- the temporary test resources were removed;
- the permanent `sail-nfs` StorageClass remains available.

Step 10: Deploy and Verify a Test Workload and Service

a. Create an nginx deployment:`kubectl create deployment nginx --image=nginx`

b. Verify the deployment: `kubectl get deployments`

c. Verify the pod and determine which node is running it: `kubectl get pods -o wide`
d. Confirm the nginx pod reaches **Running**.
e. Create a ClusterIP service:
```
kubectl expose deployment nginx --port=80 --type=ClusterIP
```
f. Verify the service: `kubectl get services`
g. Confirm an `nginx` service appears with type **ClusterIP**.

```
(base) user@MacBook-Pro setup_scripts % kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
(base) user@MacBook-Pro setup_scripts % kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
nginx   0/1     1            0           7s
(base) user@MacBook-Pro setup_scripts % kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
nginx   1/1     1            1           13s
(base) user@MacBook-Pro setup_scripts % kubectl get pods -o wide
NAME                     READY   STATUS    RESTARTS   AGE   IP          NODE        NOMINATED NODE   READINESS GATES
nginx-7f8fbb96d-rf56h    1/1     Running   0          18s   10.42.2.5   pi-node-2   <none>           <none>
(base) user@MacBook-Pro setup_scripts % kubectl expose deployment nginx --port=80 --type=ClusterIP
service/nginx exposed
(base) user@MacBook-Pro setup_scripts % kubectl get services
NAME         TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.43.0.1      <none>        443/TCP   65m
nginx        ClusterIP   10.43.52.119   <none>        80/TCP    6s
(base) user@MacBook-Pro setup_scripts %
```

Step 11: Remove the Test Workload

a. Run: `sudo kubectl delete service nginx`
`sudo kubectl delete deployment nginx`
b. Verify that the deployment and service have been removed.

```
(base) user@MacBook-Pro setup_scripts % kubectl delete service nginx
kubectl delete deployment nginx
service "nginx" deleted
deployment.apps "nginx" deleted
(base) user@MacBook-Pro setup_scripts % kubectl get deployments
kubectl get services
No resources found in default namespace.
NAME         TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.43.0.1     <none>        443/TCP   67m
```

Completion Criteria

The Kubernetes installation is complete when:

- k3s is installed on all three Raspberry Pi systems.
- All three nodes report **Ready**.
- Kubernetes system components are healthy.
- Kubernetes administration from the computer is functioning.
- The NFS CSI controller and node components are running.
- The `sail-nfs` StorageClass is available.
- Kubernetes successfully provisions storage from the SSD-backed NFS share.
- Persistent data remains available after a pod is deleted and recreated.
- A test workload deploys successfully.
- A ClusterIP service is created successfully.
- The test deployment and service are removed successfully.
- Baseline cluster information has been verified.

Section 7: Workload Design

SAIL operates multiple workloads to create an active environment for infrastructure, security, automation, and AI-assisted operations. These workloads generate ongoing activity that can be

monitored and analyzed while also providing opportunities to test failures, security events, investigation, and recovery.
The environment includes four primary workloads:

- **Application workload:** Containerized applications generate API requests, authentication activity, data processing, storage operations, errors, and other application behavior.
- **Security workload:** Security tools assess systems and assets for vulnerabilities and security weaknesses while monitoring the running environment for security-relevant behavior. Security activity produces findings and detections that can be investigated using evidence collected throughout SAIL.
- **Automated/AI workload:** Automated processes and AI agents interact with applications and services while the environment is running. These processes create sustained activity, generate different usage and security patterns, and support investigation and remediation workflows.
- **Operational workload:** Kubernetes, containers, hosts, storage, and networking generate activity through deployments, scheduled processes, configuration changes, resource usage, failures, troubleshooting, and recovery operations.

These workloads create the data used throughout SAIL. Monitoring and security systems collect logs, metrics, traces, vulnerability findings, runtime detections, application records, and system information that can be used during troubleshooting and security investigations.

Application Workload

The application workload provides functioning services for SAIL to host and operate. Applications run as containerized workloads on the Kubernetes cluster and provide real targets for deployment, monitoring, security assessment, automated activity, troubleshooting, and investigation.
The application workload includes:

1. **Interactive application activity:** Applications expose APIs or other interfaces that can receive requests and perform useful processing. Requests can involve authentication, authorization, data access, application logic, and persistent storage.
2. **Persistent application state:** Application data remains available across container and pod restarts. Database activity also provides another service dependency that can be monitored, secured, disrupted, and recovered.
3. **Security-relevant application behavior:** Authentication attempts, access-control decisions, input validation, privileged operations, configuration changes, errors, and other significant actions produce records that can be correlated with infrastructure and security telemetry.
4. **External asset interaction:** Application components can interact with authorized external assets when a workload requires information outside the SAIL cluster. This allows security-oriented services to perform useful work against assets such as owned websites or repositories rather than operating only against simulated targets.

The exact functionality of the initial application is determined after the complete SAIL workload is defined. Its design needs to provide useful application behavior while also supplying enough activity and security context for the other SAIL workloads.

Security Workload

The security workload provides SAIL with active security assessment, detection, and investigation capabilities.

1. **Assessment:** SAIL examines systems and assets for vulnerabilities and security weaknesses. Assessments can cover SAIL infrastructure and applications along with external assets owned or administered by the operator. Assessments can identify vulnerable software, exposed services, insecure configurations, application weaknesses, and other security findings. Results are recorded so findings can be reviewed, investigated, remediated, and compared with later assessments.
2. **Detection:** SAIL monitors running systems and applications for security-relevant behavior. This includes authentication failures, unauthorized access attempts, unexpected processes, container activity, configuration changes, suspicious requests, unusual application activity, and other behaviors identified by the security tooling.
3. **Investigation:** Security findings and detections are supported with evidence collected across the environment. Investigations can use application activity, system state, vulnerability results, logs, metrics, traces, runtime security information, and related records to determine what occurred and assess its significance. AI-assisted investigation capabilities can use this evidence to correlate activity and support analysis.
4. **Verification:** Security assessments can be repeated after remediation to determine whether an identified weakness has been corrected. Detection and monitoring data can also confirm whether affected systems have returned to their expected state.

The security workflow follows:

Assess → Detect → Collect Evidence → Investigate → Remediate → Verify

This workflow provides a common process for handling security findings and events throughout SAIL. It connects the security workload with the application, automated/AI, and operational workloads so activity occurring in one part of the environment can be examined using evidence produced elsewhere in the system.

Automated/AI Workload

The automated/AI workload provides sustained activity and AI-assisted operational capabilities across SAIL. Automation performs repeatable work while agents make bounded decisions based on the state of the environment and the information available to them.

The workload includes three agent roles:

1. **Activity Agent:** Generates sustained interaction with applications and services. The agent can represent different patterns of normal and security-relevant behavior, vary the frequency and sequence of its actions, and create enough activity for monitoring systems to establish patterns and detect changes.
2. **Investigation Agent:** Examines security findings, alerts, and operational problems by retrieving relevant evidence from SAIL. The agent can correlate application records, logs, metrics, traces, runtime detections, vulnerability results, and current system state to develop an evidence-supported assessment.

3. **Remediation Agent:** Uses investigation results to prepare corrective actions. Remediation activity is recorded and followed by verification so the outcome can be compared with the original finding or incident.

Agent activity is also observable. Requests, tool calls, decisions, investigation inputs, outputs, and remediation actions generate records that can be reviewed alongside the systems the agents operate against.

Operational Workload

The operational workload represents the normal administration and behavior of the SAIL infrastructure. It provides the system-level activity required to operate applications and security services across the physical hosts and Kubernetes cluster.

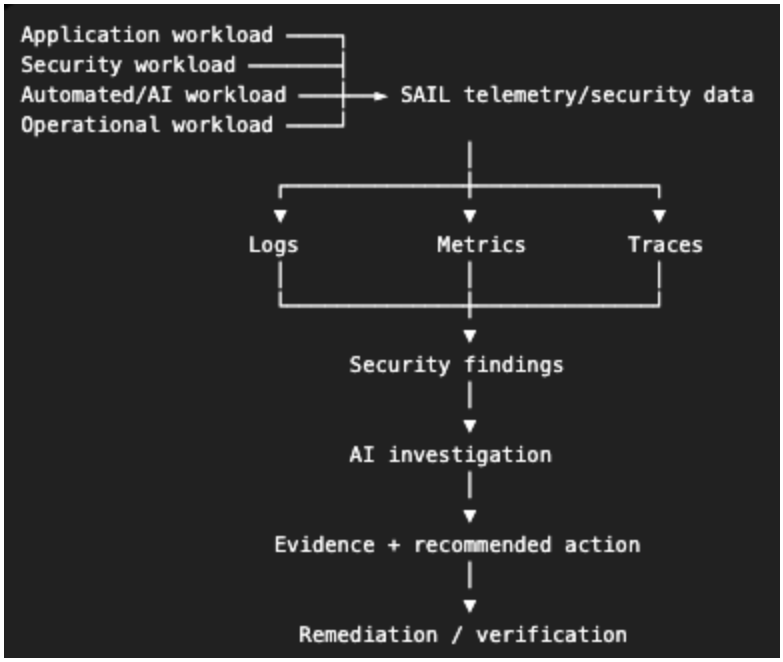
The workload includes:

1. **Cluster operations:** Kubernetes schedules and manages application, security, monitoring, automation, and AI workloads across the available nodes. Deployments, pod creation, restarts, scheduling decisions, and service changes generate operational activity.
2. **System operations:** Hosts generate CPU, memory, storage, network, process, and service activity while SAIL is running. Changes in these resources provide infrastructure-level context for application and security events.
3. **Deployment and configuration activity:** Application releases, container images, infrastructure configuration, Kubernetes resources, and automated configuration changes create a record of how the environment changes over time.
4. **Failure and recovery activity:** Service outages, pod failures, resource pressure, storage problems, network problems, and configuration errors provide scenarios for detection and troubleshooting. Recovery activity provides evidence that corrective actions restored the affected system.

Operational data provides context for determining whether an application or security event is isolated or connected to a broader change in the environment.

Integrated Workload Flow

The four workloads operate as parts of the same environment rather than as separate demonstrations. Application and operational workloads produce normal system activity. Security systems assess assets and monitor the environment. Automated processes and AI agents interact with these systems and generate additional activity.



Section 8: Application Development and Deployment

Application Design

Orbit is a social media application that provides a realistic workload for SAIL while supporting continued development as an independent application. Users can create accounts, authenticate, maintain profiles, publish posts, comment and react to posts, follow other users, and view a chronological feed. User and administrative permissions control access to application functions and data. Orbit exposes its functionality through a REST API and stores persistent data in PostgreSQL.

Automated users can generate substantial activity by exercising authentication, content creation, feed retrieval, social relationships, authorization checks, database operations, and concurrent API requests. The volume and frequency of these interactions can be varied to create different workload levels for monitoring application performance and infrastructure behavior. Orbit also generates security-relevant activity through authentication attempts, permission checks, administrative operations, validation failures, user actions, and application errors.

The SAIL implementation focuses on the functionality needed to provide a useful application workload without requiring the complete feature set of a production social media platform. Application activity can be correlated with monitoring and security data collected elsewhere in SAIL, while the initial architecture provides a foundation for continued development of Orbit after SAIL is complete.

The SAIL implementation of Orbit includes:

- User registration and authentication.
- User profiles and account management.
- User and administrative roles and permissions.
- Creation, editing, deletion, and retrieval of posts.
- Comments and reactions.
- Follow and unfollow relationships.
- A chronological user feed.
- REST API access and request validation.
- Persistent application data.
- Application and security-relevant logging.
- Health and application status endpoints.
- Support for sustained automated user activity.

The application uses:

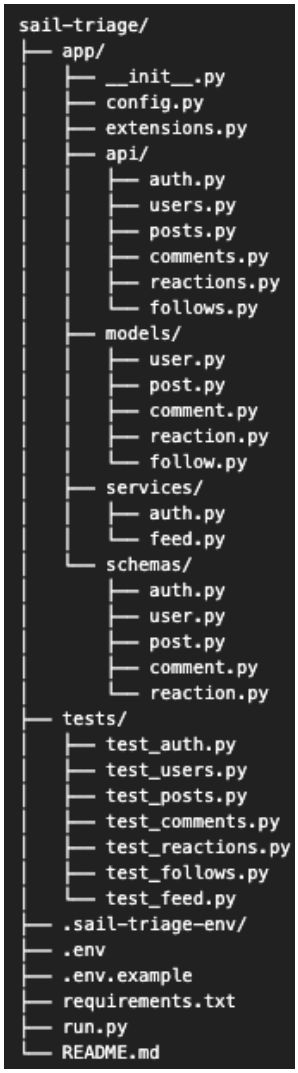
Technology	Purpose
Python	Programming language
FastAPI	Web/API framework
Pydantic	Request/response validation and schemas
Uvicorn	Runs the FastAPI application
PostgreSQL	Application database
SQLAlchemy	ORM/database access
psycopg	Connects the application to PostgreSQL
Alembic	Database schema migrations
PyJWT	Authentication tokens
pwdlib / Argon2	Password hashing
pytest	Automated testing
Black	Code formatting
Ruff	Linting/code quality
dotenv	Loads application configuration from .env files into env variables.

Frontend	Purpose
React	Frontend user interface
Vite	Frontend development server and build tooling
JavaScript	Frontend application logic

HTML	Interface structure
Tailwind CSS	Interface styling and responsive layouts
shadcn/ui	Reusable user interface components

Application Structure

The `api` package handles REST endpoints, `models` contains database models, `services` contains application logic that does not belong directly in an endpoint, `schemas` handles request and response validation, and `tests` contains automated tests for the application.



Development follows standard Python practices, including PEP 8 conventions, type hints where appropriate, automated formatting and linting, separation of application responsibilities, and automated testing.

Application Behavior

At a high level, Orbit supports the following application workflow:

1. **Register and authenticate users** using secure authentication tokens.
2. **Manage accounts and profiles** with user and administrative permissions.
3. **Create and manage posts** including publishing, retrieving, editing, and deleting.
4. **Interact with content** through comments and reactions.
5. **Manage social relationships** by categorizing connections based on relationship type or closeness.
6. **Generate a chronological feed** containing content the authenticated user is authorized to view.
7. **Validate requests and enforce authorization** before accessing protected functions or data.
8. **Store and retrieve application data** through PostgreSQL.
9. **Record security-relevant activity** including authentication, permission checks, administrative actions, validation failures, and errors.
10. **Report application health** for monitoring Orbit and its dependencies.
11. **Support sustained automated activity** at varying workload levels.

Application Deployment

The application completes in the following order:

1. Create the application folder and virtual environment. **Complete**
2. Configure VS Code to use the project environment. **Complete**
3. Configure Git for the application. **Complete**
4. Install the initial application and development dependencies. **Complete**
5. Create the Application Project Structure. **Complete**
6. Configure application settings and environment variables. **Complete**
7. Configure PostgreSQL for local development. **Complete**
8. Configure SQLAlchemy and database connectivity. **Complete**
9. Configure Alembic database migrations. **Complete**
10. Implement the user data model and database schema. **Complete**
11. Implement user registration, authentication, and password security. **Complete**
12. Implement user profiles, roles, and permissions. **Complete**
13. Run and manually verify the current backend locally. **Complete**
14. Create the Orbit frontend structure and connect it to the FastAPI backend. **Complete**
15. Build registration and login screens using the existing authentication API. **Complete**
16. GitHub Actions for Black, Ruff, and frontend ESLint **Complete**
17. Build user profile and account screens **Complete**
18. Implement Posts **Complete**
19. Add comments and reactions **Complete**
20. Add follow and unfollow relationships, i.e. data models, API endpoints, schemas, frontend functionality, and tests.

21. Implement Account Lifecycle Management
22. Build the chronological feed, including backend feed logic, API endpoints, schemas, frontend functionality, and tests.
23. Complete frontend navigation and remaining user-interface behavior.
24. Add application and security-relevant logging.
25. Add health endpoints and application error handling.
26. Complete remaining automated and integration testing.
27. Run and test the complete application locally, i.e. frontend, FastAPI, and PostgreSQL.
28. Verify FastAPI validation and OpenAPI documentation.
29. Generate sustained automated user activity and verify Orbit under a variable workload.
30. Containerize the completed application.
31. Deploy Orbit and PostgreSQL to the k3s cluster.
32. Verify the deployed application is reachable, persistent, and functioning correctly.

Create the Application Environment

Step 1: Create the application folder and virtual environment

Inside the existing `SAIL` repository, create the application folder:

```
SAIL/
├── README.md
├── docs/
└── sail-orbit/
```

a. In Terminal, from the root of the SAIL repo:

```
mkdir -p sail-orbit
cd sail-orbit
```

b. Create the virtual environment:

```
python3 -m venv .sail-orbit-env
```

c. Activate it:

```
source .sail-orbit-env/bin/activate
```

d. Verify it:

```
which python
python --version
```

The `which python` result should point to something like:

```
.../SAIL/sail-orbit/.sail-orbit-env/bin/python
```

At this point, the only thing created automatically inside `sail-orbit` is:

```
sail-orbit/
└── .sail-orbit-env/
```

Step 2: Configure VS Code to Use the Environment

a. Press `Cmd + Shift + P`.
b. Search for:

```
Python: Select Interpreter
```

c. Select: Enter interpreter path → Find.
d. Select:

```
sail-orbit/.sail-orbit-env/bin/python
```

Step 3: Configure Git for the Application

Since `SAIL` is already a Git repository, do not run `git init` again.

a. From inside `sail-orbit`, confirm are inside the existing repository:

```
git status
```

b. Add the virtual environment to the repository's `.gitignore`.

From the SAIL repository root, open or create `.gitignore` and add:

```
sail-orbit/.sail-orbit-env/
```

c. Verify Git is ignoring it:

```
git status
```

d. Create a development branch for the application work:

```
git switch -c develop
```

e. Verify the branch:

```
git branch
```

Should see:

```
* develop
main
```

f. in a .gitignore at the file base level:

```
# Python virtual environment
sail-orbit/.sail-orbit-env/

# Python generated files
__pycache__/
*.py[cod]

# Testing and tooling caches
.pytest_cache/
.ruff_cache/
.coverage
htmlcov/

# Environment variables and secrets
.env
.env.*
!.env.example

# macOS
.DS_Store

# VS Code local settings
.vscode/

# Logs
*.log
```

g. Make an initial commit for the application folder/setup once the structure wanted tracked exists. For now, if `sail-orbit` contains only the ignored virtual environment, there is nothing useful to commit yet.

Step 4: Install Application and Development Dependencies

With `.sail-orbit-env` activated and the terminal inside `sail-orbit`:

a. Create the requirements file:

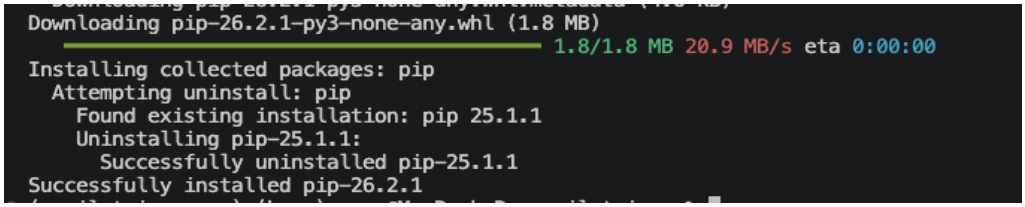
```
touch requirements.txt
```

b. Open `requirements.txt` in VS Code and add:

```
Flask
Flask-RESTX
pytest
black
ruff
```

c. Upgrade `pip`:

```
python -m pip install --upgrade pip
```



d. Install the dependencies from the requirements file:

```
pip install -r requirements.txt
```

e. Verify the installed packages:

```
pip list
```

Confirm that the list includes:

```
fastapi
pydantic
uvicorn
sqlalchemy[asyncio]
psycopg[binary]
alembic
pyjwt
pwdlib[argon2]
pytest
black
ruff
```

Once verified, put the version numbers in the requirements.txt.

f. Verify that Git now sees the application directory and requirements file:

```
git status
```

Should see `sail-orbit/requirements.txt` as an untracked file. The `.sail-orbit-env` directory should **not** appear because it is ignored.

Step 5: Create the Application Project Structure

a. From inside `sail-orbit`, create the directories:

```
mkdir -p app/api app/models app/services app/schemas tests
```

b. Create the core application files:

```
touch app/__init__.py \
      app/config.py \
      app/extensions.py \
      run.py
```

c. Create the application README:

```
touch README.md
```

d. Create the Orbit API files:

```
touch app/api/auth.py \
      app/api/users.py \
      app/api/posts.py \
      app/api/comments.py \
      app/api/reactions.py \
      app/api/follows.py
```

e. Create the database model files:

```
touch app/models/user.py \
      app/models/post.py \
      app/models/comment.py \
      app/models/reaction.py \
      app/models/follow.py
```

f. Create the service files:

```
touch app/services/auth.py \
      app/services/feed.py
```

g. Create the Pydantic schema files:

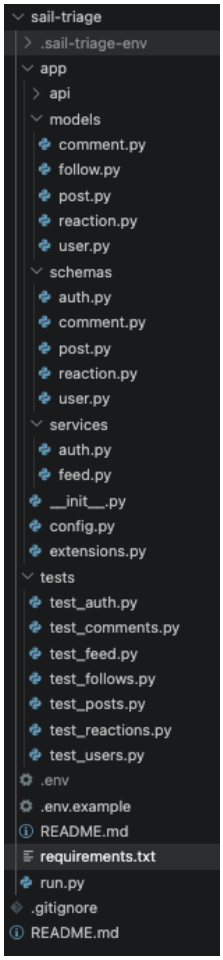
```
touch app/schemas/auth.py \
      app/schemas/user.py \
      app/schemas/post.py \
      app/schemas/comment.py \
      app/schemas/reaction.py
```

h. Create the test files:

```
touch tests/test_auth.py \
      tests/test_users.py \
      tests/test_posts.py \
      tests/test_comments.py \
      tests/test_reactions.py \
      tests/test_follows.py \
      tests/test_feed.py
```

i. Verify the structure: `find app tests -maxdepth 2 -type f | sort`

The structure should now be:



Step 6: Configure Application Settings and Environment Variables

a. Create an environment file for local development:

```
touch .env
```

b. Create a safe template that can be committed to Git:

```
touch .env.example
```

c. In `.env`, add the local application settings we know we will need:

```
FLASK_ENV=development
FLASK_DEBUG=1
```

d. Put the same variable names in `.env.example`, but without any sensitive values:

```
FLASK_ENV=development
FLASK_DEBUG=1
```

e. Confirm the root `.gitignore` includes:

```
.env
.env.*
!.env.example
```

f. In `app/config.py`, we will later load configuration from environment variables rather than hardcoding it into the application.

Step 7: Configure PostgreSQL for Development

a. First, check whether PostgreSQL is already installed on the computer:

```
psql --version
```

b. If `psql` is not found, install PostgreSQL with Homebrew:

```
brew install postgresql@17
```

```
Success. You can now start the database server using:

    '/opt/homebrew/Cellar/postgresql@17/17.11/bin/pg_ctl' -D '/opt/homebrew/var/postgresql@17' -l logfile start
```

c. Start PostgreSQL:

```
brew services start postgresql@17
```

```
(.sail-orbit-env) (base) user@MacBook-Pro sail-orbit % brew services start postgresql@17
==> Successfully started `postgresql@17` (label: homebrew.mxcl.postgresql@17)
```

d. Verify the server is running:

```
pg_isready
```

Response should be similar to:

```
/var/run/postgresql:5432 - accepting connections
```

```
(.sail-orbit-env) (base) user@MacBook-Pro sail-orbit % pg_isready
/tmp:5432 - accepting connections
```

e. Create a PostgreSQL database for Orbit:

```
createdb orbit
```

f. Verify that it exists:

```
psql -l
```

Should see `orbit` in the database list.

```
postgres=# \l
                                                  List of databases
   Name    |   Owner    | Encoding | Locale Provider |   Collate   |    Ctype    | Locale | ICU Rules | Access privileges
-----------+------------+----------+-----------------+-------------+-------------+--------+-----------+-------------------
 orbit     | triage_app | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |        |           |
 postgres  | user       | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |        |           |
 template0 | user       | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |        |           | =c/user          +
           |            |          |                 |             |             |        |           | user=CTc/user
 template1 | user       | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |        |           | =c/user          +
           |            |          |                 |             |             |        |           | user=CTc/user
(4 rows)
```

g. Open the PostgreSQL prompt:

```
psql postgres
```

Should get a prompt similar to:

```
postgres=#
```

h. Create a Orbit database user. Replace `your_password_here` with a chosen password:

```
CREATE USER orbit_app WITH PASSWORD 'your_password_here';
```

i. Make `orbit_app` the owner of the `orbit` database:

```
ALTER DATABASE orbit OWNER TO orbit_app;
```

j. Connect to the `orbit` database:

```
\c orbit
```

k. Make `orbit_app` the owner of the `public` schema (Alembic to create/modify tables later):

```
ALTER SCHEMA public OWNER TO  orbit_app;
```

l. Exit PostgreSQL:

```
\q
```

m. Test the new application account:

```
psql -U orbit_app -d orbit -h localhost
```

It should ask for the password just created. After entering it, should reach:

```
orbit=>
```

Then run:

```
SELECT current_user, current_database();
```

```
orbit=> SELECT current_user, current_database();
 current_user | current_database
--------------+------------------
 orbit_app    | orbit
(1 row)
```

n. Now exit PostgreSQL:

```
\q
```

o. Then update `DATABASE_URL` in the `.env`:

```
DATABASE_URL=postgresql+psycopg://orbit_app:orbit_et@localhost:5432/orbit
```

Keep the password only in `.env`; `.env.example` should continue to have:
DATABASE_URL=

p. To shut down the database:

```
brew services stop postgresql@17
```

Verify it stopped:

```
pg_isready
```

Should get something like:

```
localhost:5432 - no response
```

Step 8: Configure SQLAlchemy and database connectivity.

a. Open:

```
app/extensions.py
```

Add:

```
"""Application extensions for Orbit."""

from sqlalchemy import create_engine
from sqlalchemy.orm import sessionmaker

from app.config import Config


engine = create_engine(
   Config.DATABASE_URL,
   pool_pre_ping=True,
)

SessionLocal = sessionmaker(
   bind=engine,
   autoflush=False,
   autocommit=False,
)
```

b. Create a small database test file:

```
touch app/database_test.py
```

c. Add:

```
"""Verify Orbit can connect to the configured database."""

from sqlalchemy import text

from app.extensions import engine


def test_database_connection():
   """Connect to PostgreSQL and return the current database
user."""

   with engine.connect() as connection:
       result = connection.execute(
           text("SELECT current_user, current_database();")
       )

       return result.fetchone()

if __name__ == "__main__":
   print(test_database_connection())
```

d. Run it from inside `sail-orbit` with the virtual environment active:

```
python -m app.database_test
```

Expected output should look similar to:

```
('orbit_app', 'orbit')
```

e. More the file to tests and rename to a better name for explanation

```
mv app/database_test.py tests/test_database_connection.py
```

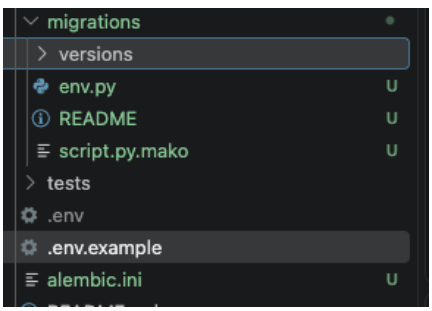
Step 9: Configure Alembic database migrations

a. From inside `sail-orbit`, initialize Alembic:

```
alembic init migrations
```

This creates:

```
alembic.ini
migrations/
├── env.py
├── script.py.mako
└── versions/
```



b. Open:

`alembic.ini`
Find:
`sqlalchemy.url =` -> Make it blank.
Want Alembic to use the same `DATABASE_URL` from `.env`

c. Open:
`migrations/env.py`
Near the top, add:
```
from app.config import Config
```
Then find:
```
config = context.config
```
Immediately after it, add:
```
config.set_main_option("sqlalchemy.url",
Config.DATABASE_URL)
```

d. For now, leave:
`target_metadata = None`
Haven't yet created the SQLAlchemy models. In the next step, replace this with the app's model metadata so Alembic can detect schema changes automatically.

e. Verify Alembic can connect to PostgreSQL:
`alembic current`

```
(.sail-triage-env) (base) user@MacBook-Pro sail-triage % alembic current
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
```

f. Verify Alembic itself is configured:
`alembic heads`

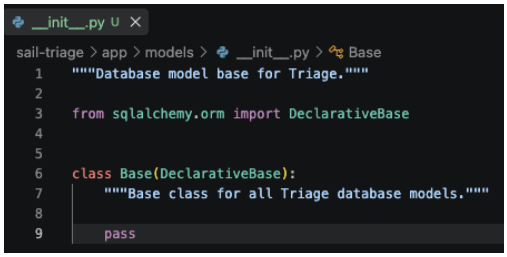
Alembic is initialized, is reading the Orbit database configuration, and can connect to PostgreSQL successfully.

Step 10: Implement the user data model and database schema

Define the SQLAlchemy base, implement this model in `app/models/user.py`, expose its metadata to Alembic, generate the first migration, apply it, and verify the `users` table was actually created in PostgreSQL.

a. Create a shared SQLAlchemy base in `app/models/__init__.py`:
`touch app/models/__init__.py`

Add:
```
"""Database model base for Orbit."""
from sqlalchemy.orm import DeclarativeBase

class Base(DeclarativeBase):
    """Base class for all Orbit database models."""
    pass
```



b. Open `app/models/user.py` and add:

```
"""User database model for Orbit."""
from datetime import datetime, timezone
from sqlalchemy import Boolean, DateTime, String, Text
from sqlalchemy.orm import Mapped, mapped_column
from app.models import Base

class User(Base):
    """Represent a Orbit user account."""
    # Name of the PostgreSQL table represented by this model.
    __tablename__ = "users"
    # Primary key used internally to uniquely identify each user.
    id: Mapped[int] = mapped_column(primary_key=True)
    # Account identity fields. Usernames and email addresses must be unique.
    username: Mapped[str] = mapped_column(
        String(50),
        unique=True,
        nullable=False,
    )
    email: Mapped[str] = mapped_column(
        String(255),
        unique=True,
        nullable=False,
    )
    # Store the password hash rather than the user's actual password.
    password_hash: Mapped[str] = mapped_column(
        String(255),
        nullable=False,
    )
    # Public profile information.
    display_name: Mapped[str] = mapped_column(
        String(100),
        nullable=False,
    )
    bio: Mapped[str | None] = mapped_column(
        Text,
        nullable=True,
    )
    # Control account permissions and whether the account can be used.
    role: Mapped[str] = mapped_column(
        String(20),
        default="user",
        nullable=False,
    )
    is_active: Mapped[bool] = mapped_column(
        Boolean,
        default=True,
        nullable=False,
    )
    # Record when the account is created and most recently updated.
    created_at: Mapped[datetime] = mapped_column(
        DateTime(timezone=True),
        default=lambda: datetime.now(timezone.utc),
        nullable=False,
    )
    updated_at: Mapped[datetime] = mapped_column(
        DateTime(timezone=True),
        default=lambda: datetime.now(timezone.utc),
        onupdate=lambda: datetime.now(timezone.utc),
        nullable=False,
    )
```

c. Wire Alembic to the Orbit model metadata.

Open `migrations/env.py`.

Add these imports near the top:

```
from app.models import Base
from app.models.user import User
```

Then replace:

```
target_metadata = None
```

with:

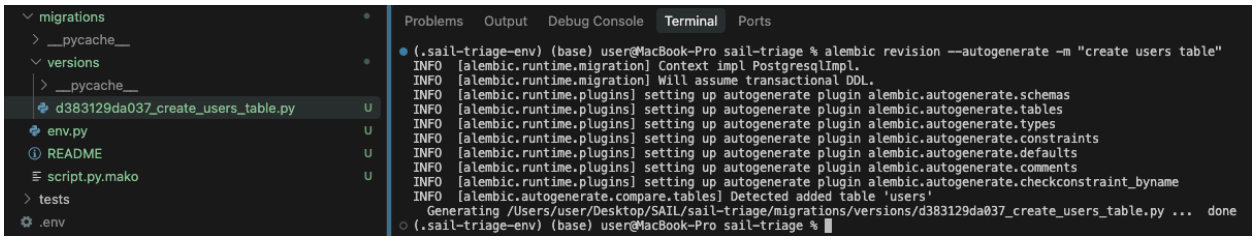
```
target_metadata = Base.metadata
```

The `User` import makes sure Alembic loads the model before inspecting `Base.metadata`.

d. Generate the first migration:

```
alembic revision --autogenerate -m "create users table"
```

Should see output indicating Alembic detected the new `users` table.



e. Inspect the migration file created under:

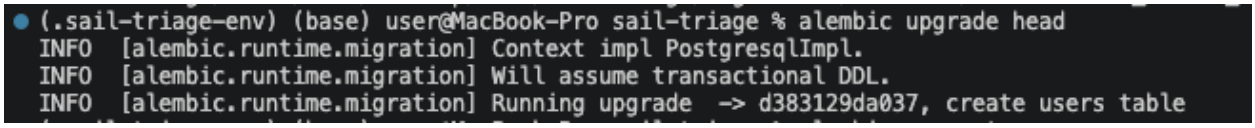
```
migrations/versions/
```

Confirm that its `upgrade()` function creates a `users` table with the fields from `User`.

```python
def upgrade() -> None:
    """Upgrade schema."""
    # ### commands auto generated by Alembic - please adjust! ###
    op.create_table('users',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('username', sa.String(length=50), nullable=False),
    sa.Column('email', sa.String(length=255), nullable=False),
    sa.Column('password_hash', sa.String(length=255), nullable=False),
    sa.Column('display_name', sa.String(length=100), nullable=False),
    sa.Column('bio', sa.Text(), nullable=True),
    sa.Column('role', sa.String(length=20), nullable=False),
    sa.Column('is_active', sa.Boolean(), nullable=False),
    sa.Column('created_at', sa.DateTime(timezone=True), nullable=False),
    sa.Column('updated_at', sa.DateTime(timezone=True), nullable=False),
    sa.PrimaryKeyConstraint('id'),
    sa.UniqueConstraint('email'),
    sa.UniqueConstraint('username')
    )
    # ### end Alembic commands ###
```

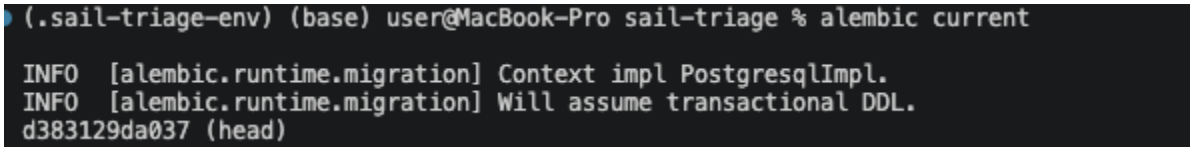
f. Apply the migration:

```
alembic upgrade head
```



g. Verify the migration state:

```
alembic current
```



It should show the revision just created and indicate `(head)`.

h. Verify the table directly in PostgreSQL:

```
psql -U orbit_app -d orbit -h localhost
```

Then run:

```
\dt
```

Should see at least:

```
alembic_version
users
```

Inspect the table:

```
\d users
```

Confirm the columns include:

```
id
username
email
password_hash
display_name
bio
role
is_active
created_at
updated_at
```

```
orbit=> \dt
          List of relations
 Schema |      Name       | Type  |   Owner
--------+-----------------+-------+-----------
 public | alembic_version | table | orbit_app
 public | users           | table | orbit_app
(2 rows)
```

```
orbit=> \d users
                                      Table "public.users"
    Column     |           Type           | Collation | Nullable |              Default
---------------+--------------------------+-----------+----------+-----------------------------------
 id            | integer                  |           | not null | nextval('users_id_seq'::regclass)
 username      | character varying(50)    |           | not null |
 email         | character varying(255)   |           | not null |
 password_hash | character varying(255)   |           | not null |
 display_name  | character varying(100)   |           | not null |
 bio           | text                     |           |          |
 role          | character varying(20)    |           | not null |
 is_active     | boolean                  |           | not null |
 created_at    | timestamp with time zone |           | not null |
 updated_at    | timestamp with time zone |           | not null |
Indexes:
    "users_pkey" PRIMARY KEY, btree (id)
    "users_email_key" UNIQUE CONSTRAINT, btree (email)
    "users_username_key" UNIQUE CONSTRAINT, btree (username)
```

Exit:

```
\q
```

Now have:

- the `User` SQLAlchemy model
- Alembic wired to the model metadata
- the first migration generated and applied
- the `users` table created in PostgreSQL
- Alembic reporting the migration at `head`

Step 11: Implement user registration, authentication, and password security

a. Create password hashing utilities

```python
"""Authentication and password security services for Orbit."""

from pwdlib import PasswordHash


PASSWORD_HASH = PasswordHash.recommended()


def hash_password(password: str) -> str:
    """Hash a plaintext password for secure storage."""
    return PASSWORD_HASH.hash(password)

                                        (parameter) hashed_password: str
def verify_password(password: str, hashed_password: str) -> bool:
    """Verify a plaintext password against a stored password hash."""
    return PASSWORD_HASH.verify(password, hashed_password)
```

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -c "from app.services.auth import hash_password, verify_password; h=hash_password
'OrbitTest123!'); print(h); print(verify_password('OrbitTest123!', h)); print(verify_password('wrong', h))"
$argon2id$v=19$m=65536,t=3,p=4$V8+TtTHpKvT999SZTLwIxg$O1u6sRfxbifU5Vh38yaH6XaqSDWHQFeXZYG4bmQwzPs
True
False
```

`auth.py` is a service module that provides Orbit's password hashing and verification functions. It was tested using `python -c` because service modules are imported and called rather than run directly.

```
tests/test_auth.py ...

============================================================== 3 passed in 0.20s ==
```

- Implemented hash_password() using pwdlib/Argon2.
- Implemented verify_password().
- Manually verified:
    - hashing works,
    - correct password → True,
    - incorrect password → False.

b. Generate a JWT secret and add `JWT_SECRET_KEY` to `.env`.

At command line:

- `python -c "import secrets; print(secrets.token_urlsafe(64))"`

paste into `.env` file

c. Create authentication/JWT utilities

Files involved:

- `app/config.py` — defines the JWT signing algorithm and token expiration settings.
- `app/services/auth.py` — creates, decodes, and validates JWT access tokens.
- `tests/test_auth.py` — tests token creation, decoding, invalid tokens, and expiration.

config.py

```python
"""Application configuration for Orbit."""

import os
from dotenv import load_dotenv

load_dotenv()

class Config:
    """Store configuration values loaded from environment variables."""

    APP_ENV = os.getenv("APP_ENV", "development")
    DEBUG = os.getenv("DEBUG", "false").lower() == "true"
    DATABASE_URL = os.getenv("DATABASE_URL")
    JWT_SECRET_KEY = os.getenv("JWT_SECRET_KEY")
    JWT_ALGORITHM = os.getenv("JWT_ALGORITHM", "HS256")
    JWT_ACCESS_TOKEN_EXPIRE_MINUTES = int(
        os.getenv("JWT_ACCESS_TOKEN_EXPIRE_MINUTES", "30")
    )
```

auth.py

```python
"""Authentication and password security services for Orbit."""

import jwt
from pwdlib import PasswordHash
from app.config import Config
from datetime import datetime, timedelta, timezone

PASSWORD_HASH = PasswordHash.recommended()

def hash_password(password: str) -> str:
    """Hash a plaintext password for secure storage."""
    return PASSWORD_HASH.hash(password)

def verify_password(password: str, hashed_password: str) -> bool:
    """Verify a plaintext password against a stored password hash."""
    return PASSWORD_HASH.verify(password, hashed_password)

def create_access_token(user_id: int) -> str:
    """Create a JWT access token for an authenticated user."""
    expires_at = datetime.now(timezone.utc) + timedelta(
        minutes=Config.JWT_ACCESS_TOKEN_EXPIRE_MINUTES
    )
    payload = {"sub": str(user_id), "exp": expires_at}

    return jwt.encode(
        payload,
        Config.JWT_SECRET_KEY,
        algorithm=Config.JWT_ALGORITHM,
    )

def decode_access_token(token: str) -> dict:
    """Decode and validate a JWT access token."""
    return jwt.decode(
        token,
        Config.JWT_SECRET_KEY,
        algorithms=[Config.JWT_ALGORITHM],
    )
```

Test run at command line uses a fake key placed in the .env file for JWT_SECRET_KEY. Once there can run:

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -c "from app.services.auth import create_access_token, decode_access_token; token=
create_access_token(1); print(token); print(decode_access_token(token))"
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxIiwiZXhwIjoxNzg4OTI0ODc2fQ.-rFf4ZI_RKbytT-3F0l7lg-Zccql5Nk5T3fbxyh7npM
{'sub': '1', 'exp': 1788924876}
```

Added tests to test_auth.py:

```python

def test_create_and_decode_access_token():
    """Verify that a valid JWT contains the correct user ID."""
    token = create_access_token(user_id=1)
    payload = decode_access_token(token)

    assert payload["sub"] == "1"
    assert "exp" in payload


def test_decode_invalid_access_token():
    """Verify that an invalid JWT is rejected."""
    with pytest.raises(jwt.InvalidTokenError):
        decode_access_token("invalid.token.value")


def test_decode_expired_access_token():
    """Verify that an expired JWT is rejected."""
    payload = {
        "sub": "1",
        "exp": datetime.now(timezone.utc) - timedelta(minutes=1),
    }
    token = jwt.encode(
        payload,
        Config.JWT_SECRET_KEY,
        algorithm=Config.JWT_ALGORITHM,
    )

    with pytest.raises(jwt.ExpiredSignatureError):
        decode_access_token(token)
```

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_auth.py
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 6 items

tests/test_auth.py ......

============================== 6 passed in 0.23s ===============================
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit %
```

d. Add registration request handling and tests

Registration establishes the API path used to create a SAIL Orbit account. Incoming registration data is validated before a `User` object is created, the plaintext password is hashed, and the new account is committed to PostgreSQL.

Files involved:

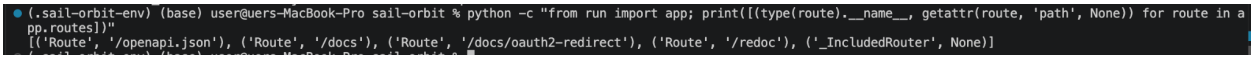
- `app/schemas/user.py` defines and validates registration fields.
- `app/extensions.py` provides a database session for the request and closes it afterward.
- `app/api/auth.py` implements `POST /auth/register` and account creation.

- `app/services/auth.py` provides password hashing.
- `app/models/user.py` defines the database representation of the account.
- `run.py` creates the FastAPI application and registers the authentication router, making `/auth/register` available through the application.

Request flow:

```
run.py registers auth router
        ↓
POST /auth/register
        ↓
UserRegistration validates request
        ↓
Password is hashed
        ↓
User model is created
        ↓
User is committed to PostgreSQL
        ↓
Non-sensitive account data returned
```

Run: `python -c "from run import app; print([(type(route).__name__, getattr(route, 'path', None)) for route in app.routes])"`



Shows app loads successfully.

Run: `python -c "from run import app; print(list(app.openapi()['paths'].keys()))"`

Shows FastAPI sees /auth/register.

Testing

Automated testing was added for the `POST /auth/register` endpoint using `httpx2.AsyncClient` and `ASGITransport`. This allows requests to be sent directly to the FastAPI application without starting a separate web server.

The test verified:

- A valid registration request returns HTTP `201 Created`.
- The response contains the expected username, email, display name, role, and account status.
- Neither the plaintext password nor `password_hash` is exposed in the API response.
- The new user is actually stored in PostgreSQL.
- The stored password is a hash rather than the submitted plaintext password.
- The stored hash successfully verifies against the original password.
- Test users are removed before and after testing so repeated test runs do not leave test data in the database.

The test uses `@pytest.mark.anyio` so pytest can execute the asynchronous HTTP requests. `httpx2` is used directly rather than Starlette's `TestClient`, avoiding the deprecated AnyIO interface encountered with `TestClient`.

Result: `1 passed in 0.39s` with no warnings.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 1 item

tests/test_users.py::test_register_user[asyncio] PASSED                  [100%]

============================== 1 passed in 0.39s ===============================
```

e. Add login request handling and tests

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py::test_login_user -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 1 item

tests/test_users.py::test_login_user[asyncio] PASSED                     [100%]

============================== 1 passed in 0.76s ===============================
```

f. Add current-user authentication dependency

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_auth.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 8 items

tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED  [ 12%]
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED [ 25%]
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED [ 37%]
tests/test_auth.py::test_create_and_decode_access_token PASSED           [ 50%]
tests/test_auth.py::test_decode_invalid_access_token PASSED              [ 62%]
tests/test_auth.py::test_decode_expired_access_token PASSED              [ 75%]
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED     [ 87%]
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED   [100%]

============================== 8 passed in 0.69s ===============================
```

g. Add duplicate username/email checks

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 4 items

tests/test_users.py::test_register_user[asyncio] PASSED                  [ 25%]
tests/test_users.py::test_login_user[asyncio] PASSED                     [ 50%]
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED [ 75%]
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED [100%]

============================== 4 passed in 0.75s ===============================
```

h. Add password validation rules

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 9 items

tests/test_users.py::test_register_user[asyncio] PASSED                  [ 11%]
tests/test_users.py::test_login_user[asyncio] PASSED                     [ 22%]
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED [ 33%]
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED [ 44%]
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED [ 55%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED [ 66%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED [ 77%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED [ 88%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED [100%]

============================== 9 passed in 0.72s ===============================
```

i. Complete registration testing

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 10 items

tests/test_users.py::test_register_user[asyncio] PASSED                  [ 10%]
tests/test_users.py::test_login_user[asyncio] PASSED                     [ 20%]
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED [ 30%]
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED [ 40%]
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED [ 50%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED [ 60%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED [ 70%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED [ 80%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED [ 90%]
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED [100%]

============================== 10 passed in 0.76s ==============================
```

j. Complete login testing

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_users.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 12 items

tests/test_users.py::test_register_user[asyncio] PASSED                  [  8%]
tests/test_users.py::test_login_user[asyncio] PASSED                     [ 16%]
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED [ 25%]
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED [ 33%]
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED [ 41%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED [ 50%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED [ 58%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED [ 66%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED [ 75%]
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED [ 83%]
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED [ 91%]
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED [100%]

============================== 12 passed in 0.95s ==============================
```

k. Complete protected authentication testing

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_auth.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 10 items

tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED  [ 10%]
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED [ 20%]
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED [ 30%]
tests/test_auth.py::test_create_and_decode_access_token PASSED           [ 40%]
tests/test_auth.py::test_decode_invalid_access_token PASSED              [ 50%]
tests/test_auth.py::test_decode_expired_access_token PASSED              [ 60%]
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED     [ 70%]
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED   [ 80%]
tests/test_auth.py::test_get_current_user_rejects_expired_token PASSED   [ 90%]
tests/test_auth.py::test_get_current_user_rejects_unknown_user PASSED    [100%]

============================== 10 passed in 0.79s ==============================
```

l. Verify Step 11 end-to-end

```
tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED  [  4%]
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED [  8%]
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED [ 13%]
tests/test_auth.py::test_create_and_decode_access_token PASSED           [ 17%]
tests/test_auth.py::test_decode_invalid_access_token PASSED              [ 21%]
tests/test_auth.py::test_decode_expired_access_token PASSED              [ 26%]
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED     [ 30%]
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED   [ 34%]
tests/test_auth.py::test_get_current_user_rejects_expired_token PASSED   [ 39%]
tests/test_auth.py::test_get_current_user_rejects_unknown_user PASSED    [ 43%]
tests/test_database_connection.py::test_database_connection PASSED       [ 47%]
tests/test_users.py::test_register_user[asyncio] PASSED                  [ 52%]
tests/test_users.py::test_login_user[asyncio] PASSED                     [ 56%]
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED [ 60%]
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED [ 65%]
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED [ 69%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED [ 73%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED [ 78%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED [ 82%]
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED [ 86%]
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED [ 91%]
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED [ 95%]
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED [100%]

============================== 23 passed in 1.05s ==============================
```

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % ruff check .
All checks passed!
```

```
All checks passed!
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % black .
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/config.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/extensions.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/models/__init__.py
reformatted                                          orbit/app/schemas/auth.py
reformatted                                          orbit/migrations/env.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/models/user.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/run.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/services/auth.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/schemas/user.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/tests/test_database_connection.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/app/api/auth.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/migrations/versions/d383129da037_create_users_table.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/tests/test_auth.py
reformatted /Users/user/Desktop/SAIL/sail-orbit/tests/test_users.py

All done! ✨ 🍰 ✨
14 files reformatted, 19 files left unchanged.
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % black --check .
All done! ✨ 🍰 ✨
33 files would be left unchanged.
```

Open file in editor (cmd + click)

Step 12: Implement user profiles, roles, and permissions

a. Create and register authenticated user profile retrieval

Create the first user endpoint in app/api/users.py. This endpoint returns the complete account/profile information appropriate for the account owner. The authenticated user's identity comes from the JWT through the existing get_current_user dependency, so the client does not supply a user ID. Then register the users router with the FastAPI application. In run.py, you should already have the authentication router registered.

Then verify that FastAPI successfully loads both routers; use FastAPI's generated OpenAPI schema:

python -c "from run import app; print(list(app.openapi()['paths'].keys()))"

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -c "from run import app; print(list(app.openapi()['paths'].keys()))"
['/auth/register', '/auth/login', '/users/me']
```

The three expected endpoints are exposed. The testing of its actual behavior next. Must verify three things:

1. A valid JWT allows access to `GET /users/me`.
2. The endpoint returns the authenticated user's expected profile/account fields.
3. The response does **not** expose `password_hash`.
4. Separately verify that access without a JWT is rejected

Also, now is a good time to refactor the test_users.py file. Include more reusable helpers and fixtures from pytest. **DONE**

1. Added 2 fixtures and 3 helpers currently
2. Tests still pass

Added `test_get_own_profile()` to test first 3, then for #4 added `test_get_own_profile_requires_authentication()`

```
tests/test_users.py::test_register_user[asyncio] PASSED
tests/test_users.py::test_login_user[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED
tests/test_users.py::test_get_own_profile[asyncio] PASSED
tests/test_users.py::test_get_own_profile_requires_authentication[asyncio] PASSED

============================================================== 14 passed in 0.93s ==============================================================
```

b. Add user profile update functionality

```
class UserProfileUpdate(BaseModel):
    """Define profile information an authenticated user may update."""

    # Both fields are optional so a user can update one profile field without
    # being required to resubmit the other.
    display_name: str | None = None
    bio: str | None = None
```

`UserProfileUpdate` deliberately creates an **allowlist** of user-editable profile fields. Even if a client submits values such as `role="admin"` or `is_active=False`, those fields are not part of this schema and therefore will not become part of the profile-update operation.

Now add the update endpoint to `app/api/users.py`.

Why **`PATCH`:** this operation supports partial profile updates. A request containing only `bio`, for example, changes only `bio`; `display_name` remains untouched.

Why **`model_dump(exclude_unset=True)`:** it distinguishes a field the user deliberately supplied from one they omitted. That is important here because `bio=None` can intentionally clear a bio, while an omitted `bio` means leave it unchanged.

```
collected 15 items

tests/test_users.py::test_register_user[asyncio] PASSED
tests/test_users.py::test_login_user[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile[asyncio] PASSED

============================================================== 15 passed in 1.02s ==============================================================
```

```
collected 30 items

tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED
tests/test_auth.py::test_create_and_decode_access_token PASSED
tests/test_auth.py::test_decode_invalid_access_token PASSED
tests/test_auth.py::test_decode_expired_access_token PASSED
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_expired_token PASSED
tests/test_auth.py::test_get_current_user_rejects_unknown_user PASSED
tests/test_database_connection.py::test_database_connection PASSED
tests/test_user_profiles.py::test_get_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_partially[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_can_clear_bio[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_cannot_change_protected_fields[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_requires_authentication[asyncio] PASSED
tests/test_users.py::test_register_user[asyncio] PASSED
tests/test_users.py::test_login_user[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED

============================== 30 passed in 1.51s ==============================
```

Implemented authenticated profile updates for `display_name` and `bio`, including partial updates and bio clearing. Profile tests were separated into `test_user_profiles.py` and verify retrieval, updates, authentication, protected fields, and sensitive-data protection; **all 7 tests pass.**

c. Add public user profile retrieval

This will allow Orbit users to view another user's public profile without exposing private account information such as email, role, account status, or password data.

Implemented public profile retrieval through `GET /users/{username}`. Public profiles expose only appropriate fields (`id`, `username`, `display_name`, `bio`, and `created_at`), exclude private account and authentication data, and return `404` for unknown users. **All 9 profile tests pass.**

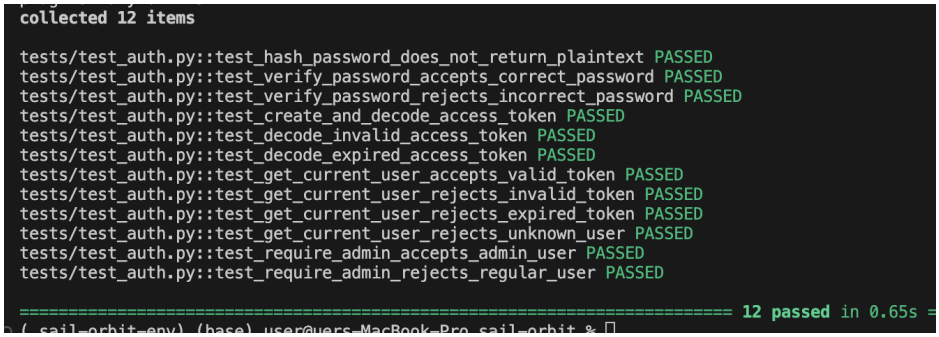
```
plugins: anyio-4.13.1
collected 9 items

tests/test_user_profiles.py::test_get_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_partially[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_can_clear_bio[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_cannot_change_protected_fields[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_get_public_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_public_profile_rejects_unknown_user[asyncio] PASSED

============================== 9 passed in 0.95s ===============================
```

d. Implement role-based authorization

Role-based authorization will be implemented by adding a reusable authorization dependency to the existing authentication service, which will allow administrator-only endpoints to verify a user's role consistently.

This creates the reusable role check for administrator-only operations. A missing or invalid login remains a **401**, while an authenticated user without the required role receives **403 Forbidden**.

Next, verify the authorization dependency itself before using it for admin endpoints. 12d authorization testing is complete: admin users are permitted, regular users receive 403 Forbidden, and all 12 authentication tests pass.



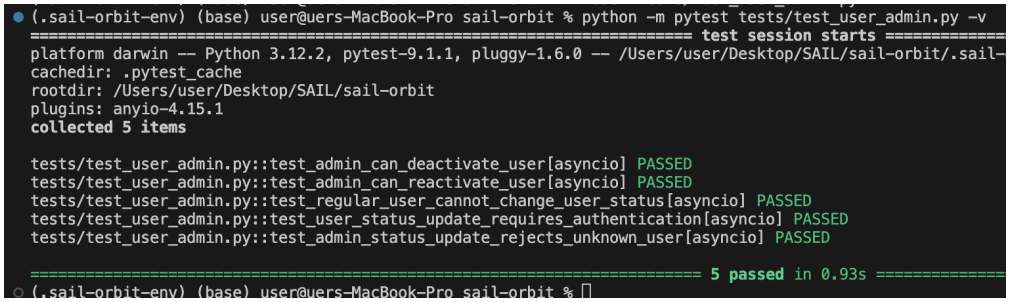
e. Add administrator-only user management, including account activation/deactivation
Start with the request schema for changing account status. Add this below `UserProfileUpdate` in `app/schemas/user.py`:

```
class UserStatusUpdate(BaseModel):
    """Define an administrator-requested change to a user's account status."""

    # Account activation is intentionally the only field accepted here.
    # Role changes and profile changes are separate operations with different
    # authorization and validation requirements.
    is_active: bool
```

Next, add the administrator endpoint that changes a user's active status. `admin_user` is intentionally present even though the function body does not use its value: resolving that dependency is what enforces administrator access before the endpoint runs.

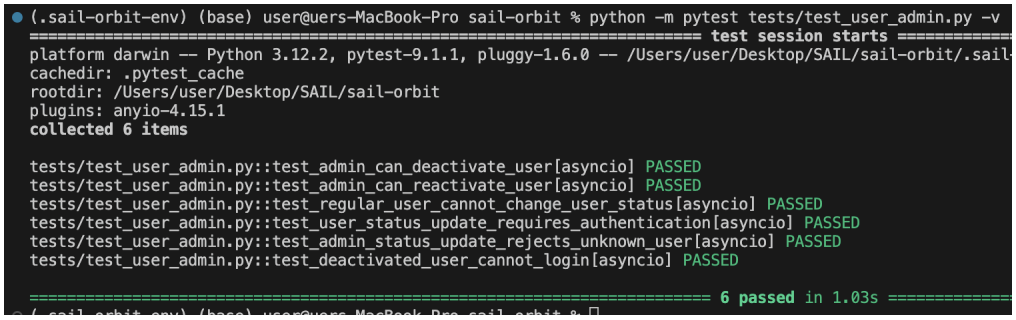
Next, create the administrator status endpoint. Since this is specifically user-management behavior, keep these tests in a new file: `touch tests/test_user_admin.py`

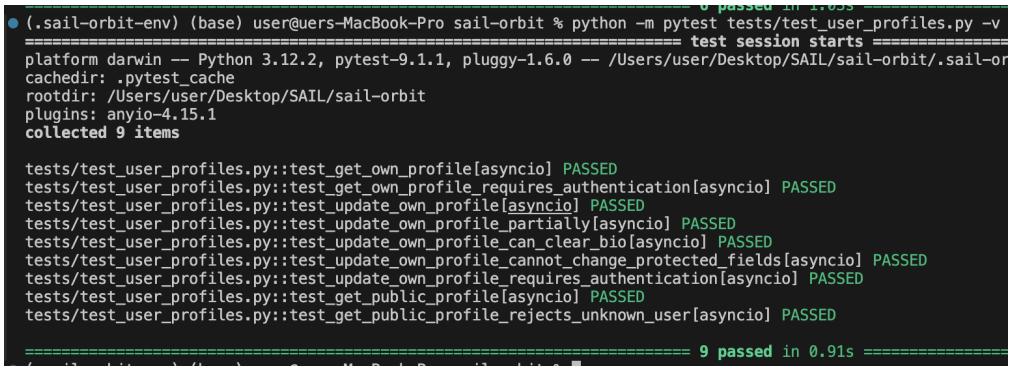
Created `tests/test_user_admin.py` to test administrator account-management functionality through the real API. The file creates isolated admin and standard-user accounts, authenticates the administrator through the normal login/JWT flow, and tests deactivation, reactivation, rejection of non-admin and unauthenticated requests, unknown-user handling, and database persistence of status changes. The tests verify that account activation and deactivation are limited to authenticated administrators and that status changes are correctly persisted.



f. Complete profile testing

Added inactive-account enforcement to `/auth/login` and a sixth admin test verifying that a deactivated user is rejected even with valid credentials. This makes account deactivation effective at authentication time instead of allowing a token to be issued and rejecting it only at protected endpoints.





g. Complete role and permission testing

Tested the `require_admin` authorization dependency and administrator account-management endpoint across permitted and rejected access paths. Verification covered admin access, `403` rejection of authenticated regular users, `401` rejection of unauthenticated requests and invalid/expired tokens, rejection of tokens for nonexistent users, administrator deactivation/reactivation of accounts, blocked login for deactivated accounts, and immediate rejection of previously issued JWTs after an account is deactivated; **all 19 tests pass.**

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest tests/test_auth.py tests/test_user_admin.py -v
collected 19 items

tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED
tests/test_auth.py::test_create_and_decode_access_token PASSED
tests/test_auth.py::test_decode_invalid_access_token PASSED
tests/test_auth.py::test_decode_expired_access_token PASSED
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_expired_token PASSED
tests/test_auth.py::test_get_current_user_rejects_unknown_user PASSED
tests/test_auth.py::test_require_admin_accepts_admin_user PASSED
tests/test_auth.py::test_require_admin_rejects_regular_user PASSED
tests/test_user_admin.py::test_admin_can_deactivate_user[asyncio] PASSED
tests/test_user_admin.py::test_admin_can_reactivate_user[asyncio] PASSED
tests/test_user_admin.py::test_regular_user_cannot_change_user_status[asyncio] PASSED
tests/test_user_admin.py::test_user_status_update_requires_authentication[asyncio] PASSED
tests/test_user_admin.py::test_admin_status_update_rejects_unknown_user[asyncio] PASSED
tests/test_user_admin.py::test_deactivated_user_cannot_login[asyncio] PASSED
tests/test_user_admin.py::test_deactivated_user_cannot_use_existing_token[asyncio] PASSED

============================== 19 passed in 1.46s ==============================
```

h. Verify Step 12 end-to-end

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest -v
collected 41 items

tests/test_auth.py::test_hash_password_does_not_return_plaintext PASSED
tests/test_auth.py::test_verify_password_accepts_correct_password PASSED
tests/test_auth.py::test_verify_password_rejects_incorrect_password PASSED
tests/test_auth.py::test_create_and_decode_access_token PASSED
tests/test_auth.py::test_decode_invalid_access_token PASSED
tests/test_auth.py::test_decode_expired_access_token PASSED
tests/test_auth.py::test_get_current_user_accepts_valid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_invalid_token PASSED
tests/test_auth.py::test_get_current_user_rejects_expired_token PASSED
tests/test_auth.py::test_get_current_user_rejects_unknown_user PASSED
tests/test_auth.py::test_require_admin_accepts_admin_user PASSED
tests/test_auth.py::test_require_admin_rejects_regular_user PASSED
tests/test_database_connection.py::test_database_connection PASSED
tests/test_user_admin.py::test_admin_can_deactivate_user[asyncio] PASSED
tests/test_user_admin.py::test_admin_can_reactivate_user[asyncio] PASSED
tests/test_user_admin.py::test_regular_user_cannot_change_user_status[asyncio] PASSED
tests/test_user_admin.py::test_user_status_update_requires_authentication[asyncio] PASSED
tests/test_user_admin.py::test_admin_status_update_rejects_unknown_user[asyncio] PASSED
tests/test_user_admin.py::test_deactivated_user_cannot_login[asyncio] PASSED
tests/test_user_admin.py::test_deactivated_user_cannot_use_existing_token[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_partially[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_can_clear_bio[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_cannot_change_protected_fields[asyncio] PASSED
tests/test_user_profiles.py::test_update_own_profile_requires_authentication[asyncio] PASSED
tests/test_user_profiles.py::test_get_public_profile[asyncio] PASSED
tests/test_user_profiles.py::test_get_public_profile_rejects_unknown_user[asyncio] PASSED
tests/test_users.py::test_register_user[asyncio] PASSED
tests/test_users.py::test_login_user[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_username[asyncio] PASSED
tests/test_users.py::test_register_rejects_duplicate_email[asyncio] PASSED
tests/test_users.py::test_register_rejects_short_password[asyncio] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-orbittest123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-ORBITTEST123!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTestPass!] PASSED
tests/test_users.py::test_register_rejects_password_missing_required_character_type[asyncio-OrbitTest1234] PASSED
tests/test_users.py::test_register_rejects_invalid_email[asyncio] PASSED
tests/test_users.py::test_login_rejects_incorrect_password[asyncio] PASSED
tests/test_users.py::test_login_rejects_unknown_username[asyncio] PASSED

============================== 41 passed in 2.26s ==============================
```

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -c "from run import app; print(list(app.openapi()['paths'].keys()))"
['/auth/register', '/auth/login', '/users/me', '/users/{username}', '/users/{username}/status']
```

Completed end-to-end verification of user profiles, roles, permissions, authentication, and administrator account management. The full test suite passes with **41 tests**, Ruff and Black checks pass, and all expected authentication and user API routes are registered.

Step 13: Run and manually verify the current backend locally.

a. Start the Orbit Application

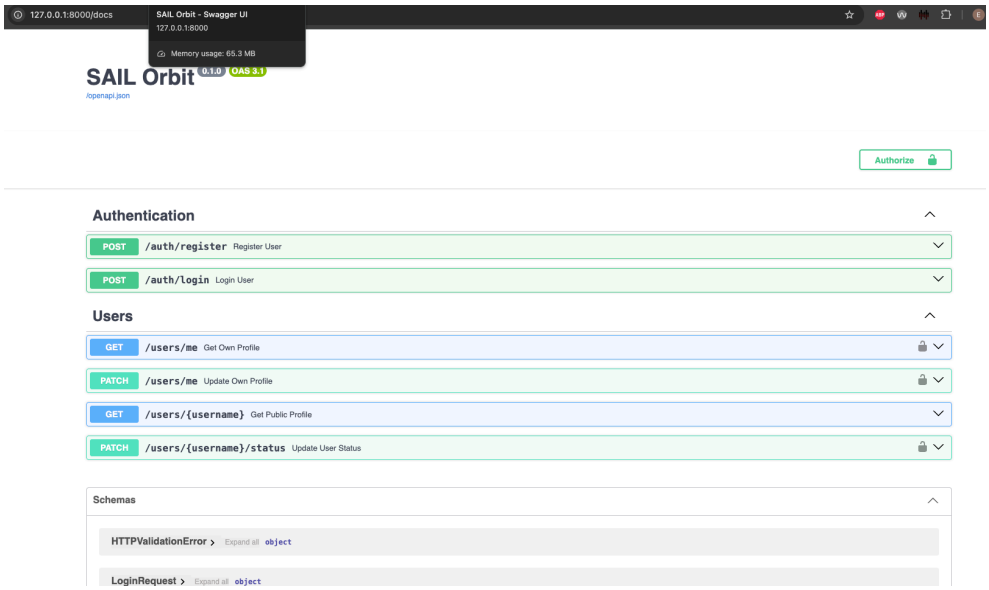
1. Start the FastAPI application: `uvicorn run:app --reload`

2. Verify the terminal reports that Uvicorn is running at `http://127.0.0.1:8000`.

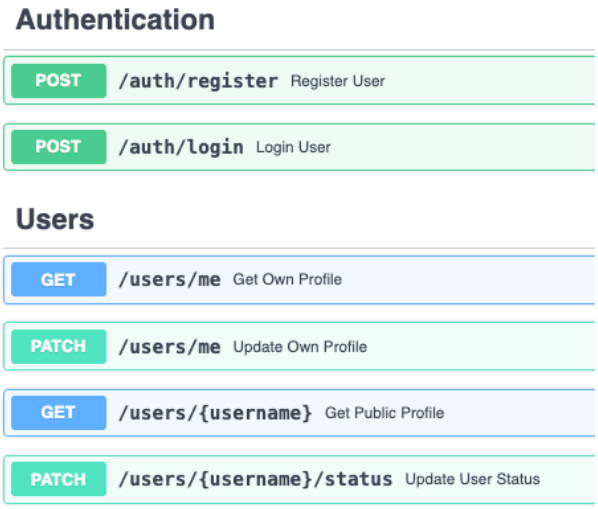
```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % uvicorn run:app --reload
INFO:     Will watch for changes in these directories: ['/Users/user/Desktop/SAIL/sail-orbit']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

b. Open the FastAPI Browser Interface

1. Open a web browser.
2. Navigate to: `127.0.0.1:8000/docs`



3. Verify the interactive API documentation loads and displays these endpoints:
   - `POST /auth/register`
   - `POST /auth/login`
   - `GET /users/me`
   - `PATCH /users/me`
   - `GET /users/{username}`
   - `PATCH /users/{username}/status`



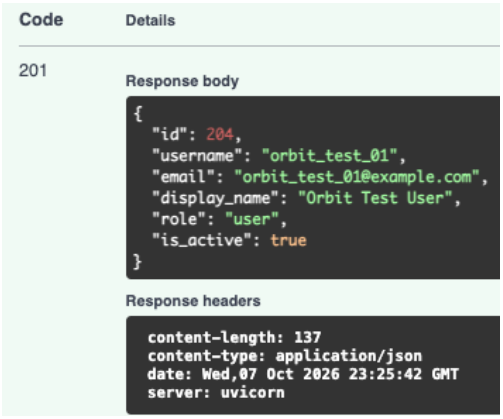
c. Verify Registration and Authentication

Use the interactive API documentation to register a test user, log in, and obtain an authentication token.
Verify that registration succeeds, login returns a token, and invalid credentials are rejected.

1. In Swagger at 127.0.0.1:8000/docs, click into auth/registration and click try it out.
2. Replace example values with:

```
{
  "username": "orbit_test_01",
  "email": "orbit_test_01@example.com",
  "password": "OrbitTest123!",
  "display_name": "Orbit Test User"
}
```

3. Click Execute



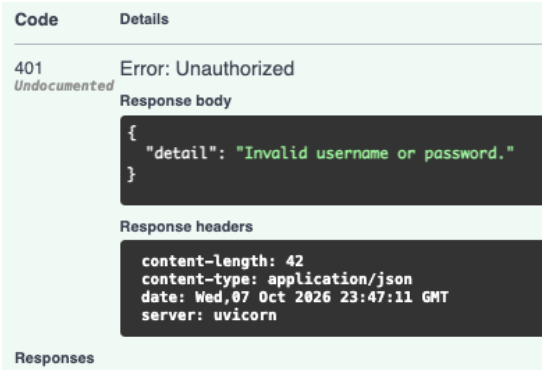
4. Repeat steps for login. Replace example code there with

```
{
  "username": "orbit_test_01",
  "password": "OrbitTest123!"
}
```

5. Click Execute

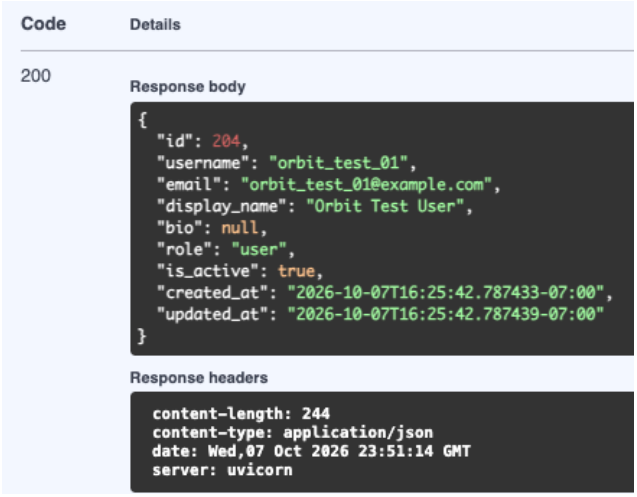
6. Repeat login Try it out steps again, using the below example:

```
{
  "username": "orbit_test_01",
  "password": "WrongPassword123!"
}
```



d. Verify User Profiles
Use the authenticated API endpoints to retrieve and update the test user's profile. Verify that changes persist in PostgreSQL and that public profile retrieval works.

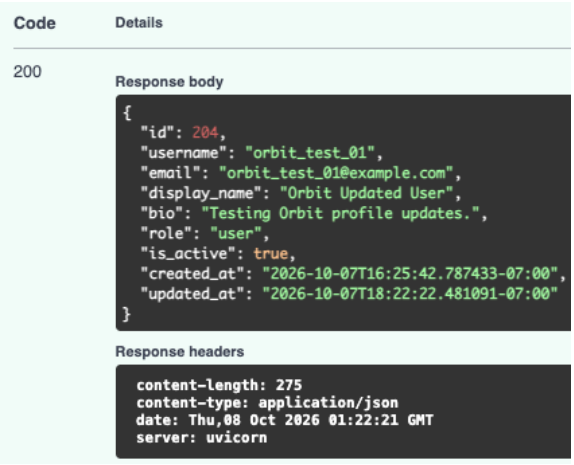
1. In Swagger UI, click Authorize at the top right.
2. Enter the access token returned by your successful login.
3. Click Authorize, then Close.
4. Expand `GET /users/me`, click Try it out, then Execute.
5. Verify the response is `200 OK` and contains the profile for `orbit_test_01`.



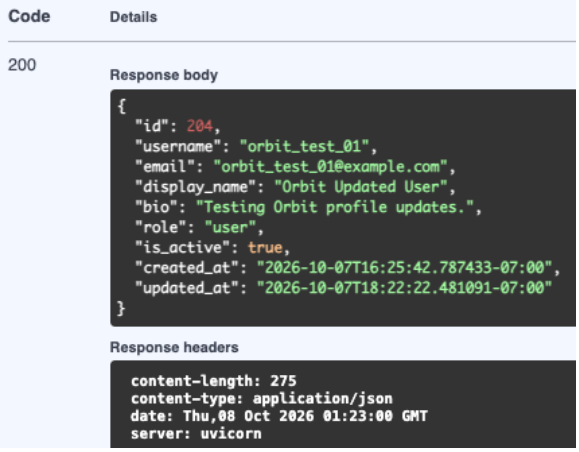
6. Verify Profile Updates
   a. Expand `PATCH /users/me` in Swagger UI.
   b. Click Try it out.
   c. Enter:

```
{
  "display_name": "Orbit Updated User",
  "bio": "Testing Orbit profile updates."
}
```

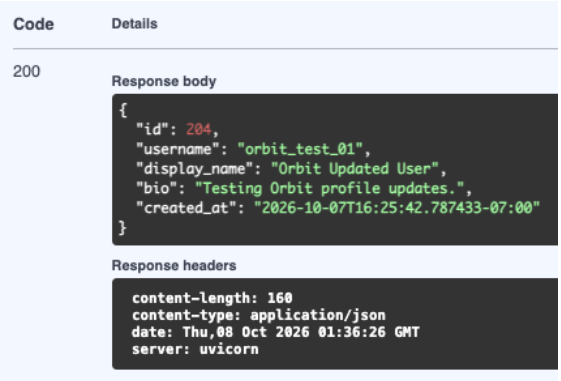
   d. Click Execute and verify the response is `200 OK`.



e. Run `GET /users/me` again to verify both changes were saved.

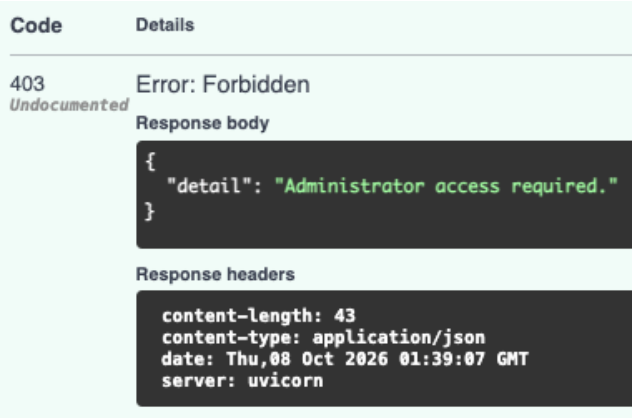


f. Run `GET /users/username` with `orbit_test_01:`



e. Verify Permissions and Account Status
Verify that regular users cannot access administrator-only operations.
Using an administrator account, test account deactivation and reactivation. Confirm that deactivated users cannot authenticate or access protected endpoints.

1. Regular user:
    a. Run `PATCH /users/{username}/status` with username `orbit_test_01` and body `{"is_active": false}` The false value tests an attempt to deactivate an account. The true value tests an attempt to reactivate one. For either, expect `403 Forbidden`.

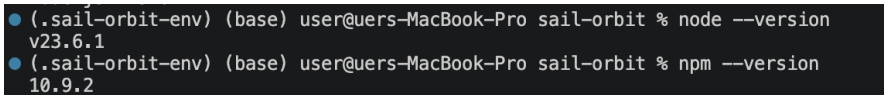


Step 14: Create the Orbit frontend structure and connect it to the FastAPI backend.

a. Create the Frontend Project

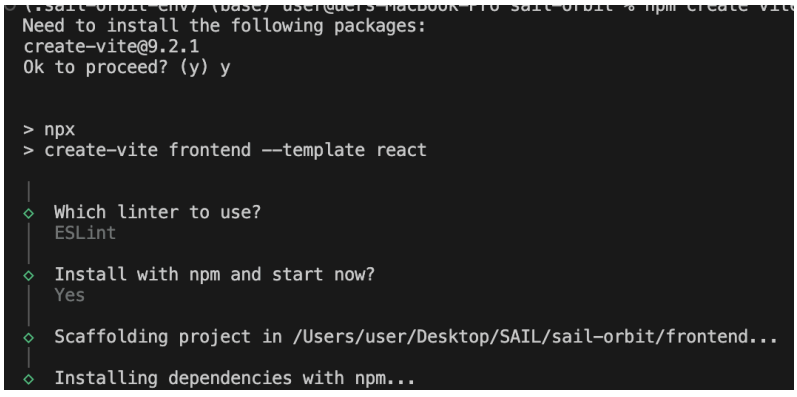
1. Open a second terminal in VS Code, separate from the terminal running Uvicorn. Navigate to `SAIL/sail-orbit`.
2. Verify Node.js and npm:

```
node --version
npm --version
```



3. Create the `frontend` directory and React/Vite project:

```
npm create vite@latest frontend -- --template react
```



b. Configure Frontend Styling and Components

1. Install Tailwind CSS and its Vite integration.
   From `SAIL/sail-orbit/frontend`, run:

```
npm install tailwindcss @tailwindcss/vite
```

2. Configure Tailwind in `frontend/vite.config.js`.

Replace the file contents with:

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'
import path from 'node:path'

export default defineConfig({
  plugins: [react(), tailwindcss()],
  resolve: {
    alias: {
      '@': path.resolve(import.meta.dirname, './src'),
    },
  },
})
```

3. Configure the stylesheet.

Replace the contents of `frontend/src/index.css` with:

```
@import "tailwindcss";
```

4. Configure the import alias.

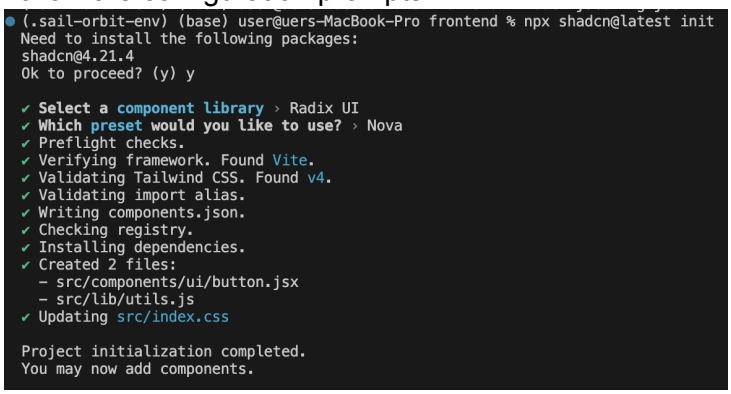
Create `frontend/jsconfig.json`:

```
{
  "compilerOptions": {
    "baseUrl": ".",
    "paths": {
      "@/*": ["./src/*"]
    }
  }
}
```

5. Initialize shadcn/ui.

From `frontend`, run: `npx shadcn@latest init`

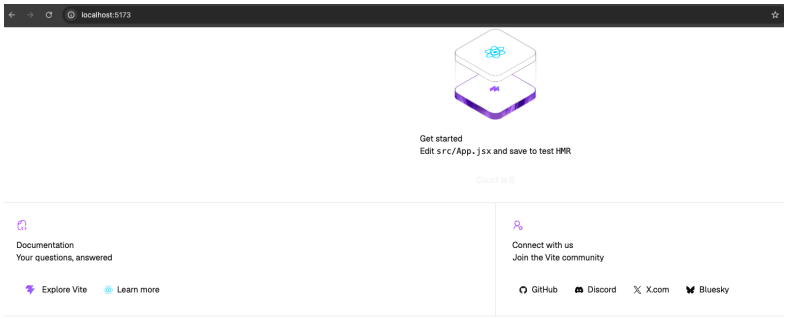
Follow the configuration prompts.



6. Verify the frontend starts.

```
npm run dev
```

Open the local URL displayed by Vite, typically `http://localhost:5173`, and confirm the application loads. The default Vite starter page is expected at this stage.



c. Connect React to the FastAPI Backend

Establishes communication between the React frontend and the existing FastAPI backend using the public user profile endpoint.

1. Configure the frontend API URL

Create this file: `SAIL/sail-orbit/frontend/.env`

Add: `VITE_API_BASE_URL=http://127.0.0.1:8000`

2. Enable CORS in FastAPI

Open `SAIL/sail-orbit/run.py`.

Add this import:

```
from fastapi.middleware.cors import CORSMiddleware
```

Immediately after:

```
app = FastAPI(title="SAIL Orbit")
```

Add:

```
app.add_middleware(
     CORSMiddleware,
     allow_origins=[
          "http://localhost:5173",
          "http://127.0.0.1:5173",
          ],
     allow_credentials=True,
     allow_methods=["*"],
     allow_headers=["*"],
)
```

Keep the existing router registrations unchanged. CORS allows the browser-based frontend on port 5173 to communicate with FastAPI on port 8000.

3. Create the frontend API client

Create:

`SAIL/sail-orbit/frontend/src/api.js`

Add:

```
const API_BASE_URL = import.meta.env.VITE_API_BASE_URL

export async function getPublicProfile(username) {
  const response = await fetch(
    `API_BASE_URL/users/{encodeURIComponent(username)}`
  )

  if (!response.ok) {
```

```
    throw new Error(`Profile request failed: ${response.status}`)
  }

  return response.json()
}
```

4. Connect the React application

Open:

`SAIL/sail-orbit/frontend/src/App.jsx`

Replace its contents with:

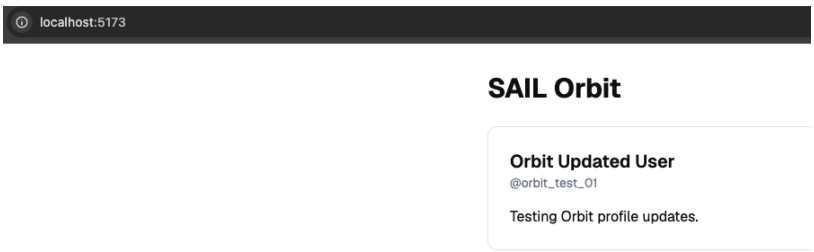
```
import { useEffect, useState } from 'react'
import { getPublicProfile } from './api'

function App() {
  const [profile, setProfile] = useState(null)
  const [error, setError] = useState(null)

  useEffect(() => {
    getPublicProfile('orbit_test_01')
      .then(setProfile)
      .catch((err) => setError(err.message))
  }, [])

  return (
    <main className="mx-auto max-w-2xl p-8">
      <h1 className="mb-6 text-3xl font-bold">SAIL Orbit</h1>

      {error && <p className="text-red-600">{error}</p>}

      {!profile && !error && <p>Loading profile...</p>}

      {profile && (
        <section className="rounded-lg border p-6">
          <h2 className="text-xl font-semibold">
            {profile.display_name}
          </h2>
          <p className="text-sm text-gray-500">
            @{profile.username}
          </p>
          <p className="mt-4">{profile.bio}</p>
        </section>
      )}
    </main>
  )
}

export default App
```

This is a temporary connection test. The actual profile interface belongs to Step 16.

5. Verify the connection

Confirm FastAPI is running at `http://127.0.0.1:8000`.
Restart Vite to load the new `.env` file: `npm run dev`
Open `http://localhost:5173`.
Expected result:
SAIL Orbit
Orbit Updated User
@orbit_test_01
Testing Orbit profile updates.

The profile data should come from FastAPI and PostgreSQL, not hardcoded frontend values.



The screenshot confirms React successfully retrieved and displayed the existing user profile through FastAPI.

Step 15: Build registration and login screens using the existing authentication API.
a. Confirm the authentication API contract
Before writing the React forms, we need to verify the exact request and response formats already implemented.

1. Keep FastAPI and Vite running.
2. Open `http://127.0.0.1:8000/docs`.
3. Expand these endpoints:
   - `POST /auth/register`
   - `POST /auth/login`
4. Confirm the Request body and Responses schemas for both endpoints correct

Endpoint	Request fields	Successful response
`POST /auth/register`	username, email, password, display_name	HTTP 201, string
`POST /auth/login`	username, password	HTTP 200, access_token, token_type

This makes sure the frontend sends the correct fields, handles authentication tokens correctly, and matches the existing backend.

b. Add authentication functions to `api.js`
Below `getPublicProfile()` function:
Add `registerUser()` and `loginUser()` to `frontend/src/api.js`.

Both functions use `fetch()` to send JSON requests to the existing FastAPI endpoints. They check whether the request succeeded, report errors, and return the server's response.

These functions provide a reusable connection between the React registration/login screens and the backend. Registration creates an account; login retrieves a JWT for authenticated requests.

c. Build the Registration and Login Screens

Create two React components using the authentication functions added previously.

1. Create `frontend/src/components/RegisterForm.jsx`

   Creates the registration form for username, email, display name, and password. React's useState manages form values, errors, and submission status. On submission, the component calls `registerUser()` in `api.js`. Successful registration returns the user to the login screen; errors are displayed in the form.

2. Create `frontend/src/components/LoginForm.jsx`

   Creates the login form for username and password. The component uses `useState` to manage inputs and calls `loginUser()` to authenticate through FastAPI. Successful authentication returns a JWT access token to `App.jsx`; failed requests display an error.

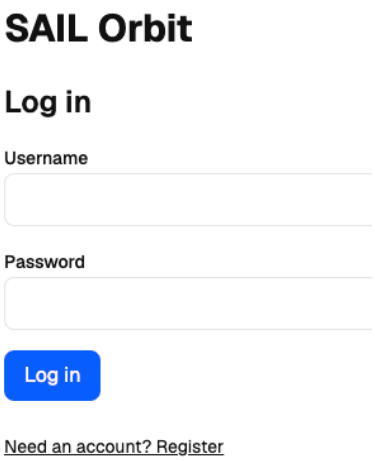
3. Update `frontend/src/App.jsx`

   Coordinates the registration and login components using React state. The `screen` variable controls which form appears, while `accessToken` stores the JWT after login. Logging out clears the token. The token is lost when the page refreshes.

4. Verify the screens

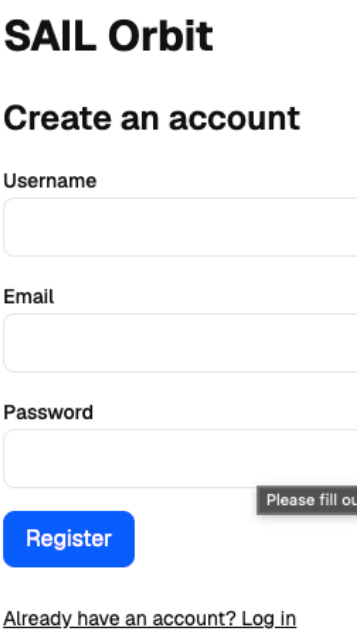
   With FastAPI and Vite running, open http://localhost:5173.

   a. Confirm the login form appears.

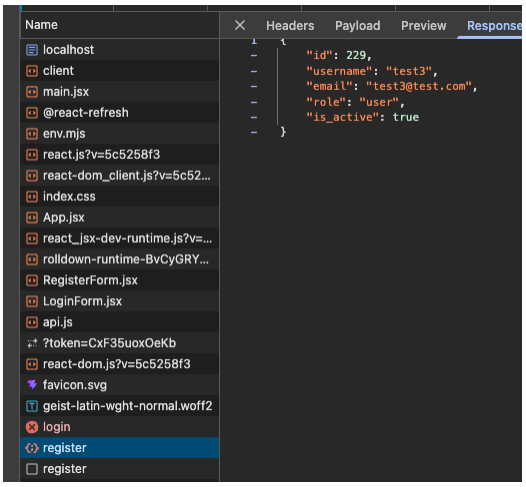


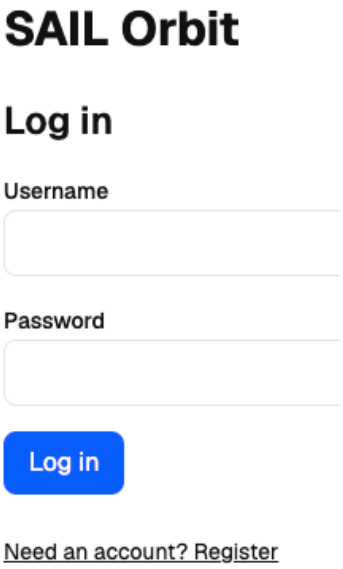
   b. Select Need an account? Register.

   At this step, a display name was recognized as being the same as a username and the display name field was removed. See the appendix, Section 8: SAIL Orbit: Removal of display_name, for details.

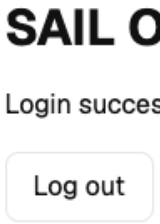


c. Register a new test user with a unique username and email and password. Confirm the application returns to the login screen.





d. Log in using the new credentials. Confirm Login successful appears.



e. Select Log out and confirm the login screen returns. **CONFIRMED**

Step 16: GitHub Actions for Black, Ruff, and frontend ESLint
a. Verify the Git repository configuration
From the SAIL repository root:
git branch --show-current
git status --short
git remote -v
Confirmed:

- Current branch: `develop`
- Working tree: clean before changes
- GitHub remote: `origin`
- GitLab remote: `gitlab`

b. Verify the existing Python and frontend configurations
cat sail-orbit/frontend/package.json
cat sail-orbit/ruff.toml
cat sail-orbit/requirements.txt
Confirmed:

- Black: `26.5.1`
- Ruff: `0.16.3`
- ESLint command: `npm run lint`

- Python application directory: `sail-orbit/`
- Frontend directory: `sail-orbit/frontend/`

c. Create the GitHub Actions workflow
From the SAIL repository root:
mkdir -p .github/workflows
Create `.github/workflows/ci.yml`:

```
name: SAIL Orbit CI

on:
  push:
    branches: [develop]
  pull_request:
    branches: [main]

jobs:
  python-checks:
    runs-on: ubuntu-latest
    continue-on-error: true
    defaults:
      run:
        working-directory: sail-orbit

    steps:
      - uses: actions/checkout@v4

      - uses: actions/setup-python@v5
        with:
          python-version: "3.12"
          cache: pip
          cache-dependency-path: sail-orbit/requirements.txt

      - name: Install Python dependencies
        run: python -m pip install -r requirements.txt

      - name: Check Black formatting
        run: black --check .

      - name: Run Ruff
        run: ruff check .

  frontend-checks:
    runs-on: ubuntu-latest
    continue-on-error: true
    defaults:
      run:
        working-directory: sail-orbit/frontend

    steps:
      - uses: actions/checkout@v4
```

```
      - uses: actions/setup-node@v4
        with:
          node-version: "22"
          cache: npm
          cache-dependency-path: sail-orbit/frontend/package-lock.json

      - name: Install frontend dependencies
        run: npm ci

      - name: Run ESLint
        run: npm run lint
```

This runs on pushes to `develop` and pull requests targeting `main`.

d. Create the GitLab CI/CD configuration

From the SAIL repository root, create `.gitlab-ci.yml`:

```
stages:
  - lint

black:
  stage: lint
  image: python:3.12
  allow_failure: true
  script:
    - cd sail-orbit
    - pip install black==26.5.1
    - black --check .

ruff:
  stage: lint
  image: python:3.12
  allow_failure: true
  script:
    - cd sail-orbit
    - pip install ruff==0.16.3
    - ruff check .

eslint:
  stage: lint
  image: node:22
  allow_failure: true
  script:
    - cd sail-orbit/frontend
    - npm ci
    - npm run lint

workflow:
  rules:
    - if: '$CI_PIPELINE_SOURCE == "merge_request_event"' - if: 'CI_COMMIT_BRANCH ==
"develop" && $CI_PIPELINE_SOURCE == "push"'
```

Runs on pushes to `develop` and merge requests.

e. Verify the frontend lockfile

ls sail-orbit/frontend/package-lock.json
Confirmed that `package-lock.json` exists. Both CI configurations use `npm ci` to install the dependencies recorded in this file.

f. Run and correct Python checks

From `sail-orbit`:
black --check .
ruff check .
Initial results:

- Black: 8 files require formatting.
- Ruff: 6 lint errors.

Run the automatic corrections:
ruff check . --fix
black .
Results:

- Ruff: 7 issues fixed, 0 remaining.
- Black: 7 files reformatted, 29 unchanged.

Verify:
ruff check .
black --check .
PASS

g. Run ESLint

From `sail-orbit/frontend`:
npm run lint
ESLint identified two errors in `src/components/ui/button.jsx`:

- Unused `React` import.
- `react-refresh/only-export-components` rule triggered by the `buttonVariants` export.

Defer these issues until the frontend components are developed further.

Step 17: Build user profile and account screens

a. Add API client functions for retrieving and updating the authenticated user's profile and activating or deactivating accounts.
b. Create the authenticated user's profile screen to display username, email, bio, role, and account status.
Create `frontend/src/components/UserProfile.jsx`.
This component will retrieve the logged-in user's profile using `getOwnProfile()` and display their username, email, biography, role, and account status.
c. Add profile editing functionality to update and save the user's bio.
Update `frontend/src/components/UserProfile.jsx` to support biography editing.

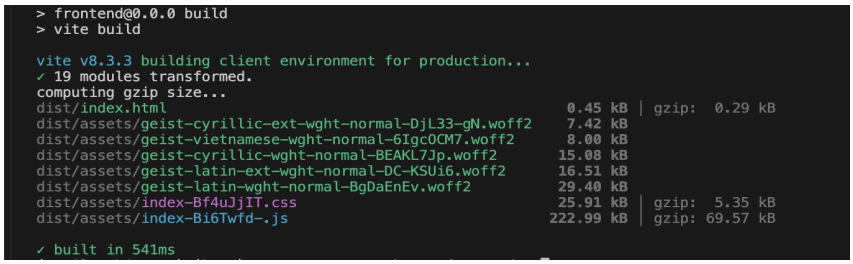
1. Added React state variables (`bio`, `editing`, `saving`, `saveError`, and `saveSuccess`) to manage the editor, API requests, and feedback messages.
2. Updated `loadProfile()` to initialize the biography editor with the existing profile data returned by `getOwnProfile()`.
3. Added the documented `handleSave()` function to submit biography changes through `updateOwnProfile()`, update the displayed profile with the saved response, and error handling.
4. Replaced the static biography display with a conditional interface containing a text area, Save button, and Cancel button.

5. Added loading protection against duplicate submissions, error messages, and confirmation of successful updates. Cancel restores the previously saved biography.
    a. `setSaving(true)` marks the save operation as in progress.
    b. `disabled={saving}` disables both Save and Cancel while the request runs.
    c. `setSaving(false)` re-enables the buttons when the request finishes.
6. Formatted `UserProfile.jsx` using Prettier:
   `npx prettier --write sail-orbit/frontend/src/components/UserProfile.jsx`

The component supports biography editing through the existing `PATCH /users/me` endpoint.

d. Create public profile screens to display usernames, biographies, and account creation dates.

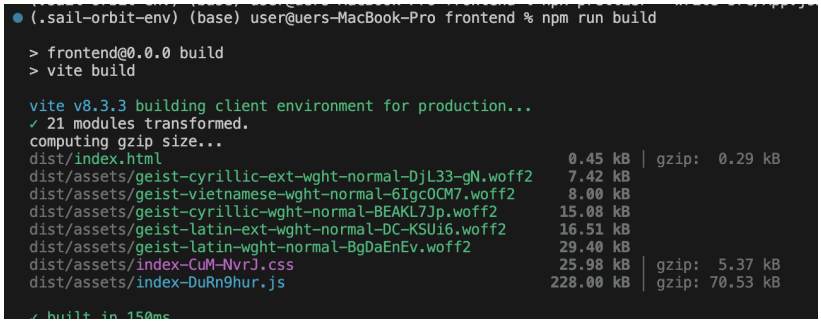
1. Create `frontend/src/components/PublicProfile.jsx`.
2. Import `useEffect`, `useState`, and the existing `getPublicProfile()` API function.
3. Add state variables for profile data, loading status, and errors.
4. Implement `loadProfile()` to retrieve public profile data using the supplied username.
5. Add cleanup logic to prevent outdated API responses from updating the component.
6. Display the username, biography, and account creation date.
7. Add loading, error, and unavailable-profile messages.
8. Format the file using Prettier:
   `npx prettier --write sail-orbit/frontend/src/components/PublicProfile.jsx`
9. Verify the frontend build:
   ```
   cd sail-orbit/frontend
   npm run build
   ```



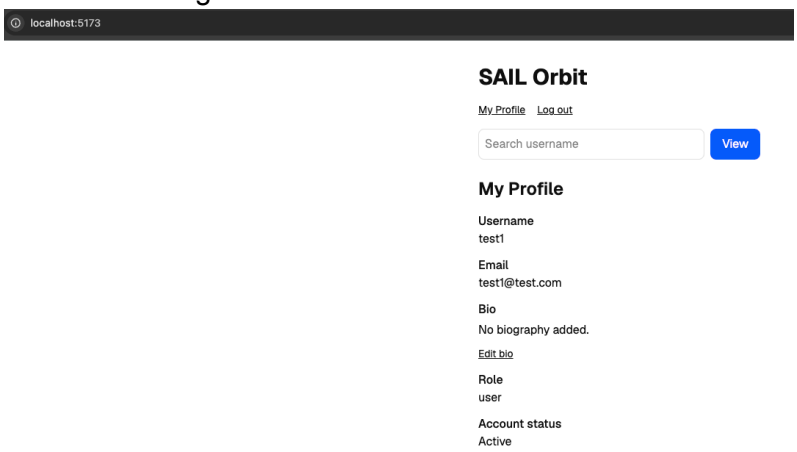
e. Add navigation between the authenticated user's profile and public profiles.

1. Update `frontend/src/App.jsx` to import `UserProfile` and `PublicProfile`.
2. Add state to track the active screen and the username of the selected public profile.
3. Add navigation controls for switching between the authenticated user's profile and public profiles.
4. Add a username input to allow users to look up public profiles.
5. Pass the JWT access token to `UserProfile` and the selected username to `PublicProfile`.
6. Add a Back to My Profile button on public profile screens.
7. Preserve the existing login, registration, and logout behavior when switching screens.

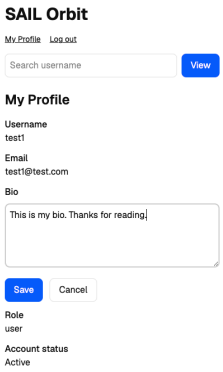
8. Format the updated file using Prettier.
9. Run `npm run build` to verify the frontend compiles.



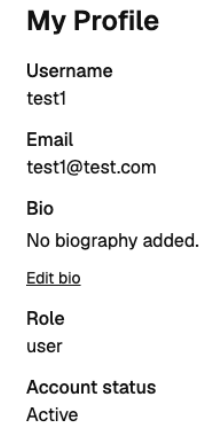
10. Test navigation between the authenticated user's profile and public profiles when the application is running locally.
    From `sail-orbit`, start FastAPI: `uvicorn app.run:app --reload`
    In a second terminal, from `sail-orbit/frontend`: `npm run dev`
    i. Successful login



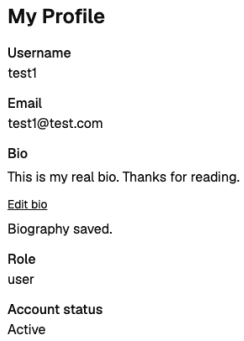
ii. Edit bio button works



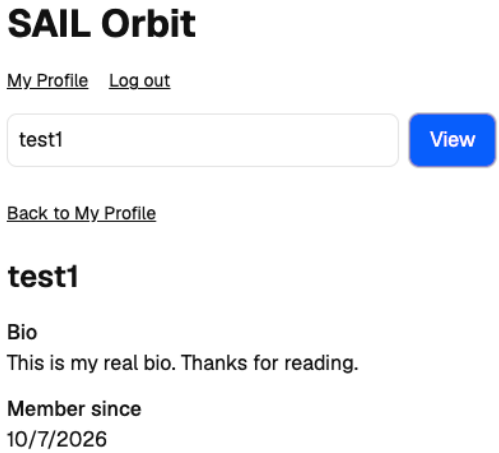
iii. Cancel closes edit without saving



iv. Edit and Save works



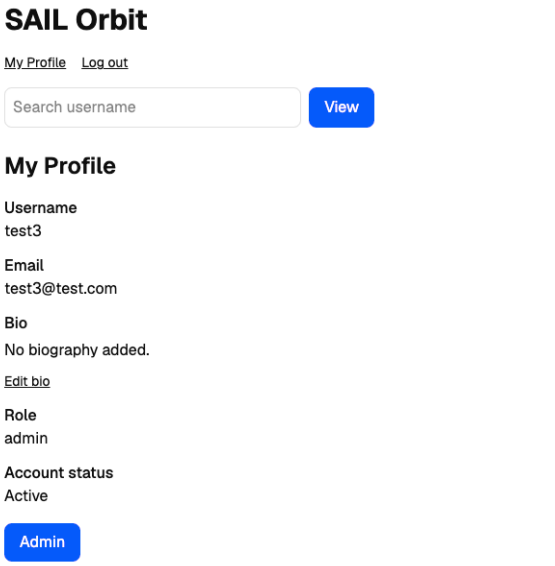
v. Search profile does return the updated profile

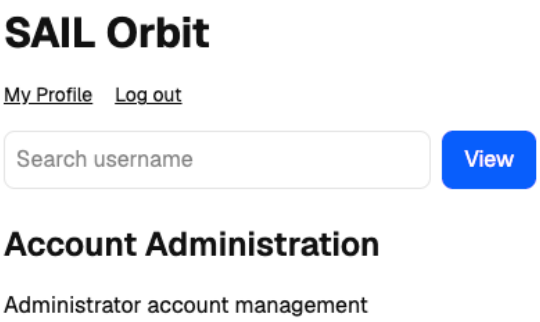


f. Implement role-based interface behavior to display administrator controls only for administrator accounts.

1. Use the existing `role` field returned by `getOwnProfile()` to determine whether the authenticated user is an administrator.
2. Update `UserProfile.jsx` to display an Admin control when `profile.role === "admin"`, using an `onOpenAdmin` callback for navigation.
3. Update `App.jsx` to support the administration screen alongside existing own-profile and public-profile screens, preserving profile lookup, My Profile, and logout.

4. Inspect the backend user model and database configuration, then create an administrator test account using the existing registration flow and assign its `admin` role in PostgreSQL.
5. Format the modified files with Prettier and run `npm run build`.
6. Test with regular and administrator accounts to verify role-based visibility, administration screen navigation, and existing profile and authentication functionality.





Updated `UserProfile.jsx` and `App.jsx` to support administrator-only navigation using the authenticated user's existing `role` field.
Configured the PostgreSQL test accounts:

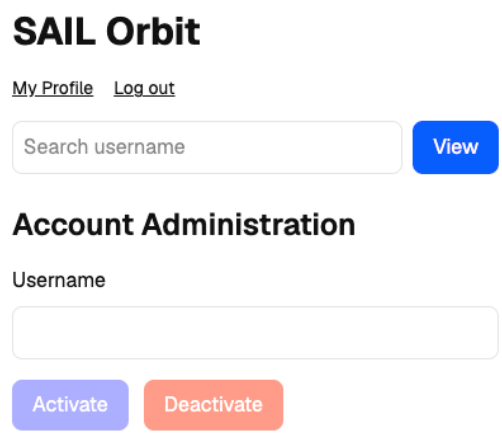
- `test3`: administrator (`admin`)
- `test1`, `test2`, `orbit_test_01`: regular users (`user`)

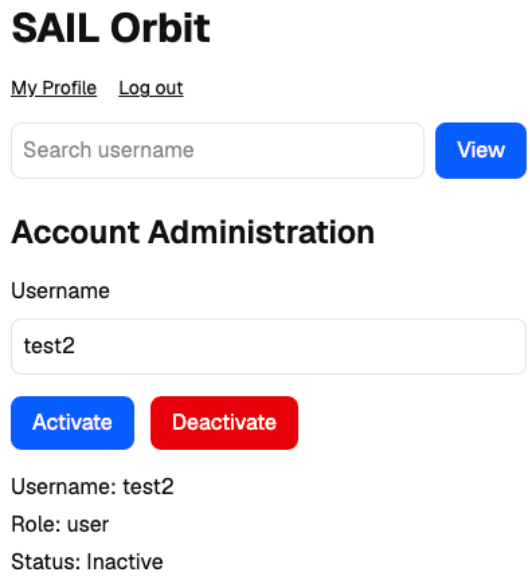
Verified that the Admin button appears for administrators, opens the Account Administration screen, and remains hidden for regular users. Existing profile navigation was also verified.
g. Create the administrator account management interface to activate and deactivate user accounts.

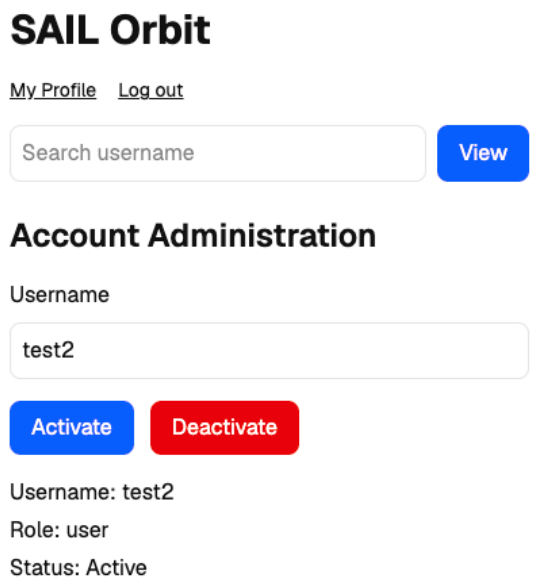
1. run: `grep -n -A 35 'export async function updateUserStatus' frontend/src/api.js` to inspect existing API function
2. create `frontend/src/components/AdminAccounts.jsx` and connect it to the Admin screen in `App.jsx`
    a. Inspect the existing backend endpoint: `grep -n -B 12 -A 55 'def update_user_status' app/api/users.py`

The backend confirms that administrators can activate or deactivate accounts, including currently inactive accounts. It also returns the username, role, and updated status. This interface uses the existing API to change an account's status. It displays the account's confirmed status after an update, rather than claiming to know its status beforehand.

a. Add this import immediately after the existing PublicProfile import:
```
import AdminAccounts from "./components/AdminAccounts";
```
b. Find this existing block in App.jsx:
```
      ) : profileScreen === "admin" ? (
        <section className="space-y-4">
          <h2 className="text-2xl font-semibold">Account Administration</h2>
          <p>Administrator account management</p>
        </section>
```
c. Replace it with:
```
      ) : profileScreen === "admin" ? (
        <AdminAccounts token={accessToken} />
```
d. Format and build from sail-orbit/frontend:
```
npx prettier --write src/App.jsx src/components/AdminAccounts.jsx
npm run build
```



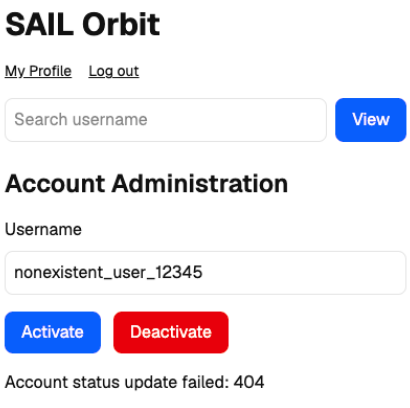


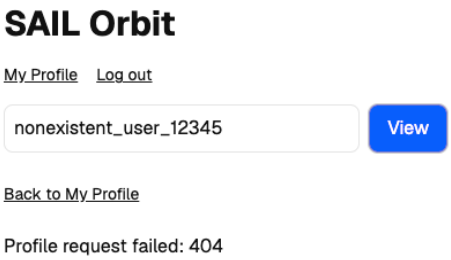


Deactivate: test2 successfully changed to Inactive.
Activate: test2 successfully returned to Active.
Administrator authorization: test3 successfully performed both operations.

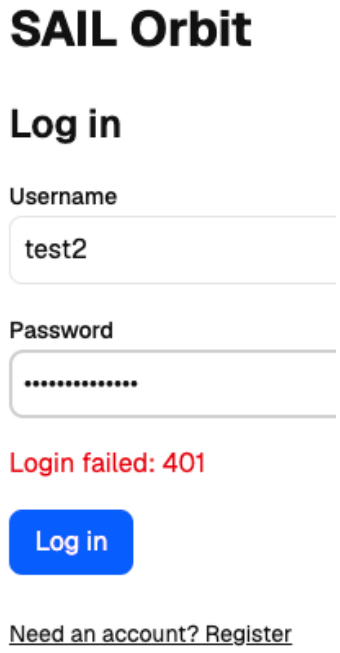
h. Add loading, success, and error handling for profile and account operations.

Entered a nonexistent username in the administrator screen and confirmed that deactivation returned a 404 error, then searched for a nonexistent public profile, confirmed another 404 error, and verified that Back to My Profile still worked.

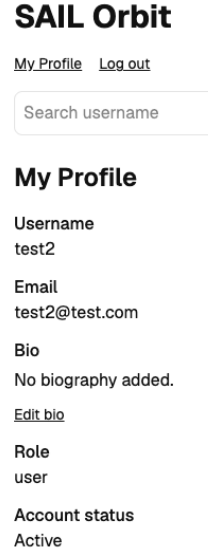




i. Verify authentication integration and complete local testing.
Logged in as admin (test3) and deactivated test2. Verified test2 cannot log in.



Reactivated and confirmed logged in as test2



NOTE: Some vulnerabilities found due to some depreciation of dependencies. See Appendix.

j. Frontend automated testing

The backend's 27 relevant tests already pass. Vitest and React Testing Library are installed, so we can finish by configuring the test runner and testing the Step 17 React components.

1. Configure Vitest

From `sail-orbit/frontend`, create `vitest.config.js`:

```
import { defineConfig } from "vitest/config";
import react from "@vitejs/plugin-react";

export default defineConfig({
```

```
  plugins: [react()],
  esbuild: {
    jsx: "automatic",
  },
  test: {
    environment: "jsdom",
    globals: true,
    clearMocks: true,
  },
});
```

2. Add the test command

In `package.json`, add `"test": "vitest run"` to the existing `scripts` section:

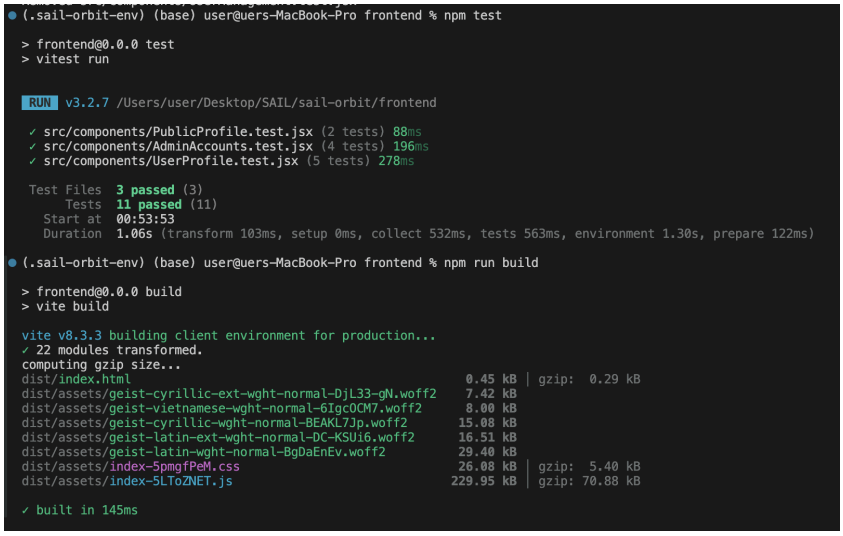
```
"scripts": {
  "dev": "vite",
  "build": "vite build",
  "lint": "eslint .",
  "preview": "vite preview",
  "test": "vitest run"
}
```

3. Get the component code

To write tests that match the actual components rather than guessing their markup, run:
cat src/components/UserProfile.jsx src/components/PublicProfile.jsx src/components/AdminAccounts.jsx

4. Initially all 11 tests were put into UserProfile.test.jsx, but needed to split into more files since it got too long. See docs -> python_scripts ->test_separation_script.py for script run in CLI at frontend dir

Tests passed



Step 18: Implement Posts

a. Database Model

- Create the post model and establish its relationship to users.
    - Update `app/models/post.py`

    - Update `app/models/user.py`
    - Register the models for Alembic in `migrations/env.py`
- Generate the database migration.
    - From the `sail-orbit` repository root, with your Python environment activated:
      `alembic revision --autogenerate -m "create posts table"`
- Apply the migration
    - From the sail-orbit repository root, with PostgreSQL running:
      `alembic upgrade head`
- Then verify the database is at the expected revision:
      `alembic current`

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % alembic upgrade head
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.runtime.migration] Running upgrade 62ce8ce120c2 -> 6c05f1b634f6, create posts table
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % alembic current
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
6c05f1b634f6 (head)
```

b. Pydantic Schemas

Created `app/schemas/post.py` to define validation and return post data.

- PostContent: Shared validation rules for creating and editing posts.
- PostCreate: Validates new post submissions.
- PostUpdate: Validates changes to existing posts.
- PostResponse: Defines the information returned by the API, including post ID, author ID, content, and timestamps.

Validation rules implemented:

- Require content between 1 and 5,000 characters.
- Reject empty and whitespace-only posts.
- Automatically remove leading and trailing whitespace.
- Reject missing content.

Both creation and editing use the same validation rules through inheritance.

c. API Endpoints

- Implement post creation, retrieval, editing, and deletion.
- Enforce authentication, ownership, and appropriate permissions.

Method	Endpoint	Function
POST	/posts	Create a post
GET	/posts	List posts, newest first
GET	/posts/{post_id}	Retrieve a post
PATCH	/posts/{post_id}	Edit a post
DELETE	/posts/{post_id}	Delete a post

Method	Endpoint	Access
POST	/posts	Authenticated users
GET	/posts	Public
GET	/posts/{post_id}	Public
PATCH	/posts/{post_id}	Post author only
DELETE	/posts/{post_id}	Post author only

d. Frontend API Integration

- Add API client functions for post operations.
- Handle authentication and API errors.

Function	Purpose
createPost()	Submit a new post
getPosts()	Retrieve all posts
getPost()	Retrieve an individual post
updatePost()	Edit an existing post
deletePost()	Delete a post

e. Frontend Functionality

- Build the post creation and editing interface.
- Display posts and provide deletion controls.
- Implement loading, success, and error states.

Created a React interface for viewing, creating, editing, and deleting posts. Added Posts navigation, ownership-based editing controls, content validation, and error handling.

f. Automated Testing

- Test post models, API endpoints, validation, and authorization.
- Test frontend rendering and interactions.
- Verify post persistence and error handling.

Added 14 integration tests covering post creation, retrieval, editing, deletion, authentication, authorization, and content validation.

Step 19: Add comments and reactions

a. Define Comment and Reaction Behavior

Comments

- Logged-in users can comment on posts and reply to comments.
- Anyone can read comments and replies.
- Users can edit or delete only their own comments and replies.
- Comments and replies appear chronologically.
- Apply existing content validation rules.
- Deleting a post also deletes its comments and replies.

Reactions

- Support Like and Dislike only, with future expansion based on user feedback.
- Logged-in users can react to posts.
- Limit users to one reaction per post.
- Users can change or remove their reactions.
- Display Like and Dislike counts separately.
- Deleting a post removes its reactions.

Authorization

- Reuse existing JWT authentication and ownership checks.
- Return appropriate errors for unauthorized actions and nonexistent resources.

Scope

- Support nested replies to comments.
- Reactions apply to posts only.
- Notifications and reaction history are outside the current scope.

b. Create Database Models and Migration

- Create `Comment` with references to the author, post, and optional parent comment for nested replies.

- Create `Reaction` with references to the user, post, and reaction type (`like` or `dislike`).
- Add relationships to the existing `User` and `Post` models.
- Enforce one reaction per user per post using a database constraint.
- Configure deletion of posts - can delete a post but comments/replies stay.
- Generate one Alembic migration for both tables.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % alembic revision --autogenerate -m "Add comments and reactions with soft d
eletion"
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.schemas
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.tables
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.types
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.constraints
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.defaults
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.comments
INFO  [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.checkconstraint_byname
INFO  [alembic.autogenerate.compare.tables] Detected added table 'comments'
INFO  [alembic.autogenerate.compare.constraints] Detected added index 'ix_comments_author_id' on '('author_id',)'
INFO  [alembic.autogenerate.compare.constraints] Detected added index 'ix_comments_parent_id' on '('parent_id',)'
INFO  [alembic.autogenerate.compare.constraints] Detected added index 'ix_comments_post_id' on '('post_id',)'
INFO  [alembic.autogenerate.compare.tables] Detected added table 'reactions'
INFO  [alembic.autogenerate.compare.constraints] Detected added index 'ix_reactions_post_id' on '('post_id',)'
INFO  [alembic.autogenerate.compare.constraints] Detected added index 'ix_reactions_user_id' on '('user_id',)'
INFO  [alembic.ddl.postgresql] Detected sequence named 'posts_id_seq' as owned by integer column 'posts(id)', assuming SERIAL and omit
ting
INFO  [alembic.ddl.postgresql] Detected sequence named 'users_id_seq' as owned by integer column 'users(id)', assuming SERIAL and omit
ting
INFO  [alembic.autogenerate.compare.tables] Detected added column 'posts.is_deleted'
  Generating /Users/user/Desktop/SAIL/sail-orbit/migrations/versions/70d79c6ef1c8_add_comments_and_reactions_with_soft_.py ...  done
```

  Alembic detected:
  - Both new tables: `comments` and `reactions`.
  - Three indexes for comments and two for reactions.
  - New `is_deleted` column on the existing `posts` table.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % alembic upgrade head && alembic current
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.runtime.migration] Running upgrade 6c05f1b634f6 -> 70d79c6ef1c8, Add comments and reactions with soft deletion
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
70d79c6ef1c8 (head)
```

- Verify the migration and database constraints with automated tests.
  - Add tests to `test_comments.py` and `test_reactions.py`
  - Run `python -m pytest -v tests/test_comments.py tests/test_reactions.py`

All 11 automated tests passed in 0.31 seconds.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % python -m pytest -v tests/test_comments.py tests/test_reactions.py
============================================== test session starts ==============================================
platform darwin -- Python 3.12.2, pytest-9.1.1, pluggy-1.6.0 -- /Users/user/Desktop/SAIL/sail-orbit/.sail-orbit-env/bin/python
cachedir: .pytest_cache
rootdir: /Users/user/Desktop/SAIL/sail-orbit
plugins: anyio-4.15.1
collected 11 items

tests/test_comments.py::test_comments_table_and_columns PASSED                                          [  9%]
tests/test_comments.py::test_comment_foreign_keys PASSED                                                [ 18%]
tests/test_comments.py::test_comment_indexes PASSED                                                     [ 27%]
tests/test_comments.py::test_comment_foreign_keys_do_not_cascade PASSED                                 [ 36%]
tests/test_reactions.py::test_reactions_table_and_columns PASSED                                        [ 45%]
tests/test_reactions.py::test_reaction_unique_constraint PASSED                                         [ 54%]
tests/test_reactions.py::test_reaction_type_constraint PASSED                                           [ 63%]
tests/test_reactions.py::test_reaction_foreign_keys PASSED                                              [ 72%]
tests/test_reactions.py::test_reaction_indexes PASSED                                                   [ 81%]
tests/test_reactions.py::test_reaction_foreign_keys_do_not_cascade PASSED                               [ 90%]
tests/test_reactions.py::test_posts_soft_deletion_column PASSED                                         [100%]

============================================== 11 passed in 0.31s ===============================================
```

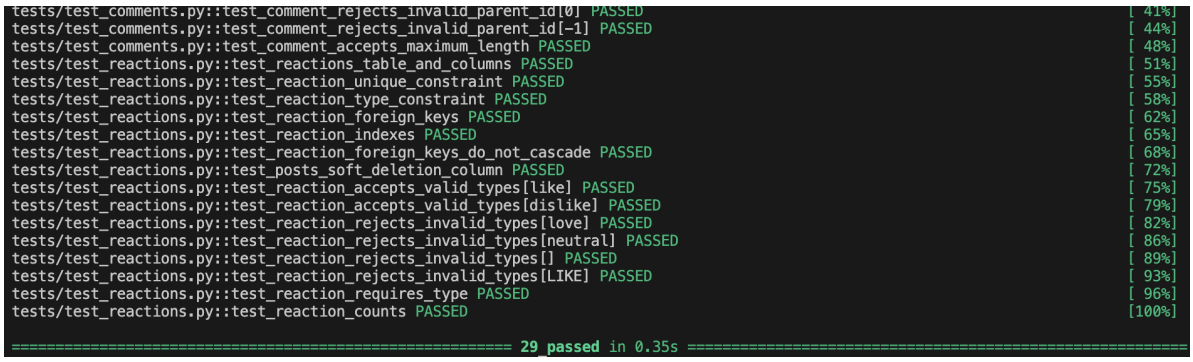
Verified:
- Comments and reactions tables, columns, and indexes
- Foreign key relationships
- Reaction uniqueness and Like/Dislike constraints
- Soft-deletion column on posts
- No database-level cascading deletion

c. Create Pydantic Schemas

Created `app/schemas/comment.py` and `app/schemas/reaction.py` to define and validate API request and response data.

- Implemented schemas for creating, editing, and retrieving comments and replies.
- Added comment validation for length, whitespace, and optional parent IDs.
- Restricted reactions to `like` and `dislike`.
- Defined response schemas for comments, reactions, and separate reaction counts.
- Added automated Pydantic validation tests to the existing test files.

Verification: All 29 database and validation tests passed. Python compilation and Ruff linting also passed.



d. Implement Backend API Endpoints

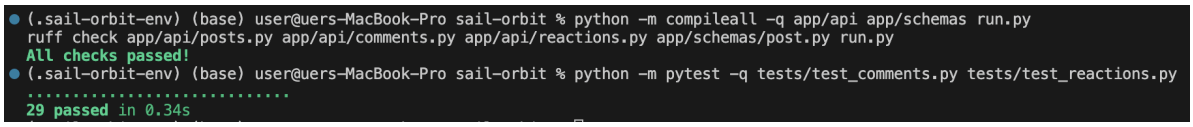
implement two routers:

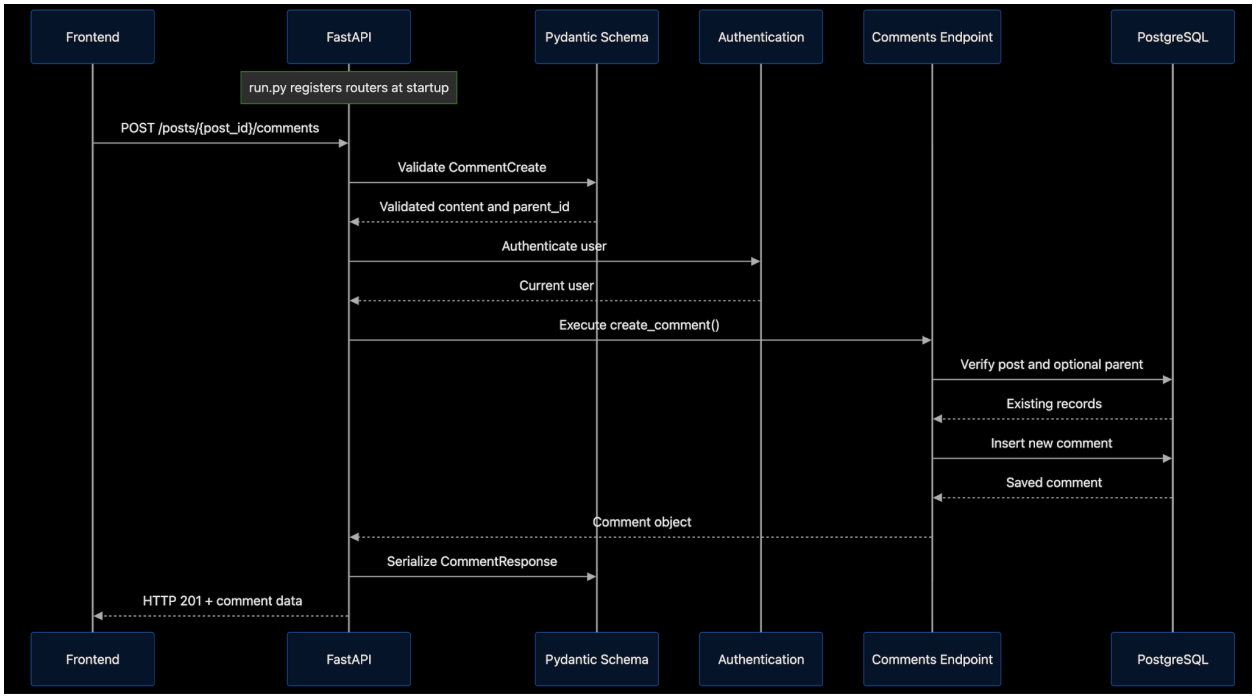
- `app/api/comments.py`: Create comments and replies, retrieve comments, edit owned comments, and soft-delete comments.
- `app/api/reactions.py`: Add, remove, or switch Like/Dislike reactions and retrieve reaction counts.

Implemented the backend endpoints for creating, retrieving, editing, and soft-deleting comments and replies, along with adding, switching, removing, and counting post reactions.

Five files were created or updated, each serving a specific purpose:

File	Purpose
run.py	Registers the new comment and reaction routers with FastAPI.
app/api/comments.py	Handles comment and reply requests, validates ownership, and soft deletion.
app/api/reactions.py	Handles Like/Dislike operations and calculates reaction counts.
app/api/posts.py	Updates existing post endpoints to preserve deleted posts and return [Deleted].
app/schemas/post.py	Adds is_deleted to post API responses.





Above illustrates how SAIL Orbit processes a comment creation request. FastAPI receives the request and uses Pydantic to validate the submitted data. Authentication identifies the user, and the comments endpoint verifies the post and optional parent comment before saving the new comment to PostgreSQL through SQLAlchemy. Finally, Pydantic formats the response, and FastAPI returns HTTP 201 with the created comment. Also distinguishes application startup, when `run.py` registers the API routes, from the steps executed for each request.

e. Integrate Frontend API Functions

1. Extended and reorganized `frontend/src/api.js`

Reorganized the API functions into separate modules based on functionality.

File	Purpose
`api/auth.js`	User registration and login
`api/users.js`	User profiles and account administration
`api/posts.js`	Create, retrieve, update, and delete posts
`api/comments.js`	Create, retrieve, edit, and soft-delete comments and replies
`api/reactions.js`	Add, switch, or remove Like/Dislike reactions and retrieve reaction counts
`api.js`	Re-exports all API functions so existing components continue working

The existing authentication, user, and post functions were preserved. Six new functions were added for comments and reactions which communicate with the FastAPI backend using `fetch`, following Orbit's existing conventions for JWT authentication, JSON request bodies, response parsing, and HTTP error handling.

2. Implemented automated API tests

Created two test files:

- `api/comments.test.js`: 9 tests.
- `api/reactions.test.js`: 6 tests.

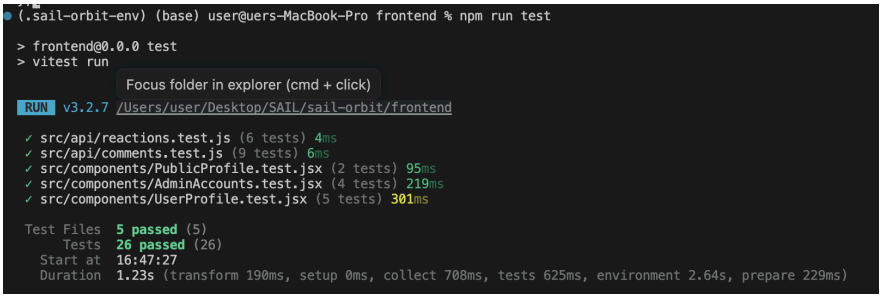
The tests use Vitest and mocked `fetch` responses, allowing us to verify frontend API behavior without running FastAPI or PostgreSQL.

What we test	What it verifies
Comment creation	Correct POST endpoint, JWT, and content
Nested replies	Parent comment ID is included
Comment retrieval	GET request without authentication
Comment editing	Correct PATCH request and updated content
Comment deletion	Correct DELETE request and handling of HTTP 204
Like/Dislike	Correct PUT request and reaction type
Reaction removal	Client correctly handles a `null` response
Reaction counts	Retrieves separate Like and Dislike totals
Error handling	Unsuccessful HTTP responses produce errors

These verify that the frontend API client constructs requests and handles responses correctly.

File tree under `api.js` now includes multiple files and removal of the spaghetti code:

- Verified API integration through automated tests.



f. Implement Frontend Comments and Reactions

- Add the GET endpoint to `app/api/reactions.py`.
- Add `getOwnReaction()` to `frontend/src/api/reactions.js` and export it through `api.js`.

1. Backend: New reaction retrieval endpoint

   File: `app/api/reactions.py`

   Highlight:

   ```
   @router.get(
       "/{post_id}/reaction",
       response_model=ReactionResponse | None,
   )
   def get_own_reaction(
       post_id: int,
       current_user: User = Depends(get_current_user),
       database: Session = Depends(get_db),
   ):
   ```

   The query that filters by both `post_id` and `current_user.id`.

   What it demonstrates: The endpoint retrieves the authenticated user's reaction rather than another user's reaction.

2. Frontend: API integration

   File: `frontend/src/api/reactions.js`

   Highlight the complete `getOwnReaction()` function, particularly:

   ```
   const response = await fetch(`API_BASE_URL/posts/{postId}/reaction`, {
     headers: {
         Authorization: `Bearer ${token}`,
     },
   });
   ```

   And:

   ```
   return response.json();
   ```

   Demonstrated the frontend can request the user's existing reaction and process either a reaction object or `null`.

3. Frontend: Automated API testing

   File: `frontend/src/api/reactions.test.js`

   Highlight these tests:

   - `retrieves the authenticated user's existing Like reaction`
   - `returns null when the authenticated user has not reacted`
   - `reports an HTTP error when retrieving the user's reaction fails`

   The new frontend function handles the important response scenarios.

4. Test results Frontend Vitest

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro frontend % npm run test

> frontend@0.0.0 test
> vitest run


 RUN  v3.2.7 /Users/user/Desktop/SAIL/sail-orbit/frontend

 ✓ src/api/comments.test.js (9 tests) 5ms
 ✓ src/api/reactions.test.js (10 tests) 7ms
 ✓ src/components/PublicProfile.test.jsx (2 tests) 94ms
 ✓ src/components/AdminAccounts.test.jsx (4 tests) 217ms
 ✓ src/components/UserProfile.test.jsx (5 tests) 307ms

 Test Files  5 passed (5)
      Tests  30 passed (30)
   Start at  18:27:17
   Duration  1.31s (transform 184ms, setup 0ms, collect 758ms, tests 630ms, environment 2.81s, prepare 325ms)
```

Test results Backend Pytest

```
python -m pytest tests/test_reaction_api.py -q
.......
7 passed in 1.39s
```

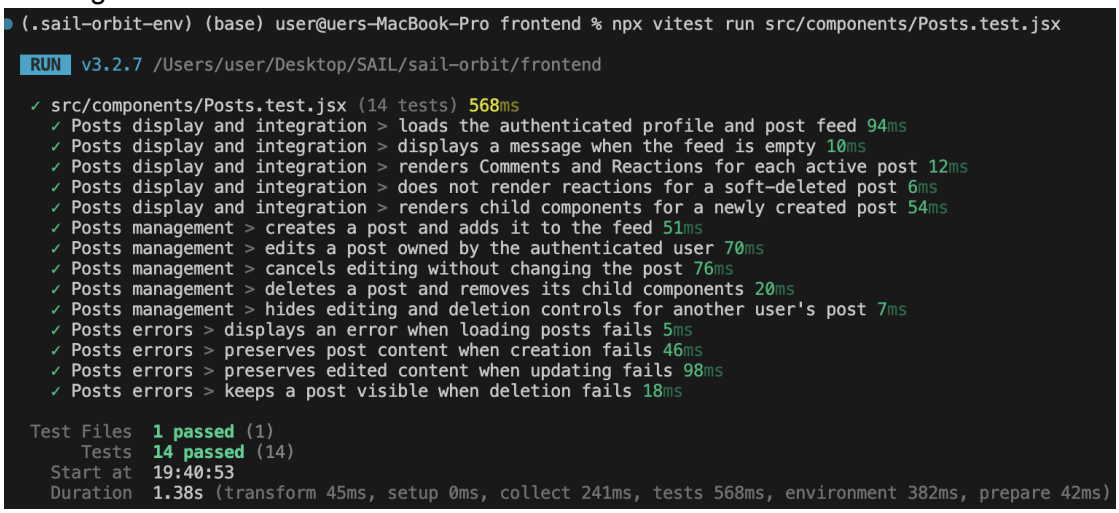
5. `Comments.jsx`: Display comments and nested replies; allow users to create, edit, and soft-delete their own comments.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro frontend % npx vitest run src/components/Comments.test.jsx

 RUN  v3.2.7 /Users/user/Desktop/SAIL/sail-orbit/frontend

 ✓ src/components/Comments.test.jsx (12 tests) 486ms
   ✓ Comments display > displays a message when no comments exist 26ms
   ✓ Comments display > displays comments and replies in their parent-child hierarchy 98ms
   ✓ Comments display > supports reply depth beyond three levels 9ms
   ✓ Comments display > collapses replies while keeping the parent visible 45ms
   ✓ Comments display > preserves soft-deleted comments and their replies 4ms
   ✓ Comments management > creates a top-level comment 64ms
   ✓ Comments management > creates a nested reply linked to its parent 76ms
   ✓ Comments management > edits a comment owned by the authenticated user 76ms
   ✓ Comments management > soft-deletes a comment without removing its replies 24ms
   ✓ Comments management > does not display editing controls for another user's comment 9ms
   ✓ Comments errors > displays an error when comment retrieval fails 5ms
   ✓ Comments errors > preserves comment text when creation fails 49ms

 Test Files  1 passed (1)
      Tests  12 passed (12)
   Start at  19:32:10
   Duration  1.23s (transform 56ms, setup 0ms, collect 167ms, tests 486ms, environment 383ms, prepare 44ms)
```

6. `Reactions.jsx`: Display Like/Dislike counts, retrieve the user's selected reaction, and support adding, switching, or removing reactions.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro frontend % npx vitest run src/components/Reactions.test.jsx

 RUN  v3.2.7 /Users/user/Desktop/SAIL/sail-orbit/frontend

 ✓ src/components/Reactions.test.jsx (9 tests) 184ms
   ✓ Reactions display > loads the user's reaction and reaction counts 65ms
   ✓ Reactions display > displays no selected reaction when the user has not reacted 14ms
   ✓ Reactions display > restores a previously selected Dislike 9ms
   ✓ Reactions management > adds a Like and refreshes reaction counts 25ms
   ✓ Reactions management > removes a Like when the selected reaction is clicked again 17ms
   ✓ Reactions management > switches from Like to Dislike 17ms
   ✓ Reactions management > adds a Dislike 15ms
   ✓ Reactions errors > displays an error when initial reaction retrieval fails 5ms
   ✓ Reactions errors > preserves the existing reaction when updating fails 17ms

 Test Files  1 passed (1)
      Tests  9 passed (9)
   Start at  19:34:20
   Duration  893ms (transform 39ms, setup 0ms, collect 149ms, tests 184ms, environment 370ms, prepare 38ms)
```

7. `Posts.jsx` integration: Add both components to each post without changing existing post-management behavior.



g. Complete Automated Backend Testing

Created and refined two integration test files to verify the Comments and Reactions APIs against the FastAPI application and PostgreSQL database.

1. Comment API testing

Created `tests/test_comment_api.py` with 11 tests covering:

- Comment creation, retrieval, and nested replies.
- Comment editing and soft deletion, including preservation of replies.
- Authentication, author ownership, and authorization.
- Rejection of edits to deleted comments.
- Missing posts and attempts to comment on deleted posts.

2. Reaction API testing

Refactored `tests/test_reaction_api.py` into five focused tests, removing redundant scenarios and unnecessary complexity while retaining the essential coverage:

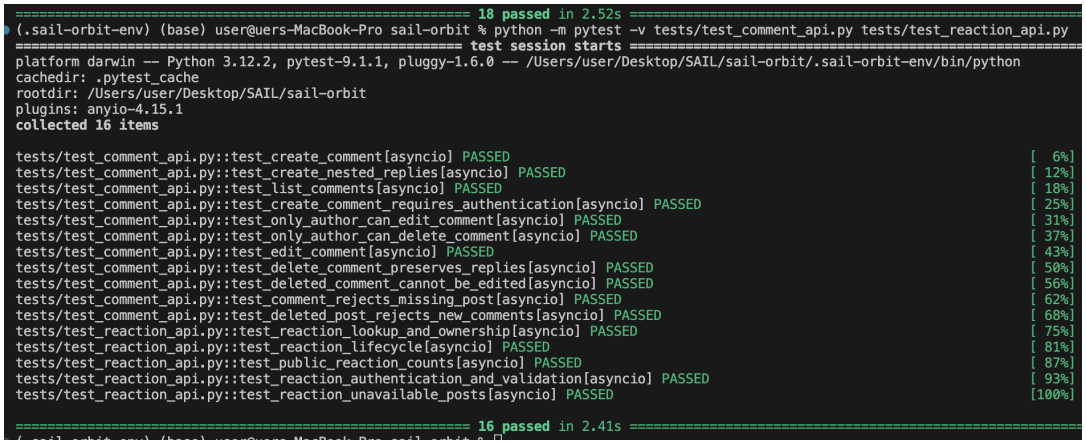
- Authenticated reaction retrieval and user-specific ownership.
- Creating, switching, and removing Like/Dislike reactions.
- Database persistence of reaction changes.
- Public reaction counts across multiple users.
- Authentication, invalid requests, and unavailable posts.

Existing database constraint and Pydantic validation tests remain separate to avoid duplicating their coverage.

3. Integration test verification

Executed both test files together:

```
python -m pytest -v tests/test_comment_api.py tests/test_reaction_api.py
```



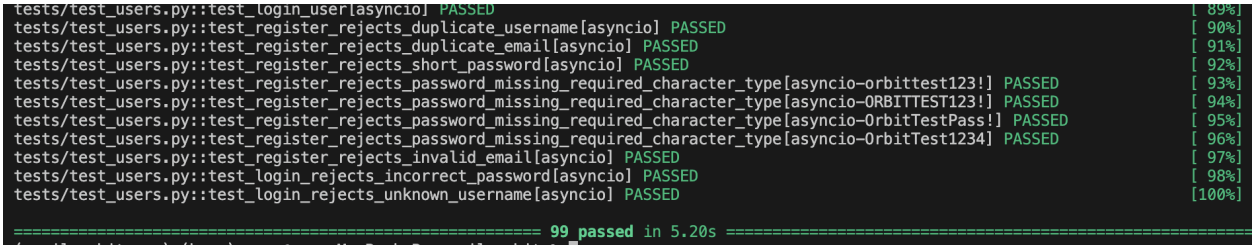
The tests use `pytest`, `httpx2`, SQLAlchemy, and PostgreSQL to verify API behavior and database operations.

Completed and verified the automated integration tests for comments and reactions. The reaction test suite was simplified to five tests without losing its principal API coverage.

h. Final Verification and Documentation

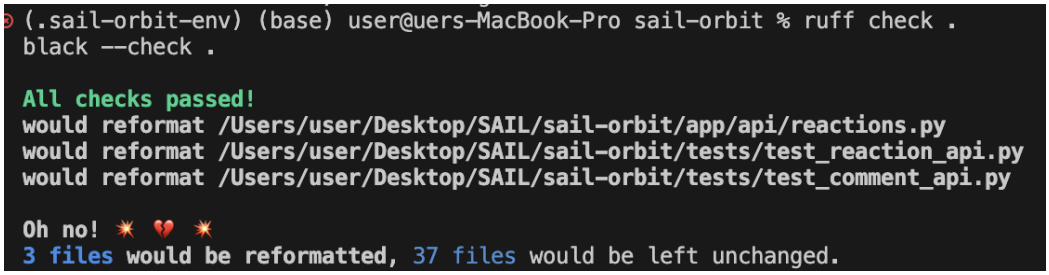
From `sail-orbit`, run:

1. Complete backend test suite: `python -m pytest -v`



2. Backend linting and formatting

`ruff check .`



`black --check .`

To fix, run: `ruff check . --fix`

`black app/api/reactions.py tests/test_reaction_api.py tests/test_comment_api.py`

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % ruff check . --fix
black app/api/reactions.py tests/test_reaction_api.py tests/test_comment_api.py
All checks passed!
reformatted app/api/reactions.py
reformatted tests/test_reaction_api.py
reformatted tests/test_comment_api.py

All done! ✨ 🍰 ✨
3 files reformatted.
(.sail-orbit-env) (base) user@uers-MacBook-Pro sail-orbit % ruff check .
black --check .
All checks passed!
All done! ✨ 🍰 ✨
40 files would be left unchanged.
```

3. Complete frontend test suite

```
cd frontend
npm test -- --run
```

4. Frontend linting and production build

```
npm run lint
npm run build
```

- Created `frontend/src/components/ui/button-variants.js` to separate button styling configurations from the React `Button` component in `button.jsx`.
- The original file exported both the component and `buttonVariants`, triggering the ESLint `react-refresh/only-export-components` error. Moving the styling definitions into a separate module resolved the error while preserving all existing button appearances, sizes, and functionality.
- Verification: ESLint passed with zero errors, the production build succeeded, and all 65 frontend tests passed.

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro frontend % npm run lint
npm run build
npm test -- --run

> frontend@0.0.0 lint
> eslint .


> frontend@0.0.0 build
> vite build

vite v8.3.3 building client environment for production...
✓ 30 modules transformed.
computing gzip size...
dist/index.html                                             0.45 kB | gzip:  0.29 kB
dist/assets/geist-cyrillic-ext-wght-normal-DjL33-gN.woff2   7.42 kB
dist/assets/geist-vietnamese-wght-normal-6IgcOCM7.woff2     8.00 kB
dist/assets/geist-cyrillic-wght-normal-BEAKL7Jp.woff2      15.08 kB
dist/assets/geist-latin-ext-wght-normal-DC-KSUi6.woff2     16.51 kB
dist/assets/geist-latin-wght-normal-BgDaEnEv.woff2         29.40 kB
dist/assets/index-BPyYHxUL.css                             27.07 kB | gzip:  5.63 kB
dist/assets/index-DdFtob9C.js                             241.73 kB | gzip: 72.94 kB

✓ built in 145ms

> frontend@0.0.0 test
> vitest run --run


 RUN  v3.2.7 /Users/user/Desktop/SAIL/sail-orbit/frontend

 ✓ src/api/comments.test.js (9 tests) 15ms
 ✓ src/api/reactions.test.js (10 tests) 9ms
 ✓ src/components/PublicProfile.test.jsx (2 tests) 106ms
 ✓ src/components/Reactions.test.jsx (9 tests) 233ms
 ✓ src/components/AdminAccounts.test.jsx (4 tests) 236ms
 ✓ src/components/UserProfile.test.jsx (5 tests) 321ms
 ✓ src/components/Comments.test.jsx (12 tests) 513ms
 ✓ src/components/Posts.test.jsx (14 tests) 558ms

 Test Files  8 passed (8)
      Tests  65 passed (65)
   Start at  22:02:09
   Duration  1.69s (transform 427ms, setup 0ms, collect 1.93s, tests 1.99s, environment 4.84s, prepare 680ms)
```

Step 20: Add follow and unfollow relationships, i.e. data models, API endpoints, schemas, frontend functionality, and tests.
Step 21: Implement Account Lifecycle Management
Step 22: Build the chronological feed, including backend feed logic, API endpoints, schemas, frontend functionality, and tests.
Step 23: Complete frontend navigation and remaining user-interface behavior.
Step 24: Add application and security-relevant logging.
Step 25: Add health endpoints and application error handling.
Step 26: Complete remaining automated and integration testing.
Step 27: Run Black and Ruff against the complete application.
Step 28: Run and test the complete application locally, i.e. frontend, FastAPI, and PostgreSQL.
Step 29: Verify FastAPI validation and OpenAPI documentation.

Step 30: Develop an AI Activity Agent

This will simulate user interactions with SAIL Orbit, generate sustained application and security-relevant activity, and evaluate application behavior under variable workloads.

The agent will use SAIL Orbit's REST API to simulate user registration, authentication, posting, commenting, reactions, social relationships, and feed activity. It will generate varying activity patterns, concurrent requests, and controlled security-relevant events to support application testing, observability, and security investigations.

Step 31: Containerize the completed application.
Step 32: Deploy Orbit and PostgreSQL to the k3s cluster.
Step 33: Verify the deployed application is reachable, persistent, and functioning correctly.

Section 9: Monitoring, Logging, & Observability

Overview

At this juncture, SAIL establishes centralized monitoring and observability for the Raspberry Pi nodes, Kubernetes cluster, Orbit application, and PostgreSQL database. Prometheus, Grafana, Loki, and OpenTelemetry provide metrics, logs, and traces to monitor system health, analyze performance, investigate failures, and identify abnormal activity.

The implementation begins with cluster verification and resource assessment, followed by deployment and configuration of monitoring components using Helm. Grafana provides centralized access to operational data, while Alertmanager handles notifications. Persistent storage, retention policies, and operating baselines support ongoing troubleshooting and the investigation workflows in Section 12.

Tools

- **Prometheus:** Collects and stores infrastructure, Kubernetes, application, and database metrics.
- **Grafana:** Provides dashboards and tools for exploring metrics, logs, and traces.
- **Node Exporter:** Exposes Raspberry Pi CPU, memory, disk, and network metrics.
- **kube-state-metrics:** Exposes Kubernetes resource and workload metrics.
- **Loki:** Stores and queries centralized logs.
- **Log collector:** Forwards Kubernetes container and system logs to Loki.
- **OpenTelemetry:** Instruments applications and collects and exports traces.
- **Trace backend:** Stores and retrieves application traces.
- **PostgreSQL exporter:** Exposes database performance and operational metrics.
- **Alertmanager:** Processes alerts and delivers notifications.
- **Helm:** Deploys and manages monitoring components in Kubernetes.

Implementation

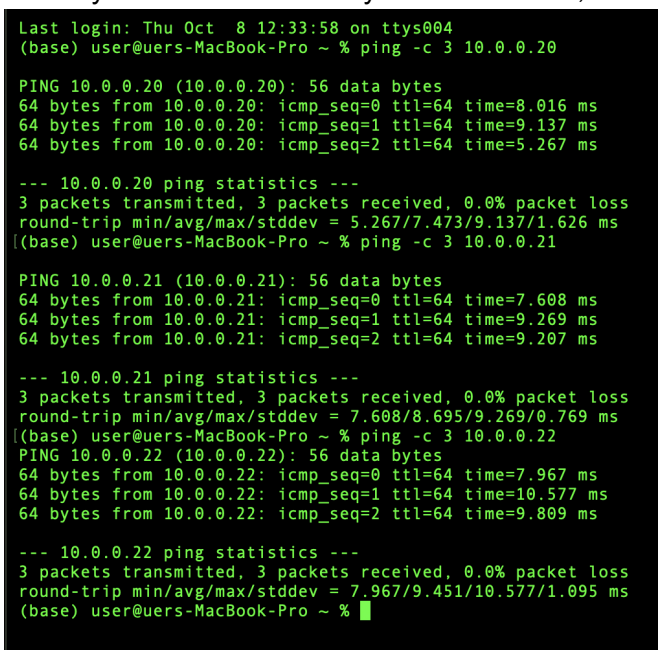
Step 1: Power on the SAIL cluster and verify network connectivity

a. Confirm the Samsung T7 SSD is connected to pi-node-1.

b. Verify all three Raspberry Pis are connected to the network switch using Ethernet.

c. Power on the SAIL tower and allow all three nodes to boot.

d. Verify network connectivity to `10.0.0.20`, `10.0.0.21`, and `10.0.0.22` using `ping`.



Step 2: Verify SSH access, k3s services, and Kubernetes connectivity

a. Connect to each Raspberry Pi using SSH with the `sailadmin` username and password.

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "hostname"
ssh sailadmin@10.0.0.21 "hostname"
ssh sailadmin@10.0.0.22 "hostname"
sailadmin@10.0.0.20's password:
pi-node-1
sailadmin@10.0.0.21's password:
pi-node-2
sailadmin@10.0.0.22's password:
pi-node-3
(base) user@uers-MacBook-Pro ~ %
```

b. Verify the `k3s` service is active on pi-node-1.

c. Verify the `k3s-agent` service is active on pi-node-2 and pi-node-3.

```
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.22 "sudo systemctl is-active k3s-agent"
sailadmin@10.0.0.22's password:
[sudo] password for sailadmin:
active
Connection to 10.0.0.22 closed.
(base) user@uers-MacBook-Pro ~ %
```

d. Verify the Kubernetes API server is accessible using `kubectl cluster-info`.

```
(base) user@uers-MacBook-Pro ~ % kubectl cluster-info
Kubernetes control plane is running at https://10.0.0.20:6443
CoreDNS is running at https://10.0.0.20:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy
Metrics-server is running at https://10.0.0.20:6443/api/v1/namespaces/kube-system/services/https:metrics-server:https/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

e. Confirm all three nodes report `Ready` using `kubectl get nodes`.

`kubectl get nodes -o wide`

```
(base) user@uers-MacBook-Pro ~ % kubectl get nodes -o wide
NAME        STATUS   ROLES           AGE   VERSION        INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION            CONTAINER-RUNTIME
pi-node-1   Ready    control-plane   63d   v1.36.3+k3s1   10.0.0.20     <none>        Debian GNU/Linux 13 (trixie)   6.18.34+rpt-rpi-v8 (arm64)   containerd://2.3.2-k3s2
pi-node-2   Ready    <none>          63d   v1.36.3+k3s1   10.0.0.21     <none>        Debian GNU/Linux 13 (trixie)   6.18.34+rpt-rpi-v8 (arm64)   containerd://2.3.2-k3s2
pi-node-3   Ready    <none>          63d   v1.36.3+k3s1   10.0.0.22     <none>        Debian GNU/Linux 13 (trixie)   6.18.34+rpt-rpi-v8 (arm64)   containerd://2.3.2-k3s2
```

`kubectl get pods -n kube-system -o wide`

```
(base) user@uers-MacBook-Pro ~ % kubectl get pods -n kube-system -o wide
NAME                                      READY   STATUS      RESTARTS       AGE   IP           NODE        NOMINATED NODE   READINESS GATES
coredns-54996dc9b4-cmvhx                  1/1     Running     2 (50d ago)    63d   10.42.0.16   pi-node-1   <none>           <none>
csi-nfs-controller-65b4f9877-5qbjz        5/5     Running     13 (30m ago)   63d   10.0.0.22    pi-node-3   <none>           <none>
csi-nfs-node-27nvf                        3/3     Running     6 (30m ago)    63d   10.0.0.22    pi-node-3   <none>           <none>
csi-nfs-node-f76m8                        3/3     Running     6 (30m ago)    63d   10.0.0.21    pi-node-2   <none>           <none>
csi-nfs-node-zccfd                        3/3     Running     6 (50d ago)    63d   10.0.0.20    pi-node-1   <none>           <none>
helm-install-traefik-crd-m5j9c            0/1     Completed   0              63d   <none>       pi-node-1   <none>           <none>
helm-install-traefik-fq8fn                0/1     Completed   1 (63d ago)    63d   <none>       pi-node-1   <none>           <none>
local-path-provisioner-58d557dc48-s7jg8   1/1     Running     2 (50d ago)    63d   10.42.0.14   pi-node-1   <none>           <none>
metrics-server-6dc596dfb8-726qs           1/1     Running     2 (50d ago)    63d   10.42.0.15   pi-node-1   <none>           <none>
svclb-traefik-a496dd55-cvjbp              2/2     Running     4 (30m ago)    63d   10.42.2.7    pi-node-2   <none>           <none>
svclb-traefik-a496dd55-hp67j              2/2     Running     4 (30m ago)    63d   10.42.4.4    pi-node-3   <none>           <none>
svclb-traefik-a496dd55-s66hl              2/2     Running     4 (50d ago)    63d   10.42.0.17   pi-node-1   <none>           <none>
traefik-59b7647586-z6xsr                  1/1     Running     2 (50d ago)    63d   10.42.0.18   pi-node-1   <none>           <none>
(base) user@uers-MacBook-Pro ~ %
```

Step 3: Verify SSD, NFS, and Kubernetes storage

a. Verify the external SSD is mounted at `/mnt/sail-storage` on pi-node-1.

`ssh sailadmin@10.0.0.20 "lsblk -f"`

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "lsblk -f"
sailadmin@10.0.0.20's password:
NAME        FSTYPE FSVER LABEL       UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       swap   1
sda
└─sda1      ext4   1.0   SAIL_STORAGE bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a
mmcblk0
├─mmcblk0p1 vfat   FAT32 bootfs      B2F0-82D2                             428.7M    15% /boot/firmware
└─mmcblk0p2 ext4   1.0   rootfs      15f4c6be-1102-4331-9904-f78e78afd1fd   21.4G    20% /
zram0       swap   1     zram0       4b2380c0-83d8-42a0-ab8f-dea5fd78fcf3                [SWAP]
```

ssh sailadmin@10.0.0.20 "grep -nE 'SAIL_STORAGE|bebf7baf|sail-storage' /etc/fstab"

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "grep -nE 'SAIL_STORAGE|bebf7baf|sail-storage' /etc/fstab"
sailadmin@10.0.0.20's password:
4:UUID=bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a /mnt/sail-storage ext4 defaults,nofail 0 2
```

UUID=bebf7baf-d72d-4e9a-9d56-3f8cad05fd9a /mnt/sail-storage ext4 defaults,nofail 0 2

Confirms:

- The UUID matches the connected Samsung T7 SSD.
- The configured mount point is `/mnt/sail-storage`.
- The filesystem is `ext4`.
- The `nofail` option allows the Raspberry Pi to boot even if the SSD cannot be mounted.

```
ssh -t sailadmin@10.0.0.20 "sudo mount /mnt/sail-storage"
```

```
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "sudo mount /mnt/sail-storage"
sailadmin@10.0.0.20's password:
Connection to 10.0.0.20 closed.
```

The SSH connection closed normally, and no mount error was displayed. This suggests the mount succeeded, but still need to verify it.

```
ssh sailadmin@10.0.0.20 "findmnt /mnt/sail-storage"
```

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "findmnt /mnt/sail-storage"
sailadmin@10.0.0.20's password:
TARGET           SOURCE    FSTYPE OPTIONS
/mnt/sail-storage /dev/sda1 ext4   rw,relatime,stripe=8191
```

The Samsung T7 SSD is now successfully mounted.

Property	**Verified value**
Mount point	`/mnt/sail-storage`
Device	`/dev/sda1`
Filesystem	`ext4`
Access	Read/write (`rw`)

b. Check available SSD capacity using `df -h`.

```
ssh sailadmin@10.0.0.20 "df -h /mnt/sail-storage"
```

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "df -h /mnt/sail-storage"
sailadmin@10.0.0.20's password:
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       916G  2.1M  870G   1% /mnt/sail-storage
```

The Samsung T7 SSD has sufficient available capacity for the planned monitoring deployment.

Metric	**Result**
Total capacity	916 GB
Used	21 MB
Available	870 GB
Utilization	1%

Mount point `/mnt/sail-storage`

c. Confirm the NFS server is active and `/mnt/sail-storage/kubernetes` is exported.

```
ssh sailadmin@10.0.0.20 "systemctl is-active nfs-kernel-server"
```

```
/dev/sda1       916G  2.1M  870G   1% /mnt/sail-storage
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "systemctl is-active nfs-kernel-server"
sailadmin@10.0.0.20's password:
Connection closed by 10.0.0.20 port 22
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "systemctl is-active nfs-kernel-server"
```

```
ssh -t sailadmin@10.0.0.20 "systemctl is-active nfs-kernel-server"
```

```
inactive
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "systemctl is-active nfs-kernel-server"
sailadmin@10.0.0.20's password:
inactive
Connection to 10.0.0.20 closed.
```

```
ssh -t sailadmin@10.0.0.20 "sudo systemctl status nfs-kernel-server --no-pager -l"
```

```
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "sudo systemctl status nfs-kernel-server --no-pager -l"
sailadmin@10.0.0.20's password:
○ nfs-server.service - NFS server and services
     Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
    Drop-In: /run/systemd/generator/nfs-server.service.d
             └─order-with-mounts.conf
     Active: inactive (dead) since Tue 2026-09-08 18:35:17 PDT; 1 month 19 days ago
   Duration: 12.242s
 Invocation: 17eb819485f3439eabaf3a5e9af37b73
       Docs: man:rpc.nfsd(8)
             man:exportfs(8)
    Process: 872 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SUCCESS)
    Process: 875 ExecStart=/bin/sh -c /usr/sbin/nfsdctl autostart || /usr/sbin/rpc.nfsd (code=exited, status=0/SUCCESS)
    Process: 970 ExecStop=/bin/sh -c /usr/sbin/nfsdctl threads 0 || /usr/sbin/rpc.nfsd 0 (code=exited, status=0/SUCCESS)
    Process: 974 ExecStopPost=/usr/sbin/exportfs -au (code=exited, status=0/SUCCESS)
    Process: 976 ExecStopPost=/usr/sbin/exportfs -f (code=exited, status=0/SUCCESS)
   Main PID: 875 (code=exited, status=0/SUCCESS)
   Mem peak: 1.9M
        CPU: 64ms

Sep 08 18:35:04 pi-node-1 systemd[1]: Starting nfs-server.service - NFS server and services...
Sep 08 18:35:04 pi-node-1 sh[878]: nfsdctl: lockd configuration failure
Sep 08 18:35:05 pi-node-1 systemd[1]: Finished nfs-server.service - NFS server and services.
Sep 08 18:35:17 pi-node-1 systemd[1]: Stopping nfs-server.service - NFS server and services...
Sep 08 18:35:17 pi-node-1 systemd[1]: nfs-server.service: Deactivated successfully.
Sep 08 18:35:17 pi-node-1 systemd[1]: Stopped nfs-server.service - NFS server and services.
Connection to 10.0.0.20 closed.
```

```
ssh -t sailadmin@10.0.0.20 "sudo systemctl start nfs-server"
```

```
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "sudo systemctl start nfs-server"
sailadmin@10.0.0.20's password:
Connection to 10.0.0.20 closed.
```

```
ssh -t sailadmin@10.0.0.20 "systemctl is-active nfs-server"
```

```
Connection to 10.0.0.20 closed.
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "systemctl is-active nfs-server"
sailadmin@10.0.0.20's password:
active
Connection to 10.0.0.20 closed.
```

d. Verify the `sail-nfs` Kubernetes StorageClass exists.

```
ssh -t sailadmin@10.0.0.20 "sudo exportfs -v"
```

```
Connection to 10.0.0.20 closed.
(base) user@uers-MacBook-Pro ~ % ssh -t sailadmin@10.0.0.20 "sudo exportfs -v"
sailadmin@10.0.0.20's password:
/mnt/sail-storage/kubernetes
                10.0.0.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
Connection to 10.0.0.20 closed.
```

The NFS export is configured and active.

Property	**Verified value**
Export directory	`/mnt/sail-storage/kubernetes`

Allowed network	`10.0.0.0/24`
Access	Read/write (`rw`)
Root squashing	Enabled (`root_squash`)
Write synchronization	Enabled (`sync`)

e. Review existing persistent volumes and persistent volume claims for errors.

```
kubectl get storageclass
```

```
(base) user@uers-MacBook-Pro ~ % kubectl get storageclass
NAME                   PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
local-path (default)   rancher.io/local-path   Delete          WaitForFirstConsumer   false                  63d
sail-nfs               nfs.csi.k8s.io          Retain          Immediate              true                   63d
```

Step 4: Verify internet access and install or verify Helm

a. Confirm internet connectivity from the administration computer.

curl -I https://get.helm.sh

```
(base) user@uers-MacBook-Pro ~ % curl -I https://get.helm.sh
HTTP/2 404
date: Thu, 29 Oct 2026 03:44:25 GMT
x-ms-request-id: 98b841d2-301e-006a-5557-672d1c000000
x-azure-ref: 20261029T034424Z-164d66d4487b74phhC1PHX8nt00000000a40000000005p8y
x-fd-int-roxy-purgeid: 0
x-cache: TCP_MISS
```

Successfully connected to `get.helm.sh` over HTTPS

b. Verify internet connectivity from pi-node-1 and the worker nodes.

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20 "curl -I https://get.helm.sh"
ssh sailadmin@10.0.0.21 "curl -I https://get.helm.sh"
ssh sailadmin@10.0.0.22 "curl -I https://get.helm.sh"
sailadmin@10.0.0.20's password:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0HTTP/2 404
date: Thu, 29 Oct 2026 03:46:21 GMT
x-ms-request-id: c05e4adf-701e-006b-2258-6772c0000000
x-azure-ref: 20261029T034621Z-164d66d4487ctvjzhC1PHX26ps00000006dg00000000d90b
x-fd-int-roxy-purgeid: 0
x-cache: TCP_MISS

  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
sailadmin@10.0.0.21's password:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0HTTP/2 404
date: Thu, 29 Oct 2026 03:46:28 GMT
x-ms-request-id: d8f8fea1-b01e-0006-7858-67c68b000000
x-azure-ref: 20261029T034628Z-r1f4c8f767cr5vknhC1PHXb36400000002w000000000vpyk
x-fd-int-roxy-purgeid: 0
x-cache: TCP_MISS

  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
sailadmin@10.0.0.22's password:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0HTTP/2 404
date: Thu, 29 Oct 2026 03:46:34 GMT
x-ms-request-id: 4db14c72-c01e-0051-4f58-6768b8000000
x-azure-ref: 20261029T034634Z-164d66d44877xb7lhC1PHXy3a400000009v000000000t9ta
x-fd-int-roxy-purgeid: 0
x-cache: TCP_MISS

  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
```

All three SAIL nodes successfully reached the Helm server over HTTPS.

Node	IP address	HTTP response	Result
pi-node-1	10.0.0.20	404	Passed
pi-node-2	10.0.0.21	404	Passed
pi-node-3	10.0.0.22	404	Passed

c. Check whether Helm is installed using `helm version`.
helm version

```
(base) user@uers-MacBook-Pro ~ % helm version
zsh: command not found: helm
```

d. Install Helm if necessary and verify the installation.
`brew install helm`

```
For more information, see:
  https://docs.brew.sh/Tap-Trust
==> Downloading bottle manifests
✔ Bottle Manifest helm (4.3.1)
==> Fetching downloads for: helm
✔ Bottle helm (4.3.1)                                    Downloaded   13.1KB/ 13.1KB
==> Pouring helm--4.3.1.arm64_tahoe.bottle.tar.gz        Downloaded   18.8MB/ 18.8MB
🍺  /opt/homebrew/Cellar/helm/4.3.1: 69 files, 63.7MB
==> Caveats
zsh completions have been installed to:
  /opt/homebrew/share/zsh/site-functions
```

`helm version`

```
(base) user@uers-MacBook-Pro ~ % helm version
version.BuildInfo{Version:"v4.3.1", GitCommit:"c0d08d0d6d9424d4a14febb5e8a3e51b0dd00dff", GitTreeState:"dirty", GoVersion:"go1.27.2", KubeClientVersion:"v1.37"}
```

e. Review existing Helm releases using `helm list -A`.

```
(base) user@uers-MacBook-Pro ~ % helm list -A
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                           APP VERSION
traefik         kube-system     1               2026-08-26 04:59:50.968924846 +0000 UTC deployed        traefik-40.1.4+up40.1.0         v3.7.1
traefik-crd     kube-system     1               2026-08-26 04:59:36.527300868 +0000 UTC deployed        traefik-crd-40.1.4+up40.1.0     v3.7.1
```

The `helm list -A` command executed successfully and identified two existing Helm releases.

Release	Namespace	Status	Application version
`traefik`	`kube-system`	deployed	v3.7.1
`traefik-crd`	`kube-system`	deployed	v3.7.1

Step 5: Inspect cluster resources, Metrics Server, and existing workloads
a. List running Kubernetes pods, deployments, and services.
`kubectl top nodes`

```
(base) user@uers-MacBook-Pro ~ % kubectl top nodes
NAME         CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
pi-node-1    209m         5%     1401Mi          17%
pi-node-2    82m          2%     580Mi           7%
pi-node-3    97m          2%     796Mi           10%
```

b. Verify Metrics Server is available in the cluster.
`kubectl get deployment metrics-server -n kube-system`

```
(base) user@uers-MacBook-Pro ~ % kubectl get deployment metrics-server -n kube-system
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
metrics-server   1/1     1            1           63d
```

c. Review CPU and memory utilization using `kubectl top nodes` and `kubectl top pods -A`.

```
(base) user@uers-MacBook-Pro ~ % kubectl top pods -A
NAMESPACE     NAME                                      CPU(cores)   MEMORY(bytes)
kube-system   coredns-54996dc9b4-cmvhx                  7m           72Mi
kube-system   csi-nfs-controller-65b4f9877-5qbjz        13m          215Mi
kube-system   csi-nfs-node-27nvf                        1m           75Mi
kube-system   csi-nfs-node-f76m8                        2m           85Mi
kube-system   csi-nfs-node-zccfd                        1m           85Mi
kube-system   local-path-provisioner-58d557dc48-s7jg8   1m           49Mi
kube-system   metrics-server-6dc596dfb8-726qs           12m          79Mi
kube-system   svclb-traefik-a496dd55-cvjbp              0m           2Mi
kube-system   svclb-traefik-a496dd55-hp67j              0m           2Mi
kube-system   svclb-traefik-a496dd55-s66hl              0m           2Mi
kube-system   traefik-59b7647586-z6xsr                  1m           121Mi
```

d. Inspect node capacity, allocatable resources, and current resource requests.
`kubectl describe nodes`

Node	Allocatable CPU	Allocatable memory	CPU requested	Memory requested
pi-node-1	4 cores	7.64 GiB	230m (5%)	200 MiB (2%)
pi-node-2	4 cores	7.64 GiB	30m (0%)	60 MiB (0%)
pi-node-3	4 cores	7.64 GiB	80m (2%)	160 MiB (2%)

e. Identify existing workloads and available resources for monitoring components.
kubectl get pods,deployments,services -A

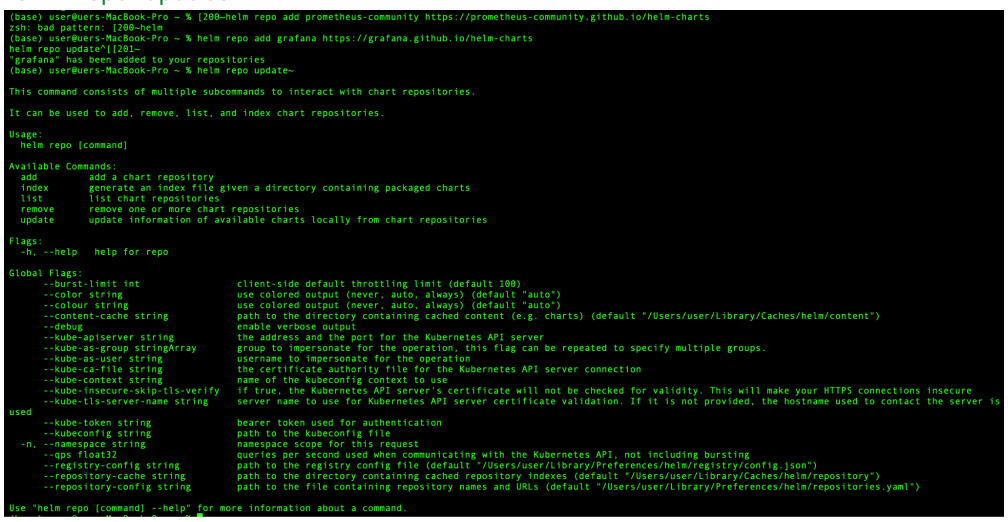
```
(base) user@uers-MacBook-Pro ~ % kubectl get pods,deployments,services -A
NAMESPACE     NAME                                          READY   STATUS      RESTARTS        AGE
kube-system   pod/coredns-54996dc9b4-cmvhx                  1/1     Running     2 (50d ago)     63d
kube-system   pod/csi-nfs-controller-65b4f9877-5qbjz        5/5     Running     13 (129m ago)   63d
kube-system   pod/csi-nfs-node-27nvf                        3/3     Running     6 (129m ago)    63d
kube-system   pod/csi-nfs-node-f76m8                        3/3     Running     6 (129m ago)    63d
kube-system   pod/csi-nfs-node-zccfd                        3/3     Running     6 (50d ago)     63d
kube-system   pod/helm-install-traefik-crd-m5j9c            0/1     Completed   0               63d
kube-system   pod/helm-install-traefik-fq8fn                0/1     Completed   1 (63d ago)     63d
kube-system   pod/local-path-provisioner-58d557dc48-s7jg8   1/1     Running     2 (50d ago)     63d
kube-system   pod/metrics-server-6dc596dfb8-726qs           1/1     Running     2 (50d ago)     63d
kube-system   pod/svclb-traefik-a496dd55-cvjbp              2/2     Running     4 (129m ago)    63d
kube-system   pod/svclb-traefik-a496dd55-hp67j              2/2     Running     4 (129m ago)    63d
kube-system   pod/svclb-traefik-a496dd55-s66hl              2/2     Running     4 (50d ago)     63d
kube-system   pod/traefik-59b7647586-z6xsr                  1/1     Running     2 (50d ago)     63d

NAMESPACE     NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
kube-system   deployment.apps/coredns                  1/1     1            1           63d
kube-system   deployment.apps/csi-nfs-controller       1/1     1            1           63d
kube-system   deployment.apps/local-path-provisioner   1/1     1            1           63d
kube-system   deployment.apps/metrics-server           1/1     1            1           63d
kube-system   deployment.apps/traefik                  1/1     1            1           63d

NAMESPACE     NAME                     TYPE           CLUSTER-IP      EXTERNAL-IP                     PORT(S)                      AGE
default       service/kubernetes       ClusterIP      10.43.0.1       <none>                          443/TCP                      63d
kube-system   service/kube-dns         ClusterIP      10.43.0.10      <none>                          53/UDP,53/TCP,9153/TCP       63d
kube-system   service/metrics-server   ClusterIP      10.43.69.212    <none>                          443/TCP                      63d
kube-system   service/traefik          LoadBalancer   10.43.220.136   10.0.0.20,10.0.0.21,10.0.0.22   80:32485/TCP,443:30504/TCP   63d
(base) user@uers-MacBook-Pro ~ %
```

Step 6: Plan monitoring deployment, storage, and resource allocations
a. Determine the Kubernetes namespace for monitoring components.
`kubectl get namespace monitoring`
`kubectl create namespace monitoring`

```
(base) user@uers-MacBook-Pro ~ % kubectl create namespace monitoring
namespace/monitoring created
(base) user@uers-MacBook-Pro ~ % kubectl get namespace monitoring
NAME         STATUS   AGE
monitoring   Active   3s
```

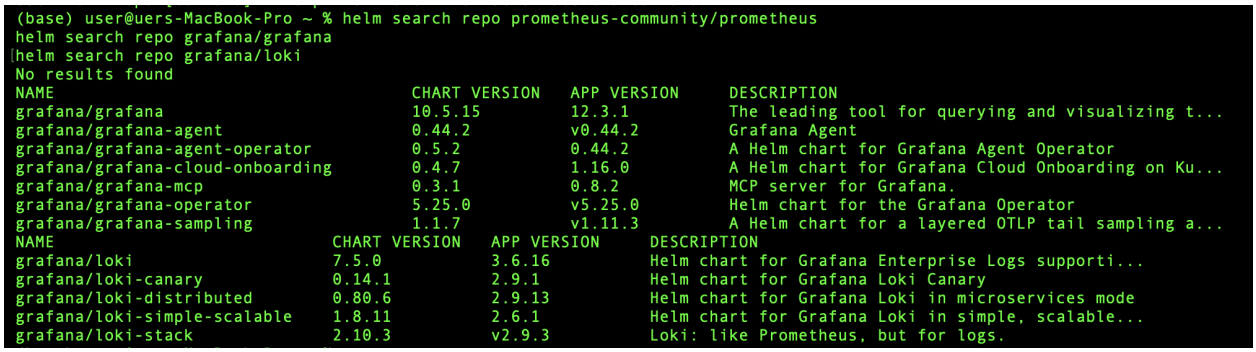
b. Select compatible Helm charts and ARM64 container images.

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo add grafana https://grafana.github.io/helm-charts
helm repo update
```



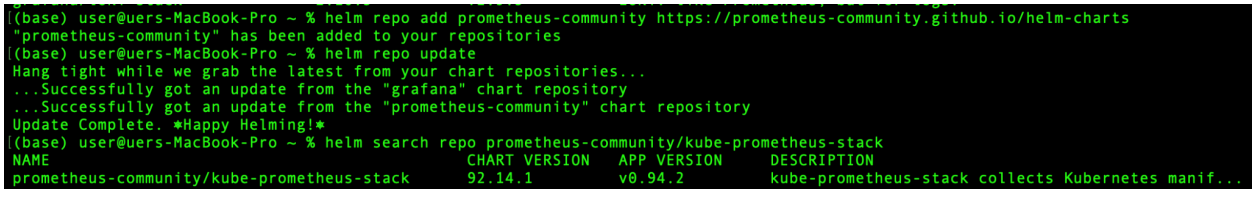
list the available charts:

```
helm search repo prometheus-community/prometheus
helm search repo grafana/grafana
helm search repo grafana/loki
```



Correct the repo setup:

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update
helm search repo prometheus-community/kube-prometheus-stack
```



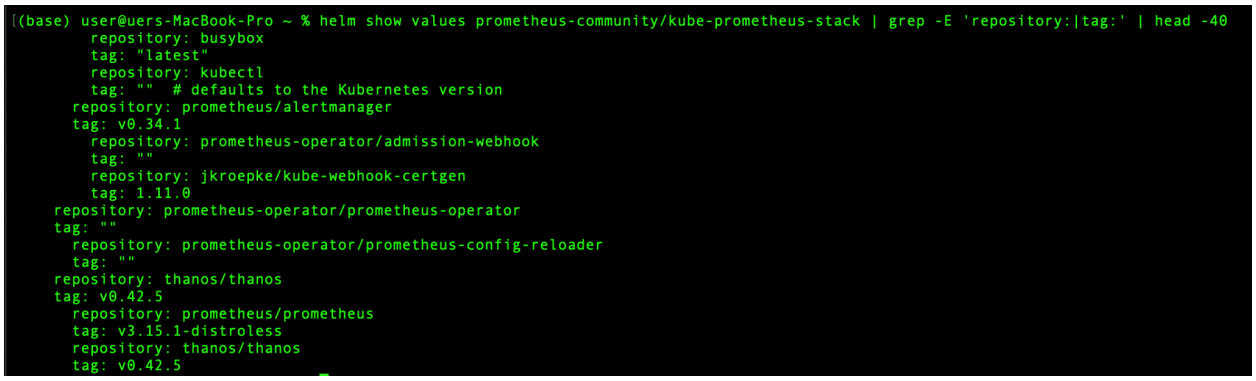
All three commands succeeded.

- Prometheus repository: Added successfully.
- Grafana repository: Updated successfully.
- Helm repository indexes: Updated successfully.

- Prometheus monitoring stack: Found.

Chart	Version
`prometheus-community/kube-prometheus-stack`	92.14.1
`grafana/grafana`	10.5.15
`grafana/loki`	7.5.0

```
helm show values prometheus-community/kube-prometheus-stack | grep -E 'repository:|tag:' | head
-40
```




```
docker manifest inspect prom/prometheus:v3.15.1-distroless | grep -A 5
'"architecture": "arm64"'
```

This confirms that `prom/prometheus:v3.15.1-distroless` supports the Raspberry Pi cluster's architecture. Also identified Helm charts for Prometheus, Grafana, and Loki.

c. Establish CPU and memory allocations for Prometheus, Grafana, Loki, and other components.

Starting with conservative resource allocations so monitoring does not interfere with SAIL Orbit when it is deployed.

Proposed initial allocations

Component	CPU request	CPU limit	Memory request	Memory limit
Prometheus	250m	1000m	512 MiB	1.5 GiB
Grafana	100m	500m	128 MiB	512 MiB
Loki	200m	1000m	256 MiB	1 GiB

Prometheus Operator	100m	500m	128 MiB	512 MiB
Alertmanager	50m	250m	64 MiB	256 MiB
Node Exporter (3 instances)	30m each	100m each	32 MiB each	128 MiB each

- Request: Resources Kubernetes reserves for scheduling.
- Limit: Maximum resources a container can consume. CPU limits throttle usage; exceeding a memory limit can cause an out-of-memory termination.
- 1000m CPU: One CPU core.
- 1024 MiB: One GiB of memory.

The proposed allocations leave substantial capacity for SAIL Orbit, PostgreSQL, and future security workloads.

d. Make storage requirements and configure persistent volumes in Helm values.

```
mkdir -p monitoring
touch prometheus.yaml
touch loki.yaml
```

```
(.sail-orbit-env) (base) user@uers-MacBook-Pro SAIL % helm template monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring \
  -f monitoring/prometheus.yaml \
  > /dev/null
(.sail-orbit-env) (base) user@uers-MacBook-Pro SAIL % helm template loki grafana/loki \
  --namespace monitoring \
  -f monitoring/loki.yaml \
  > /dev/null
```

Confirmed:

- prometheus.yaml renders successfully with the Prometheus Helm chart.
- loki.yaml renders successfully with the Loki Helm chart.

```
ssh sailadmin@10.0.0.20
sudo mkdir -p /mnt/sail-storage/monitoring/{prometheus,loki}
ls -ld /mnt/sail-storage/monitoring/*
```

```
(base) user@uers-MacBook-Pro ~ % ssh sailadmin@10.0.0.20
sailadmin@10.0.0.20's password:
Linux pi-node-1 6.18.34+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.18.34-1+rpt1 (2026-06-09) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Oct 28 20:33:58 2026 from 10.0.0.2
sailadmin@pi-node-1:~ $ sudo mkdir -p /mnt/sail-storage/monitoring/{prometheus,loki}
ls -ld /mnt/sail-storage/monitoring/*
drwxr-xr-x 2 root root 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/loki
drwxr-xr-x 2 root root 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/prometheus
sailadmin@pi-node-1:~ $
```

Create Kubernetes PersistentVolumes.

First, exit SSH to return to the MacBook:

```
exit
```

From the SAIL repository root, create:

`monitoring/storage.yaml`

Paste this configuration:

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: sail-prometheus-pv
spec:
  capacity:
    storage: 20Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: sail-monitoring-local
  local:
    path: /mnt/sail-storage/monitoring/prometheus
  nodeAffinity:
    required:
      nodeSelectorTerms:
        - matchExpressions:
            - key: kubernetes.io/hostname
              operator: In
              values:
                - pi-node-1
---
apiVersion: v1
kind: PersistentVolume
metadata:
  name: sail-loki-pv
spec:
  capacity:
    storage: 20Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: sail-monitoring-local
  local:
    path: /mnt/sail-storage/monitoring/loki
  nodeAffinity:
    required:
      nodeSelectorTerms:
        - matchExpressions:
            - key: kubernetes.io/hostname
              operator: In
              values:
                - pi-node-1
---
```

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: sail-monitoring-local
provisioner: kubernetes.io/no-provisioner
volumeBindingMode: WaitForFirstConsumer
```

Apply it:

```
kubectl apply -f monitoring/storage.yaml
```

```
Connection to 10.0.0.20 closed.
(base) user@uers-MacBook-Pro ~ % kubectl apply -f /Users/user/Desktop/SAIL/monitoring/storage.yaml
persistentvolume/sail-prometheus-pv created
persistentvolume/sail-loki-pv created
storageclass.storage.k8s.io/sail-monitoring-local created
```

Verify:

```
kubectl get pv
kubectl get storageclass
```

```
storageclass.storage.k8s.io/sail-monitoring-local created
(base) user@uers-MacBook-Pro ~ % kubectl get pv
kubectl get storageclass
NAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM                          STORAGECLASS           VOLUMEATTRIBUTESCLASS   REASON   AGE
pvc-7f421529-6d6b-432b-8ae6-1cf11410fb90   1Gi        RWX            Retain           Released    default/sail-storage-test      sail-nfs               <unset>                          63d
sail-loki-pv                               20Gi       RWO            Retain           Available                                  sail-monitoring-local  <unset>                          50s
sail-prometheus-pv                         20Gi       RWO            Retain           Available                                  sail-monitoring-local  <unset>                          50s
NAME                  PROVISIONER                    RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
local-path (default)  rancher.io/local-path          Delete          WaitForFirstConsumer   false                  64d
sail-monitoring-local kubernetes.io/no-provisioner   Delete          WaitForFirstConsumer   false                  50s
sail-nfs              nfs.csi.k8s.io                 Retain          Immediate              true                   63d
(base) user@uers-MacBook-Pro ~ %
```

```
helm template monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring \
  -f monitoring/prometheus.yaml \
  | grep -A 120 '^kind: Prometheus$' \
  | grep -A 12 'storage:'
```

```
(base) user@uers-MacBook-Pro SAIL % helm template monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring \
  -f monitoring/prometheus.yaml \
  | grep -A 120 '^kind: Prometheus$' \
  | grep -A 12 'storage:'
  storage:
    volumeClaimTemplate:
      spec:
        accessModes:
        - ReadWriteOnce
        resources:
          requests:
            storage: 20Gi
        storageClassName: sail-monitoring-local
  affinity:
    podAntiAffinity:
      preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 100
        podAffinityTerm:
          topologyKey: kubernetes.io/hostname
          labelSelector:
            matchExpressions:
            - {key: app.kubernetes.io/name, operator: In, values: [prometheus]}
            - {key: app.kubernetes.io/instance, operator: In, values: [monitoring-kube-prometheus-prometheus]}
  portName: http-web
```

Both Prometheus and Loki now generate the intended storage configurations, matching the two PersistentVolumes created on `pi-node-1`.

One remaining task for Step 6d: Verify filesystem permissions so the containers can write to their SSD directories.

```
ssh sailadmin@10.0.0.20 \
  "ls -ld /mnt/sail-storage/monitoring/{prometheus,loki}"
```

```
(base) user@uers-MacBook-Pro SAIL % ssh sailadmin@10.0.0.20 \
[  "ls -ld /mnt/sail-storage/monitoring/{prometheus,loki}"
[sailadmin@10.0.0.20's password:
drwxr-xr-x 2 root root 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/loki
drwxr-xr-x 2 root root 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/prometheus
(base) user@uers-MacBook-Pro SAIL %
```

Both directories are owned by `root:root` with `755` permissions. The monitoring containers may not be able to write to them.

Run this from your MacBook Terminal to assign ownership to the containers' expected user IDs:

```
ssh sailadmin@10.0.0.20 \
  "sudo chown 65534:65534 /mnt/sail-storage/monitoring/prometheus && \
   sudo chown 10001:10001 /mnt/sail-storage/monitoring/loki"
```

Verify:

```
ssh sailadmin@10.0.0.20 \
  "ls -ldn /mnt/sail-storage/monitoring/{prometheus,loki}"
```

```
drwxr-xr-x 2 root root 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/prometheus
(base) user@uers-MacBook-Pro SAIL % ssh sailadmin@10.0.0.20 \
  "sudo chown 65534:65534 /mnt/sail-storage/monitoring/prometheus && \
[   sudo chown 10001:10001 /mnt/sail-storage/monitoring/loki"
[sailadmin@10.0.0.20's password:
(base) user@uers-MacBook-Pro SAIL % ssh sailadmin@10.0.0.20 \
[  "ls -ldn /mnt/sail-storage/monitoring/{prometheus,loki}"
[sailadmin@10.0.0.20's password:
drwxr-xr-x 2 10001 10001 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/loki
drwxr-xr-x 2 65534 65534 4096 Oct 28 22:12 /mnt/sail-storage/monitoring/prometheus
(base) user@uers-MacBook-Pro SAIL %
```

Filesystem ownership updated successfully.

- Prometheus: `65534:65534`
- Loki: `10001:10001`
- Permissions: `755` on both directories

e: Set metrics, logs, and trace retention in those values.

Initial retention periods:

Component	Data	Retention
Prometheus	Metrics	15 days
Loki	Logs	7 days
Tracing backend (planned)	Traces	3 days

- Metrics: 15 days — Enough history to establish baselines and compare performance.

- Logs: 7 days — Controls storage growth from application and security activity.
- Traces: 3 days — Keeps recent troubleshooting data while conserving resources.

These are starting values for the Raspberry Pi cluster and can be adjusted after measuring storage usage.

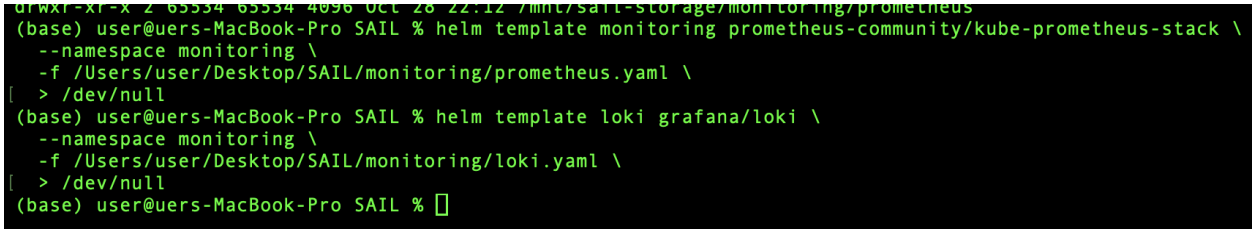
1. Update `monitoring/prometheus.yaml`
   Under `prometheus.prometheusSpec`, add:
   ```
   retention: 15d
   retentionSize: 15GB
   ```

2. Update `monitoring/loki.yaml`
   Under the existing `loki:` section, add `limits_config`:
   ```
   limits_config:
     retention_period: 168h
   ```
   Also add this under `loki:` to enable retention enforcement:
   ```
   compactor:
     retention_enabled: true
     delete_request_store: filesystem
   ```

3. Tracing retention
   Decision: 3 days.

Verify the current Helm configurations
From the SAIL repository root:

```
helm template monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring \
  -f /Users/user/Desktop/SAIL/monitoring/prometheus.yaml \
  > /dev/null

helm template loki grafana/loki \
  --namespace monitoring \
  -f /Users/user/Desktop/SAIL/monitoring/loki.yaml \
  > /dev/null
```



f: Configure browser access through Traefik

Check its service configuration:

```
kubectl get svc traefik -n kube-system
```

```
(base) user@uers-MacBook-Pro sail-orbit % kubectl get svc traefik -n kube-system
NAME      TYPE           CLUSTER-IP      EXTERNAL-IP                    PORT(S)                      AGE
traefik   LoadBalancer   10.43.220.136   10.0.0.20,10.0.0.21,10.0.0.22   80:32485/TCP,443:30504/TCP   64d
```

Traefik is already configured to accept traffic on all three Raspberry Pis.

Setting	Verified value
Service type	LoadBalancer
External IPs	10.0.0.20, 10.0.0.21, 10.0.0.22
HTTP	Port 80
HTTPS	Port 443

Verify HTTPS connectivity

```
curl -k -I https://10.0.0.20
```

```
(base) user@uers-MacBook-Pro sail-orbit % curl -k -I https://10.0.0.20
HTTP/2 404
content-type: text/plain; charset=utf-8
x-content-type-options: nosniff
content-length: 19
date: Fri, 30 Oct 2026 04:17:22 GMT
```

Configure local hostnames

Interface	Browser address
Grafana	grafana.sail.test
Prometheus	prometheus.sail.test
Alertmanager	alertmanager.sail.test

1. Open the computer hosts file:
   ```
   sudo nano /etc/hosts
   ```
   Add these lines, preserving the existing entries:
   ```
   10.0.0.20 grafana.sail.test
   10.0.0.20 prometheus.sail.test
   10.0.0.20 alertmanager.sail.test
   ```
2. Verify that all three hostnames resolve to `10.0.0.20`
   ```
   ping -c 1 grafana.sail.test
   ping -c 1 prometheus.sail.test
   ping -c 1 alertmanager.sail.test
   ```

```
(base) user@uers-MacBook-Pro sail-orbit % ping -c 1 grafana.sail.test

PING grafana.sail.test (10.0.0.20): 56 data bytes
64 bytes from 10.0.0.20: icmp_seq=0 ttl=64 time=8.321 ms

--- grafana.sail.test ping statistics ---
1 packets transmitted, 1 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 8.321/8.321/8.321/0.000 ms
(base) user@uers-MacBook-Pro sail-orbit % ping -c 1 prometheus.sail.test

PING prometheus.sail.test (10.0.0.20): 56 data bytes
64 bytes from 10.0.0.20: icmp_seq=0 ttl=64 time=5.735 ms

--- prometheus.sail.test ping statistics ---
1 packets transmitted, 1 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.735/5.735/5.735/nan ms
(base) user@uers-MacBook-Pro sail-orbit % ping -c 1 alertmanager.sail.test
PING alertmanager.sail.test (10.0.0.20): 56 data bytes
64 bytes from 10.0.0.20: icmp_seq=0 ttl=64 time=8.994 ms

--- alertmanager.sail.test ping statistics ---
1 packets transmitted, 1 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 8.994/8.994/8.994/nan ms
```

3. HTTPS certificate configuration
   Use a local certificate authority (CA) to provide HTTPS for all three monitoring interfaces without purchasing certificates.
   Check whether OpenSSL is available on your computer:
   ```
   openssl version
   ```
   ```
   (base) user@uers-MacBook-Pro sail-orbit % openssl version
   OpenSSL 3.0.18 30 Sep 2025 (Library: OpenSSL 3.0.18 30 Sep 2025)
   ```

   ```
   mkdir -p /Users/user/Desktop/SAIL/monitoring/certs
   ```
   add it to .gitignore
   ```
   (base) user@uers-MacBook-Pro sail-orbit % git -C /Users/user/Desktop/SAIL check-ignore -v monitoring/certs/test.key
   .gitignore:43:monitoring/certs/ monitoring/certs/test.key
   ```

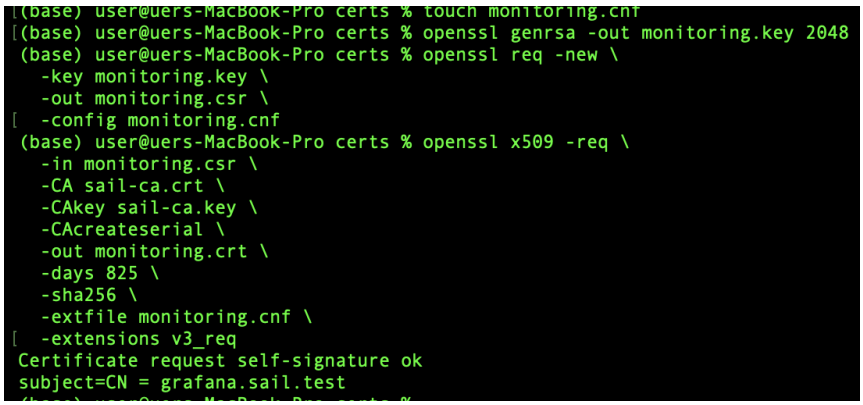
4. Generate the local certificate authority
   ```
   openssl genrsa -out sail-ca.key 4096

   openssl req -x509 -new -sha256 -days 3650 \
     -key sail-ca.key \
     -out sail-ca.crt \
     -subj "/CN=SAIL Local CA"
   ```

   Creates the private CA key and certificate to sign HTTPS certificates for monitoring.
5. Generate a TLS certificate covering all three monitoring hostnames.
   create:
   ```
   /Users/user/Desktop/SAIL/monitoring/certs/monitoring.cnf
   ```
   Add:
   ```
   [req]
   prompt = no
   distinguished_name = dn
   req_extensions = v3_req

   [dn]
   CN = grafana.sail.test
   ```

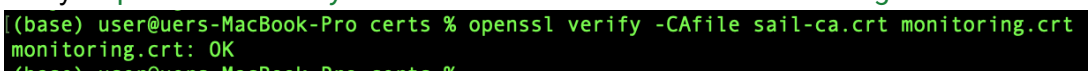
```
[v3_req]
subjectAltName = @alt_names

[alt_names]
DNS.1 = grafana.sail.test
DNS.2 = prometheus.sail.test
DNS.3 = alertmanager.sail.test
```

From `/Users/user/Desktop/SAIL/monitoring/certs`, run:

```
openssl genrsa -out monitoring.key 2048

openssl req -new \
  -key monitoring.key \
  -out monitoring.csr \
  -config monitoring.cnf

openssl x509 -req \
  -in monitoring.csr \
  -CA sail-ca.crt \
  -CAkey sail-ca.key \
  -CAcreateserial \
  -out monitoring.crt \
  -days 825 \
  -sha256 \
  -extfile monitoring.cnf \
  -extensions v3_req
```



Verify: `openssl verify -CAfile sail-ca.crt monitoring.crt`



Confirms the certificate is trusted by the local SAIL CA.

6. Create the Kubernetes TLS Secret

```
kubectl create secret tls sail-monitoring-tls \
  --cert=/Users/user/Desktop/SAIL/monitoring/certs/monitoring.crt \
  --key=/Users/user/Desktop/SAIL/monitoring/certs/monitoring.key \
  --namespace=monitoring
```

```
(base) user@uers-MacBook-Pro certs % kubectl create secret tls sail-monitoring-tls \
  --cert=/Users/user/Desktop/SAIL/monitoring/certs/monitoring.crt \
  --key=/Users/user/Desktop/SAIL/monitoring/certs/monitoring.key \
  --namespace=monitoring
secret/sail-monitoring-tls created
```

```
kubectl get secret sail-monitoring-tls -n monitoring
```

```
(base) user@uers-MacBook-Pro certs % kubectl get secret sail-monitoring-tls -n monitoring
NAME                TYPE                DATA   AGE
sail-monitoring-tls   kubernetes.io/tls   2      7s
```

Kubernetes TLS Secret verified. `sail-monitoring-tls` exists in the `monitoring` namespace with the correct type and two data entries.

7. Inspect the existing Prometheus Helm configuration

```
grep -n -A 12 -B 3 'ingress:' /Users/user/Desktop/SAIL/monitoring/prometheus.yaml
```

No output means prometheus.yaml has no lines matching ingress

   a. Under `prometheus:`, immediately before `prometheusSpec:`, add:

```
ingress:
  enabled: true
  ingressClassName: traefik
  hosts:
    - prometheus.sail.test
  paths:
    - /
  pathType: Prefix
  tls:
    - secretName: sail-monitoring-tls
      hosts:
        - prometheus.sail.test
```

   b. Under `grafana:`, immediately before `replicas: 1`, add:

```
ingress:
  enabled: true
  ingressClassName: traefik
  hosts:
    - grafana.sail.test
  path: /
  pathType: Prefix
  tls:
    - secretName: sail-monitoring-tls
      hosts:
        - grafana.sail.test
```

   c. Under `alertmanager:`, immediately before `alertmanagerSpec:`, add:

```
ingress:
   enabled: true
   ingressClassName: traefik
   hosts:
     - alertmanager.sail.test
   paths:
     - /
   pathType: Prefix
   tls:
     - secretName: sail-monitoring-tls
       hosts:
         - alertmanager.sail.test
```

Configures HTTPS routing thru Traefik using TLS Secret created in `monitoring` namespace.

8. Validate the updated Helm configuration

```
helm template sail-monitoring prometheus-community/kube-prometheus-stack \
  --version 92.14.1 \
  --namespace monitoring \
  --values /Users/user/Desktop/SAIL/monitoring/prometheus.yaml \
  > /tmp/sail-monitoring-rendered.yaml

grep -A 35 'kind: Ingress' /tmp/sail-monitoring-rendered.yaml
```

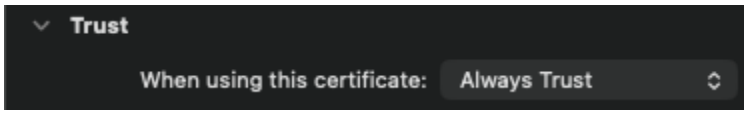
All three ingress resources rendered correctly. Helm ingress configuration is validated.

Service	Hostname	Backend port	Traefik	TLS Secret
Grafana	`grafana.sail.test`	80	Correct	Correct
Alertmanager	`alertmanager.sail.test`	9093	Correct	Correct
Prometheus	`prometheus.sail.test`	9090	Correct	Correct

9. Configure certificate trust

```
open /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt
```

```
(base) user@uers-MacBook-Pro certs % open /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt
(base) user@uers-MacBook-Pro certs % security find-certificate -c "SAIL" -a
keychain: "/Users/user/Library/Keychains/login.keychain-db"
version: 512
class: 0x80001000
attributes:
    "alis"<blob>="SAIL Local CA"
    "cenc"<uint32>=0x00000003
    "ctyp"<uint32>=0x00000001
    "hpky"<blob>=0xB4824B6A8CEF6AD4119075404EC4B7E832843678  "\264\202Kj\214\357j\324\021\220u@N\304\267\3502\2046x"
    "issu"<blob>=0x30183116301406035504030C0D5341494C204C6F63616C204341  "0\0301\0260\024\006\003U\004\003\014\015SAIL Local CA"
    "labl"<blob>="SAIL Local CA"
    "skid"<blob>=0xB4824B6A8CEF6AD4119075404EC4B7E832843678  "\264\202Kj\214\357j\324\021\220u@N\304\267\3502\2046x"
    "snbr"<blob>=0x4999C1486A9C28EC420A275500C417490A2C2B2E  "I\231\301Hj\234(\354B\012'U\000\304\027I\012.+."
    "subj"<blob>=0x30183116301406035504030C0D5341494C204C6F63616C204341  "0\0301\0260\024\006\003U\004\003\014\015SAIL Local CA"
(base) user@uers-MacBook-Pro certs %
```



Verify: `security verify-cert -c /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt`

```
(base) user@uers-MacBook-Pro certs % security verify-cert -c /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt
...certificate verification successful.
```

10. Verify that the monitoring server certificate is trusted through that CA.

```
openssl verify -CAfile /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt
/Users/user/Desktop/SAIL/monitoring/certs/monitoring.crt
```

```
(base) user@uers-MacBook-Pro certs % openssl verify -CAfile /Users/user/Desktop/SAIL/monitoring/certs/sail-ca.crt /Users/user/Desktop/SAIL/monitoring/certs/monitoring.crt
/Users/user/Desktop/SAIL/monitoring/certs/monitoring.crt: OK
```

11. Confirm LAN-only access restrictions.
kubectl get svc traefik -n kube-system -o yaml

The Traefik Service configuration confirms:

- Type: `LoadBalancer`
- Node addresses: `10.0.0.20`, `10.0.0.21`, `10.0.0.22`
- HTTP: Port 80
- HTTPS: Port 443
- Additional NodePorts: 32485 and 30504

12. Check external exposure

```
kubectl get svc -A -o wide
```

```
(base) user@uers-MacBook-Pro certs % kubectl get svc -A -o wide
NAMESPACE     NAME             TYPE           CLUSTER-IP      EXTERNAL-IP                  PORT(S)                      AGE   SELECTOR
default       kubernetes       ClusterIP      10.43.0.1       <none>                       443/TCP                      65d   <none>
kube-system   kube-dns         ClusterIP      10.43.0.10      <none>                       53/UDP,53/TCP,9153/TCP       65d   k8s-app=kube-dns
kube-system   metrics-server   ClusterIP      10.43.69.212    <none>                       443/TCP                      65d   k8s-app=metrics-server
kube-system   traefik          LoadBalancer   10.43.220.136   10.0.0.20,10.0.0.21,10.0.0.22   80:32485/TCP,443:30504/TCP   65d   app.kubernetes.io/instance=traefik
-kube-system,app.kubernetes.io/name=traefik
```

Confirms Traefik is the only externally exposed Kubernetes Service

Summary

- Configured HTTPS ingress for Grafana, Prometheus, and Alertmanager through Traefik.
- Created and validated TLS certificates.
- Installed and trusted the SAIL Local CA certificate.
- Verified all three Helm ingress configurations render correctly.
- Confirmed Traefik uses private LAN addresses (`10.0.0.20–10.0.0.22`).
- Documented the existing network limitation: Xfinity gateway administration is unavailable, preventing verification of router settings.

g: Configure Grafana authentication, Kubernetes Secrets, and access restrictions.

1. In VS Code, open:
   `/Users/user/Desktop/SAIL/monitoring/prometheus.yaml`
   Immediately below `grafana:` (line 39), add:

```
 grafana.ini:
    auth.anonymous:
      enabled: false
    users:
      allow_sign_up: false
    auth.basic:
      enabled: true
```

2. Create a Kubernetes Secret for Grafana credentials

```
kubectl create secret generic sail-grafana-admin \
  --namespace monitoring \
  --from-literal=admin-user=admin \
  --from-literal=admin-password="$(openssl rand -base64 32)"
```

```
(base) user@uers-MacBook-Pro certs % kubectl create secret generic sail-grafana-admin \
  --namespace monitoring \
  --from-literal=admin-user=admin \
  --from-literal=admin-password="$(openssl rand -base64 32)"
secret/sail-grafana-admin created
```

Verify: kubectl get secret sail-grafana-admin -n monitoring

```
(base) user@uers-MacBook-Pro certs % kubectl get secret sail-grafana-admin -n monitoring
NAME                 TYPE     DATA   AGE
sail-grafana-admin   Opaque   2      36s
```

4. Connect Grafana to the Secret

In VS Code, open:

`/Users/user/Desktop/SAIL/monitoring/prometheus.yaml`

Immediately below `grafana:`, add:

```
    admin:
      existingSecret: sail-grafana-admin
      userKey: admin-user
      passwordKey: admin-password
```

Validate the Grafana security configuration:

```
helm template sail-monitoring prometheus-community/kube-prometheus-stack \
  --version 92.14.1 \
  --namespace monitoring \
  --values /Users/user/Desktop/SAIL/monitoring/prometheus.yaml \
  > /tmp/sail-monitoring-rendered.yaml
```

no errors

Verify Grafana's administrator credentials:

```
grep -n -A 12 'name: sail-monitoring-grafana' /tmp/sail-monitoring-rendered.yaml | grep -A
12 'admin'
grep -n -B 5 -A 8 'sail-grafana-admin' /tmp/sail-monitoring-rendered.yaml
```

The rendered Helm configuration confirms:

- `GF_SECURITY_ADMIN_USER` references `sail-grafana-admin`, key `admin-user`.
- `GF_SECURITY_ADMIN_PASSWORD` references `sail-grafana-admin`, key `admin-password`.
- Grafana's provisioning containers also reference the correct Secret.

Verify Grafana login restrictions:

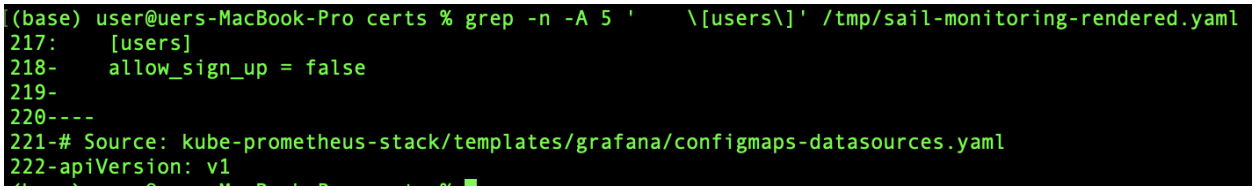
```
grep -n -A 20 'grafana.ini:' /tmp/sail-monitoring-rendered.yaml
```

The rendered configuration confirms:

- Anonymous access: Disabled (`enabled = false`)
- Username/password authentication: Enabled (`enabled = true`)
- Account registration: The `[users]` section is present, but the output cuts off before its settings.

Finish verifying login restrictions:

```
grep -n -A 5 '  \[users\]' /tmp/sail-monitoring-rendered.yaml
```



- Anonymous access: Disabled
- Username/password authentication: Enabled
- Self-registration: Disabled

Step 7: Deploy Prometheus and configure metrics collection
a. Add the selected Prometheus Helm repository.
b. Configure Prometheus resource requests and limits.
c. Configure persistent metrics storage using a storage arrangement appropriate for Prometheus.
d. Set the initial metrics retention period.
e. Deploy Prometheus to Kubernetes.
f. Verify Prometheus pods and services are running.
g. Access the Prometheus interface and confirm metrics are being collected.
Step 8: Configure Raspberry Pi and Kubernetes metrics
a. Deploy or enable Node Exporter across the three Raspberry Pis.
b. Deploy or enable kube-state-metrics.
c. Configure Prometheus to collect Kubernetes component metrics.
d. Verify CPU, memory, filesystem, and network metrics from all three nodes.
e. Verify Kubernetes node, pod, deployment, and workload metrics.
Step 9: Deploy Grafana and create monitoring dashboards
a. Deploy Grafana using a compatible Helm chart.
b. Configure persistent storage for Grafana settings and dashboards.
c. Configure browser access to Grafana.
d. Add Prometheus as a Grafana data source.
e. Create or import dashboards for Raspberry Pi resource utilization.
f. Create or import Kubernetes dashboards for nodes, pods, and workloads.
g. Include shared storage measurements where available.
h. Verify dashboard readings against actual cluster measurements.
Step 10: Deploy Loki and centralized log collection
a. Select a Loki configuration appropriate for the Raspberry Pi cluster.
b. Configure Loki resource allocations and persistent log storage.
c. Set the initial log retention period.
d. Deploy Loki to Kubernetes.
e. Deploy a compatible log collector.
f. Configure collection of Kubernetes container logs and relevant system logs.
g. Add Loki as a Grafana data source.
h. Verify logs can be searched by node, namespace, and workload.
Step 11: Review monitoring storage, retention, and persistence
a. Review the storage configurations established for Prometheus, Grafana, and Loki.
b. Verify that each component uses its intended persistent storage.
c. Review configured metrics and log retention periods.
d. Verify retained data survives relevant pod restarts.
e. Configure monitoring of storage consumption.
f. Identify any storage configuration or retention adjustments needed before adding tracing.

g. Configure backups for Grafana settings and dashboards.
h. Verify recovery of critical monitoring configurations from backup.

Step 12: Deploy OpenTelemetry and a tracing backend

a. Select a compatible trace storage backend.
b. Configure resource allocations and appropriate persistent trace storage.
c. Establish the initial trace retention period.
d. Deploy an OpenTelemetry Collector if required.
e. Configure telemetry collection and export.
f. Connect the tracing backend to Grafana.
g. Generate test traces and verify they can be retrieved.

Step 13: Instrument Orbit and PostgreSQL

a. Add FastAPI instrumentation for request counts, response times, and errors.
b. Configure structured application logging.
c. Add OpenTelemetry tracing to selected application and database operations.
d. Deploy and configure PostgreSQL metrics collection.
e. Configure Prometheus to collect Orbit and PostgreSQL metrics.
f. Create Grafana dashboards for application and database monitoring.
g. Generate application activity and verify metrics, logs, and traces are collected.

Step 14: Configure monitoring alerts

a. Configure Prometheus alerting rules and Alertmanager.
b. Create alerts for unavailable nodes and services.
c. Configure CPU, memory, and storage utilization alerts.
d. Configure Orbit error-rate and PostgreSQL availability alerts.
e. Configure alert notification delivery.
f. Test alert activation and resolution using controlled conditions.

Step 15: Establish normal operating baselines

a. Record normal CPU and memory utilization across all three Raspberry Pis.
b. Record filesystem and network activity.
c. Record Kubernetes workload resource consumption and pod behavior.
d. Review typical log activity and volume.
e. Record Orbit response times, error rates, and PostgreSQL activity.
f. Document observation periods and operating conditions for future comparison.

Step 16: Verify monitoring end-to-end

a. Confirm metrics are collected from all three Raspberry Pi nodes.
b. Verify Kubernetes, Orbit, and PostgreSQL dashboards.
c. Generate and retrieve centralized container logs.
d. Generate and retrieve application traces.
e. Verify monitoring data persistence after relevant pod restarts.
f. Trigger controlled alert conditions and verify notifications.
g. Restore normal conditions and confirm alerts resolve.
h. Record verification results and remaining monitoring gaps.

End State: Grafana provides centralized visibility into SAIL infrastructure, Kubernetes, Orbit, and PostgreSQL through metrics, logs, traces, and alerts. Normal operating baselines are available for comparison during Section 12.

Section 10: Security & Infrastructure Automation

Purpose: Add security visibility and make configuration/security work reproducible rather than manually administered.

Include:

- Falco: runtime behavior detection
- Trivy: image/configuration/vulnerability scanning
- Ansible: system/configuration automation
- Terraform/OpenTofu: IaC
- Bash: appropriate system automation
- Python: more complex automation
- Security scanning incorporated into the application lifecycle
- Establish a **known security baseline**

End state: We can identify vulnerable configurations/images and suspicious runtime activity, while demonstrating repeatable infrastructure/configuration management.

Section 11: AI-Assisted Operations & Deployment Automation

This is where several pieces we've discussed come together.

Purpose: Automate application delivery and give local AI controlled access to SAIL operational information.

Include:

- GitHub Actions: CI/CD
- Helm: Kubernetes application packaging/configuration
- Argo CD: GitOps deployment
- Ollama: local model runtime
- Open WebUI: human AI interface
- **MCP:** controlled AI access to selected SAIL data/tools
- **Deterministic verification:** verify important AI claims against actual system data
- **AI provenance:** retain evidence supplied to the AI and resulting analysis
- **Human approval gates:** AI can recommend consequential remediation; you authorize it
- **Agent observability:** record AI request success/failure and latency

End state: A code change can travel through a professional automated deployment process, while the local AI can perform evidence-based analysis of the environment without being given unrestricted control.

Section 12: Event Scenarios and Investigation Workflows

The purpose of this section is to document the operational and security scenarios used to validate the SAIL environment. Each scenario will include event generation, detection mechanisms, collected evidence, AI-assisted analysis, and remediation actions.

Planned scenarios include:

Scenario	Category
Worker node failure	Infrastructure
Pod crash loop	Infrastructure
Storage utilization threshold exceeded	Infrastructure
Unauthorized shell access to a container	Security
Deployment of a vulnerable container image	Security
Excessive resource consumption by a workload	Security / Operations
Unexpected process execution within a container	Security
Service outage and recovery validation	Operations

Each scenario will document:

1. Event description
2. Event generation method
3. Detection mechanism
4. Logs and metrics collected
5. AI-generated analysis
6. Investigation process
7. Remediation actions
8. Lessons learned

Appendix:

Section 5 Step 7: sail-nfs-client-setup.sh

```
#!/usr/bin/env bash

################################################################################
# SAIL - NFS Client Setup and Verification
#
# Description:
#   Configures NFS client support across the three-node SAIL Kubernetes cluster
#   and verifies that the worker nodes can access the shared NFS storage hosted
#   by pi-node-1.
#
#   The script:
#     1. Installs nfs-common on all three Raspberry Pi nodes.
#     2. Mounts the SAIL NFS share temporarily from pi-node-2.
#     3. Verifies pi-node-2 can write to the shared storage.
#     4. Mounts the same share temporarily from pi-node-3.
#     5. Verifies pi-node-3 can see data written by pi-node-2.
#     6. Verifies pi-node-3 can write to the shared storage.
#     7. Unmounts temporary test mounts when testing is complete.
#
# Requirements:
#   - Run from the SAIL administration MacBook.
#   - SSH access to all three Raspberry Pi nodes must already be configured.
#   - pi-node-1 must already be exporting the NFS share.
#   - The SAIL nodes must be reachable at the addresses defined below.
#
# Usage:
#   chmod +x sail-nfs-client-setup.sh
#   ./sail-nfs-client-setup.sh
#
################################################################################

set -e

# ------------------------------------------------------------------------------
# Configuration
# ------------------------------------------------------------------------------

SSH_USER="sailadmin"

NODE1="10.0.0.20"
NODE2="10.0.0.21"
NODE3="10.0.0.22"

NFS_SERVER="$NODE1"
NFS_SHARE="/mnt/sail-storage/kubernetes"
TEST_MOUNT="/mnt/sail-nfs-test"


# ------------------------------------------------------------------------------
# Functions
# ------------------------------------------------------------------------------

# Display a section heading in the terminal output.
```

```
section() {
   echo
   echo "========================================"
   echo "$1"
   echo "========================================"
}


# Install the NFS client utilities on a SAIL node.
install_nfs_client() {
   local node="$1"

   echo "Configuring ${node}..."

   ssh -t "${SSH_USER}@${node}" \
       "sudo apt update && sudo apt install -y nfs-common"

   echo "NFS client configuration completed on ${node}."
}


# ------------------------------------------------------------------------------
# NFS Client Installation
# ------------------------------------------------------------------------------

section "Installing NFS client utilities"

for node in "$NODE1" "$NODE2" "$NODE3"; do
   install_nfs_client "$node"
done


# ------------------------------------------------------------------------------
# pi-node-2 NFS Verification
# ------------------------------------------------------------------------------

section "Testing NFS access from pi-node-2"

ssh -t "${SSH_USER}@${NODE2}" "
   set -e

   # Create a temporary mount point for the NFS verification.
   sudo mkdir -p ${TEST_MOUNT}

   # Mount the SSD-backed NFS share hosted by pi-node-1.
   sudo mount -t nfs -o vers=4.1 \
       ${NFS_SERVER}:${NFS_SHARE} \
       ${TEST_MOUNT}

   echo 'NFS share mounted on pi-node-2.'

   # Display the active mount for verification.
   findmnt ${TEST_MOUNT}

   # Write a file that will later be verified from pi-node-3.
   touch ${TEST_MOUNT}/pi-node-2-test
```

```
    if [ ! -f ${TEST_MOUNT}/pi-node-2-test ]; then
        echo 'ERROR: pi-node-2 test file was not created.'
        exit 1
    fi

    echo 'pi-node-2 test file created successfully.'

    ls -l ${TEST_MOUNT}

    # The mount is required only for this verification.
    sudo umount ${TEST_MOUNT}

    echo 'pi-node-2 NFS verification completed successfully.'
"


# ------------------------------------------------------------------------------
# pi-node-3 NFS Verification
# ------------------------------------------------------------------------------

section "Testing NFS access from pi-node-3"

ssh -t "${SSH_USER}@${NODE3}" "
    set -e

    # Create a temporary mount point for the NFS verification.
    sudo mkdir -p ${TEST_MOUNT}

    # Mount the same SSD-backed NFS share.
    sudo mount -t nfs -o vers=4.1 \
        ${NFS_SERVER}:${NFS_SHARE} \
        ${TEST_MOUNT}

    echo 'NFS share mounted on pi-node-3.'

    findmnt ${TEST_MOUNT}

    # Confirm that data written from pi-node-2 is visible from pi-node-3.
    if [ ! -f ${TEST_MOUNT}/pi-node-2-test ]; then
        echo 'ERROR: pi-node-3 cannot see the file created by pi-node-2.'
        exit 1
    fi

    echo 'pi-node-3 can see pi-node-2-test.'

    # Confirm that pi-node-3 can also write to the shared storage.
    touch ${TEST_MOUNT}/pi-node-3-test

    if [ ! -f ${TEST_MOUNT}/pi-node-3-test ]; then
        echo 'ERROR: pi-node-3 test file was not created.'
        exit 1
    fi

    echo 'pi-node-3 test file created successfully.'

    ls -l ${TEST_MOUNT}
```

```
   # Remove the temporary mount after verification.
   sudo umount ${TEST_MOUNT}

   echo 'pi-node-3 NFS verification completed successfully.'
"


# ------------------------------------------------------------------------------
# Results
# ------------------------------------------------------------------------------

section "NFS client setup and verification complete"

echo "Verified:"
echo "  - nfs-common installed on all three nodes"
echo "  - pi-node-2 can mount the NFS share"
echo "  - pi-node-2 can write to shared storage"
echo "  - pi-node-3 can mount the NFS share"
echo "  - pi-node-3 can read data written by pi-node-2"
echo "  - pi-node-3 can write to shared storage"
```

Section 6 Step 2: `sail-k3s-prerequisites.sh`

```
#!/usr/bin/env bash

################################################################################
# SAIL - k3s Prerequisite Configuration
#
# Description:
#   Verifies and configures the Raspberry Pi OS cgroup settings required by
#   k3s across all three SAIL Kubernetes nodes.
#
#   The script:
#     1. Connects to each Raspberry Pi over SSH.
#     2. Checks /boot/firmware/cmdline.txt for the required memory cgroup
#        parameters.
#     3. Adds only parameters that are missing.
#     4. Preserves cmdline.txt as a single logical line.
#     5. Verifies the resulting configuration.
#     6. Reports which nodes require a reboot before k3s installation.
#
# Requirements:
#   - Run from the SAIL administration MacBook.
#   - SSH access to all three Raspberry Pi nodes must already work.
#   - Raspberry Pi OS must already be installed on the nodes.
#
# Usage:
#   chmod +x sail-k3s-prerequisites.sh
#   ./sail-k3s-prerequisites.sh
#
################################################################################

set -uo pipefail

# ------------------------------------------------------------------------------
# Configuration
# ------------------------------------------------------------------------------
```

```
SSH_USER="sailadmin"

NODES=(
   "10.0.0.20"
   "10.0.0.21"
   "10.0.0.22"
)

CMDLINE_FILE="/boot/firmware/cmdline.txt"

MODIFIED_NODES=()


# ------------------------------------------------------------------------------
# Functions
# ------------------------------------------------------------------------------

# Display a section heading in terminal output.
section() {
   echo
   echo "========================================"
   echo "$1"
   echo "========================================"
}


# Verify and configure one Raspberry Pi node in a single SSH session.
configure_node() {
   local node="$1"
   local status

   section "Checking ${node}"

   ssh -t "${SSH_USER}@${node}" \
       "CMDLINE_FILE='${CMDLINE_FILE}' bash -s" <<'REMOTE_SCRIPT'

set -e

REQUIRED_CGROUP_MEMORY="cgroup_memory=1"
REQUIRED_CGROUP_ENABLE="cgroup_enable=memory"

cmdline=(cat"CMDLINE_FILE")

echo "Current configuration:"
echo "$cmdline"
echo

missing=()

# Check each required parameter independently.
if [[ " $cmdline " != *" $REQUIRED_CGROUP_MEMORY "* ]]; then
   missing+=("$REQUIRED_CGROUP_MEMORY")
fi

if [[ " $cmdline " != *" $REQUIRED_CGROUP_ENABLE "* ]]; then
   missing+=("$REQUIRED_CGROUP_ENABLE")
fi

# No modification is necessary when both parameters are already present.
if [[ ${#missing[@]} -eq 0 ]]; then
   echo "PASS: Required cgroup parameters are already present."
   exit 0
fi
```

```
echo "Missing parameter(s): ${missing[*]}"
echo "Updating $CMDLINE_FILE..."

additions="${missing[*]}"

# Create a backup before changing the boot configuration.
sudo cp "CMDLINE_FILE""{CMDLINE_FILE}.sail-backup"

# Append only the missing parameters to the existing command line.
sudo sed -i "1 s|[[:space:]]*$| additions|""CMDLINE_FILE"

# Verify that cmdline.txt still contains exactly one logical line.
line_count=(awk'ENDprintNR'"CMDLINE_FILE")

if [[ "$line_count" -ne 1 ]]; then
   echo "ERROR: cmdline.txt no longer contains exactly one logical line."
   exit 1
fi

# Verify both required parameters are now present.
updated_cmdline=(cat"CMDLINE_FILE")

if [[ " $updated_cmdline " != *" $REQUIRED_CGROUP_MEMORY "* ]] ||
  [[ " $updated_cmdline " != *" $REQUIRED_CGROUP_ENABLE "* ]]; then
   echo "ERROR: Required cgroup parameters were not successfully configured."
   exit 1
fi

echo
echo "PASS: Required cgroup parameters are now configured."
echo "Updated configuration:"
echo "$updated_cmdline"

# Exit code 10 tells the Mac-side script that this node was modified.
exit 10

REMOTE_SCRIPT

   status=$?

   case "$status" in
       0)
           # Node was already configured.
           ;;
       10)
           MODIFIED_NODES+=("$node")
           ;;
       *)
           echo
           echo "ERROR: Prerequisite configuration failed on ${node}."
           exit "$status"
           ;;
   esac
}


# ------------------------------------------------------------------------------
# Main
# ------------------------------------------------------------------------------

section "SAIL k3s prerequisite configuration"
```

```
for node in "${NODES[@]}"; do
   configure_node "$node"
done


# ------------------------------------------------------------------------------
# Results
# ------------------------------------------------------------------------------

section "k3s prerequisite verification complete"

if [[ ${#MODIFIED_NODES[@]} -eq 0 ]]; then
   echo "All three nodes are configured correctly."
   echo "No reboot is required."
else
   echo "The following node(s) were modified and must be rebooted:"
   printf '  - %s\n' "${MODIFIED_NODES[@]}"

   echo
   echo "Reboot these nodes before installing k3s."
   echo "After they return, run this script again to confirm all nodes pass."
fi
```

Section 6 Step 3: `sail-k3s-cluster-setup.sh`

```
#!/usr/bin/env bash

################################################################################
# SAIL - k3s Cluster Setup
#
# Installs and verifies the three-node SAIL k3s cluster.
#
#   pi-node-1  10.0.0.20  k3s server/control plane
#   pi-node-2  10.0.0.21  k3s agent
#   pi-node-3  10.0.0.22  k3s agent
#
# SSH connections are reused during the script to avoid repeated SSH
# password prompts.
################################################################################

set -euo pipefail

SSH_USER="sailadmin"
SERVER="10.0.0.20"
NODE2="10.0.0.21"
NODE3="10.0.0.22"

# Reuse SSH connections for 10 minutes.
SSH_OPTS="-o ControlMaster=auto -o ControlPersist=10m -o
ControlPath=/tmp/sail-ssh-%r@%h:%p"


# ------------------------------------------------------------------------------
# Cleanup
# ------------------------------------------------------------------------------

cleanup() {
   for node in "$SERVER" "$NODE2" "$NODE3"; do
       ssh $SSH_OPTS -O exit "${SSH_USER}@${node}" 2>/dev/null || true
   done
}
```

```
trap cleanup EXIT


# ------------------------------------------------------------------------------
# Establish reusable SSH connections
# ------------------------------------------------------------------------------

echo
echo "========================================"
echo "Opening SSH connections"
echo "========================================"

for node in "$SERVER" "$NODE2" "$NODE3"; do
   echo "Connecting to ${node}..."
   ssh $SSH_OPTS -Nf "${SSH_USER}@${node}"
done


# ------------------------------------------------------------------------------
# Install k3s server on pi-node-1
# ------------------------------------------------------------------------------

echo
echo "========================================"
echo "Installing k3s on pi-node-1"
echo "========================================"

ssh -t $SSH_OPTS "${SSH_USER}@${SERVER}" \
   "curl -sfL https://get.k3s.io | sh -"


# ------------------------------------------------------------------------------
# Wait for control plane
# ------------------------------------------------------------------------------

echo
echo "Waiting for pi-node-1 to become Ready..."

for attempt in {1..36}; do

   if ssh $SSH_OPTS "${SSH_USER}@${SERVER}" \
       "sudo k3s kubectl get node pi-node-1 \
       -o jsonpath='{.status.conditions[?(@.type==\"Ready\")].status}'" \
       2>/dev/null | grep -q True; then

       echo "pi-node-1 is Ready."
       break
   fi

   if [[ "$attempt" -eq 36 ]]; then
       echo "ERROR: pi-node-1 did not become Ready."
       exit 1
   fi

   sleep 5
done


# ------------------------------------------------------------------------------
# Retrieve cluster token
# ------------------------------------------------------------------------------

echo
```

```
echo "========================================="
echo "Retrieving cluster join token"
echo "========================================="

K3S_TOKEN=$(
   ssh $SSH_OPTS "${SSH_USER}@${SERVER}" \
       "sudo cat /var/lib/rancher/k3s/server/node-token"
)

if [[ -z "$K3S_TOKEN" ]]; then
   echo "ERROR: Could not retrieve the cluster join token."
   exit 1
fi

echo "Cluster join token retrieved."


# ------------------------------------------------------------------------------
# Install worker nodes
# ------------------------------------------------------------------------------

echo
echo "========================================="
echo "Installing k3s on pi-node-2"
echo "========================================="

ssh -t $SSH_OPTS "${SSH_USER}@${NODE2}" \
   "curl -sfL https://get.k3s.io | \
   K3S_URL='https://${SERVER}:6443' \
   K3S_TOKEN='${K3S_TOKEN}' \
   sh -"


echo
echo "========================================="
echo "Installing k3s on pi-node-3"
echo "========================================="

ssh -t $SSH_OPTS "${SSH_USER}@${NODE3}" \
   "curl -sfL https://get.k3s.io | \
   K3S_URL='https://${SERVER}:6443' \
   K3S_TOKEN='${K3S_TOKEN}' \
   sh -"


# ------------------------------------------------------------------------------
# Verify cluster
# ------------------------------------------------------------------------------

echo
echo "========================================="
echo "Verifying cluster"
echo "========================================="

for attempt in {1..36}; do

   READY_COUNT=$(
       ssh $SSH_OPTS "${SSH_USER}@${SERVER}" \
           "sudo k3s kubectl get nodes --no-headers | \
           awk '\$2 == \"Ready\" {count++} END {print count+0}'"
   )

   if [[ "$READY_COUNT" -eq 3 ]]; then
```

```
        break
    fi

    echo "Ready nodes: ${READY_COUNT}/3"

    if [[ "$attempt" -eq 36 ]]; then
        echo "ERROR: Not all nodes became Ready."
        exit 1
    fi

    sleep 5
done


# ------------------------------------------------------------------------------
# Final result
# ------------------------------------------------------------------------------

ssh $SSH_OPTS "${SSH_USER}@${SERVER}" \
    "sudo k3s kubectl get nodes"

echo
echo "========================================"
echo "k3s cluster setup complete"
echo "========================================"

echo "Verified:"
echo "  - k3s server installed on pi-node-1"
echo "  - k3s agent installed on pi-node-2"
echo "  - k3s agent installed on pi-node-3"
echo "  - all three nodes joined the cluster"
echo "  - all three nodes report Ready"
```

Removal of Display_name from files and database

Section 8: SAIL Orbit: Removal of display_name

Status: Backend changes and database migration completed
Migration: `62ce8ce120c2`
Test results: 40 passed
1. Identify References to `display_name`
From the `sail-orbit` project root, ran:

```
grep -Rn --exclude-dir=node_modules --exclude-dir=.git
--exclude-dir=.sail-orbit-env "display_name" .
```

The command recursively searched for `display_name`, excluding the Node.js dependencies, Git directory, and Python virtual environment.

The search identified references in:

- `app/models/user.py`
- `app/schemas/user.py`
- `app/api/auth.py`
- `app/api/users.py`
- `frontend/src/components/RegisterForm.jsx`
- `frontend/src/api.js`
- `tests/test_auth.py`

- `tests/test_users.py`
- `tests/test_user_profiles.py`
- `tests/test_user_admin.py`
- `migrations/versions/d383129da037_create_users_table.py`

2. Remove `display_name` From Application Code
Updated the affected files:

File	Changes
`app/models/user.py`	Removed the SQLAlchemy display_name column
`app/schemas/user.py`	Removed display_name from UserRegistration and UserProfileUpdate
`app/api/auth.py`	Removed display_name from registration and its response
`app/api/users.py`	Removed display_name from profile and account responses
`frontend/src/components/RegisterForm.jsx`	Removed the display name registration field
`frontend/src/api.js`	Removed the obsolete display_name reference in the API documentation
`tests/test_auth.py`	Removed display_name references
`tests/test_users.py`	Removed display_name from registration requests and response assertions
`tests/test_user_profiles.py`	Removed display_name references and deleted test_update_own_profile_partially
`tests/test_user_admin.py`	Removed display_name references

The `username` field remains the unique login identifier and is included in public profile API responses. The `bio` field was retained.
The original migration, `d383129da037_create_users_table.py`, was left unchanged.

3. Generate the Alembic Migration
Ran:

```
alembic revision --autogenerate -m "remove display_name from users"
```

Alembic generated:

```
migrations/versions/62ce8ce120c2_remove_display_name_from_users.py
```

The output included:

```
Detected removed table 'users'
```

The generated `upgrade()` contained:

```
op.drop_table('users')
```

This would have removed the entire `users` table, so was corrected before execution.

4. Edit the Migration File

The migration file was opened in TextEdit:

```
open -e migrations/versions/62ce8ce120c2_remove_display_name_from_users.py
```

The autogenerated table removal and recreation operations were replaced.

Corrected upgrade:

```
def upgrade() -> None:
    """Remove the display_name column from the users table."""
    op.drop_column("users", "display_name")
```

Corrected downgrade:

```
def downgrade() -> None:
    """Restore the display_name column to the users table."""
    op.add_column(
        "users",
        sa.Column(
            "display_name",
            sa.String(length=100),
            nullable=False,
            server_default="",
        ),
    )
    op.alter_column(
        "users",
        "display_name",
        server_default=None,
    )
```

The downgrade uses a temporary default when restoring the non-nullable column to a table containing existing records.

5. Verify and Apply the Migration

Checked the current database revision: `alembic current`

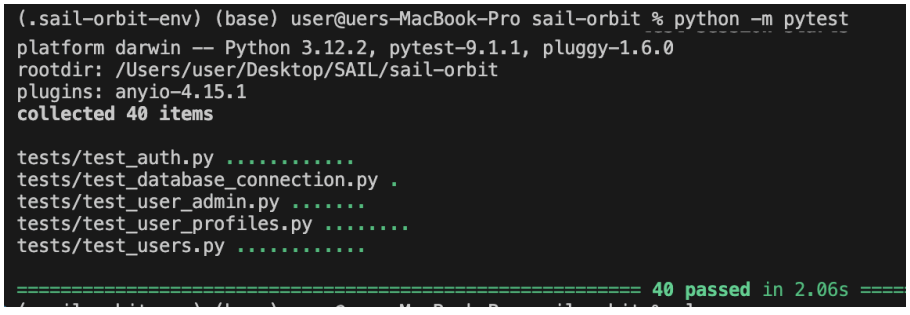
Output: `d383129da037`

Applied the corrected migration: `alembic upgrade head`

The terminal reported the upgrade from `d383129da037` to `62ce8ce120c2`.

6. Run Backend Tests

`python -m pytest`



7. Final Status

- Removed `display_name` references from the application code and tests.
- Preserved `username` and `bio`.
- Retained the original database migration.

- Corrected the autogenerated Alembic migration to drop only the `display_name` column.
- Successfully applied the migration.
- Verified that all 40 backend tests pass.

Section 8 Step 17i: Deferred Issues: Frontend Dependencies and Security

Identified: October 14, 2026
Status: Deferred until the application is ready for deployment.

1. Node.js compatibility

- Current version: Node.js 23.6.1.
- ESLint 10 requires Node.js 20.19+, 22.13+, or 24+.
- npm reported `EBADENGINE` warnings during installation.
- Recommended resolution: Move to a supported Node.js LTS version.

2. Dependency vulnerabilities

- npm initially reported 10 vulnerabilities: 1 moderate, 7 high, and 2 critical.
- A production-only audit (`npm audit --omit=dev`) reported 7 high-severity vulnerabilities.
- The production findings originate from the `shadcn` dependency chain, including `braces`, `micromatch`, and `fast-glob`.
- The reported `braces` vulnerability involves potential denial of service through deeply nested patterns.
- npm suggested a potentially breaking downgrade to `shadcn@1.0.0` using `npm audit fix --force`.
- No corrective changes were made.

3. Dependency classification

- `shadcn` is currently listed under production dependencies.
- A search found no direct imports of `shadcn` in the frontend's `src` directory.
- Moving `shadcn` to development dependencies may reduce production exposure, but this has not been implemented or verified.

Decision

Defer remediation while Orbit remains under active development. The vulnerabilities have not been demonstrated to affect Step 17 functionality.

Before deployment: Reassess dependency vulnerabilities, confirm which packages are required at runtime, upgrade Node.js to a supported version, apply appropriate fixes, and rerun security audits and application tests.